NETWORK INFRASTRUCTURE, SECURITY & OSINT

UFCD 5892 | Modelos de Gestão de Redes e Suporte a Clientes

Relatório técnico de laboratório

Pedro Gonçalves | Portefólio pessoal

Visão geral

Este projeto documenta a implementação de uma infraestrutura virtual que combina serviços de rede Windows e Linux, firewall, controlo de acessos e testes de segurança em ambiente de laboratório. Inclui evidências visuais, configurações, procedimentos de diagnóstico e exercícios de recolha de informação pública (OSINT).

Objetivos e âmbito

Projetar uma rede interna, configurar serviços de identidade e resolução de nomes, implementar distribuição dinâmica de endereços, estudar continuidade operacional e verificar controlos de acesso. Os testes ofensivos apresentados destinam-se exclusivamente a sistemas do laboratório autorizado.

Infraestrutura de referência

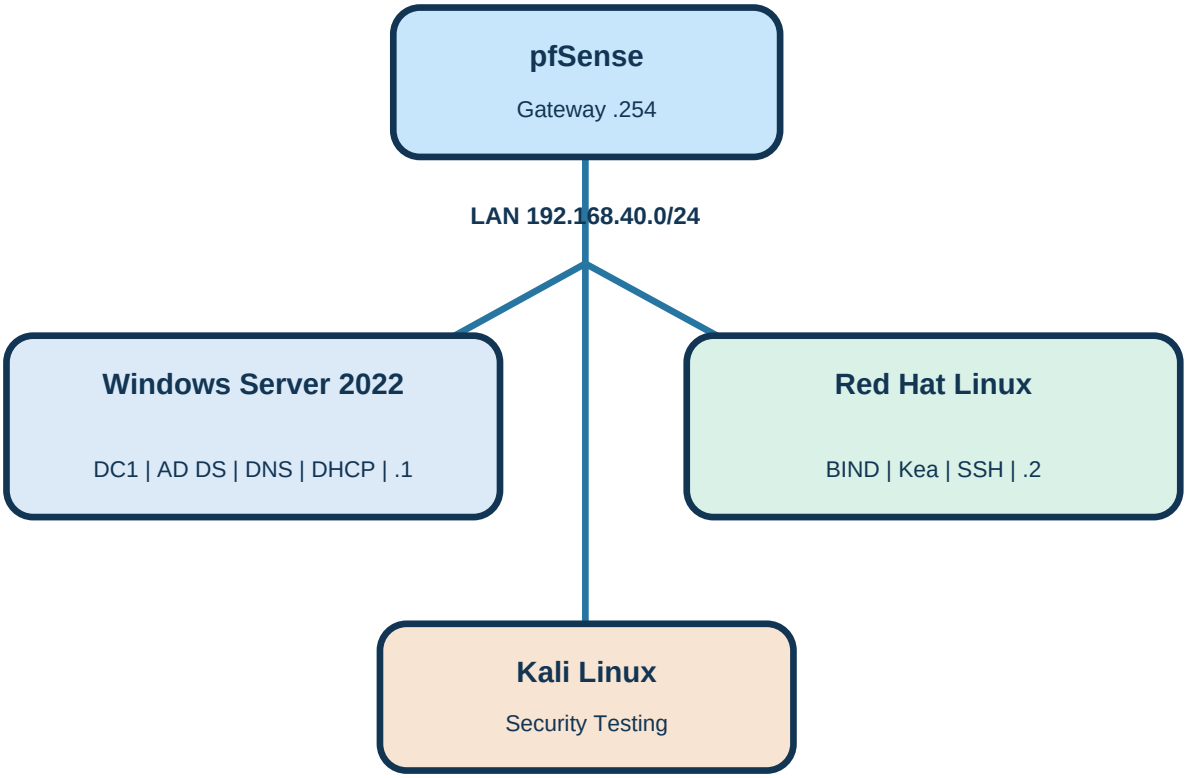
Componente	Função / endereçamento
pfSense CE	Firewall e gateway LAN: 192.168.40.254
Windows Server 2022 (DC1)	AD DS, DNS, DHCP: 192.168.40.1
Red Hat Enterprise Linux	DNS BIND, DHCP Kea, SSH: 192.168.40.2
Kali Linux	Diagnóstico e testes de segurança autorizados
Rede / domínio	192.168.40.0/24 | layer8.pt
DHCP	192.168.40.45 a 192.168.40.89

Percurso técnico

1. Configuração da rede, gateway e serviços de domínio. 2. Certificados e administração HTTPS no pfSense. 3. DNS, DHCP e redundância com Red Hat. 4. Automatização e diagnóstico com Python e systemd. 5. Administração SSH e regras NAT. 6. Testes de segurança e análise dos resultados. 7. Exercícios OSINT com pesquisa geográfica e análise de metadados.

Topologia lógica do laboratório

Representação esquemática original. As ligações indicam conectividade lógica e não representam a cablagem física.



Notas de interpretação

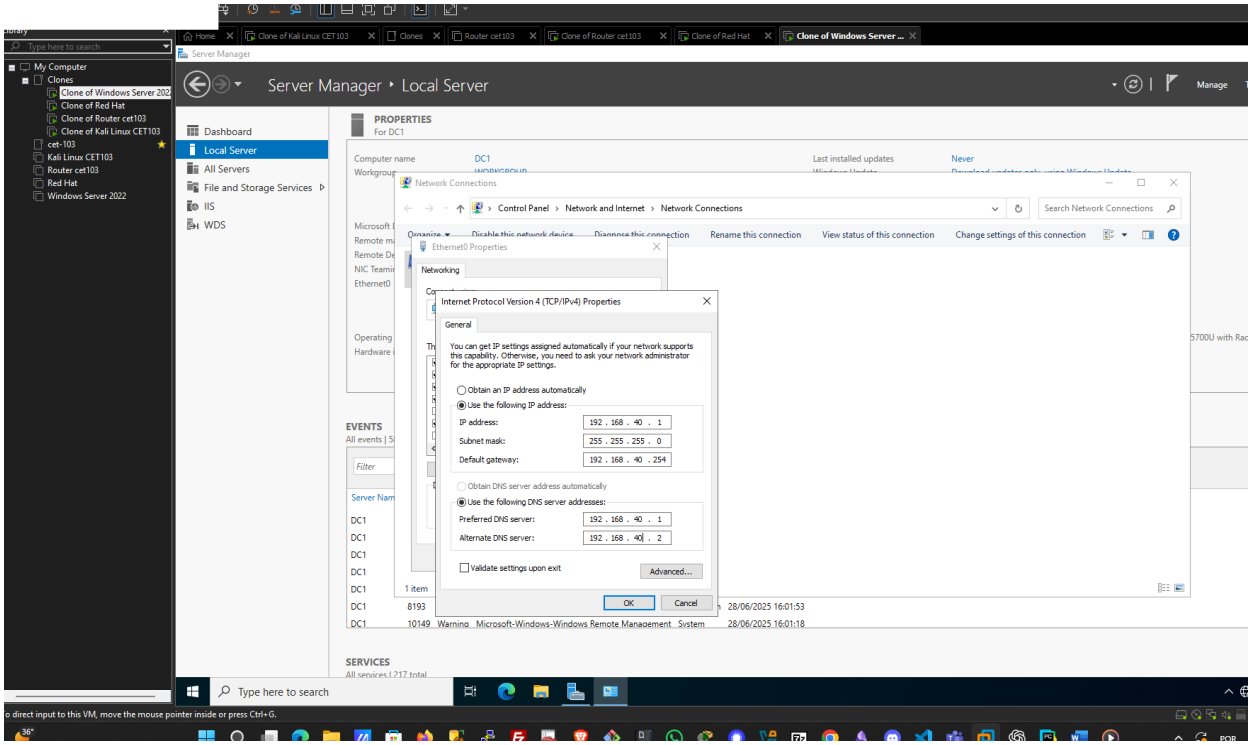
O documento original contém registos de configuração, tentativas falhadas e inconsistências de endereçamento ou portas. Estes elementos foram preservados como evidência histórica de aprendizagem e não devem ser interpretados como validação de uma implementação de produção.

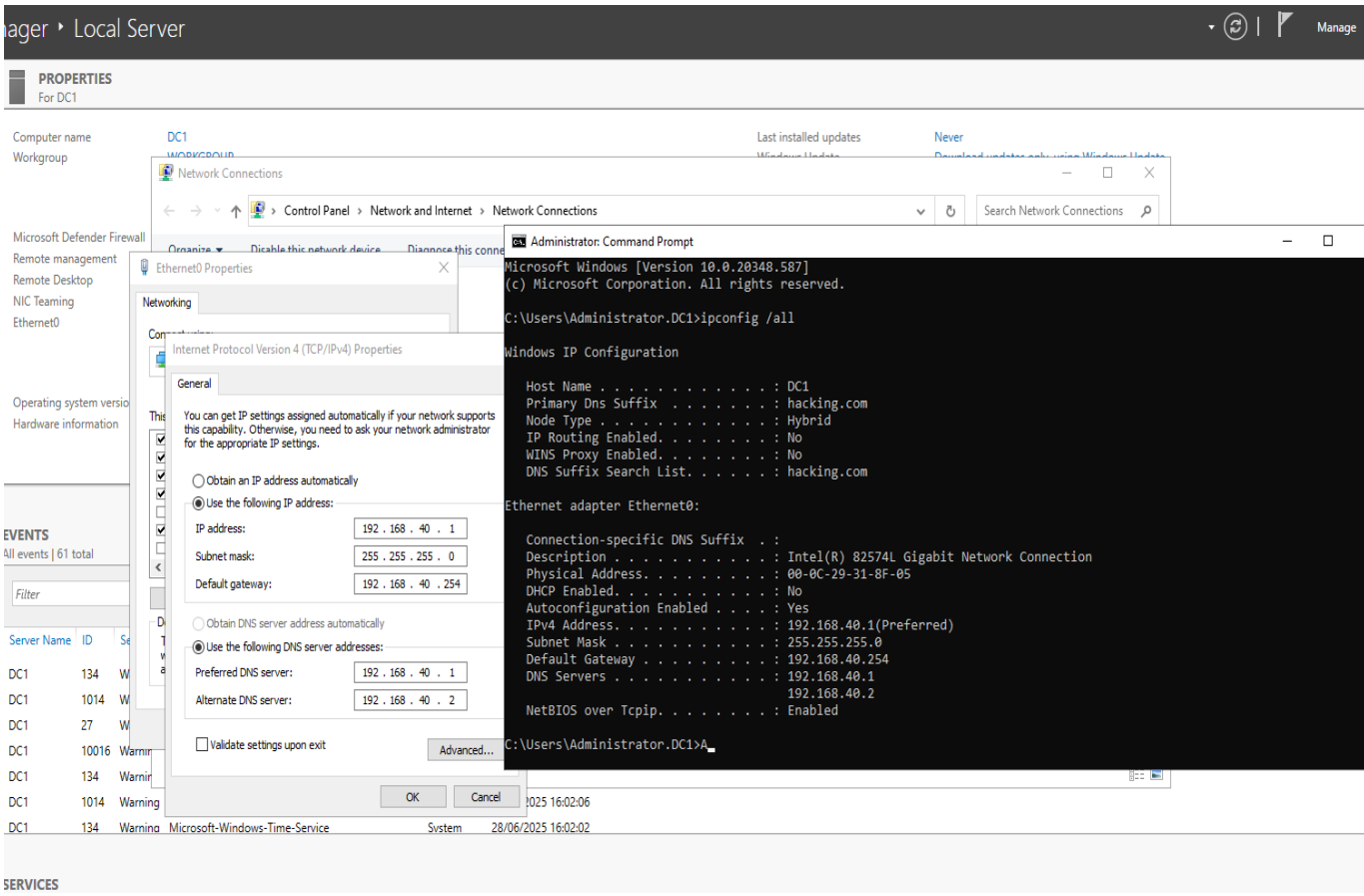
Segurança e confidencialidade

As palavras-passe explícitas identificadas foram ocultadas na edição pública. Os testes de autenticação, varrimento e interceção de tráfego devem ser executados apenas em ambientes autorizados. Alterar uma porta SSH não substitui autenticação forte, limites de tentativas ou políticas de acesso.

Organização do relatório

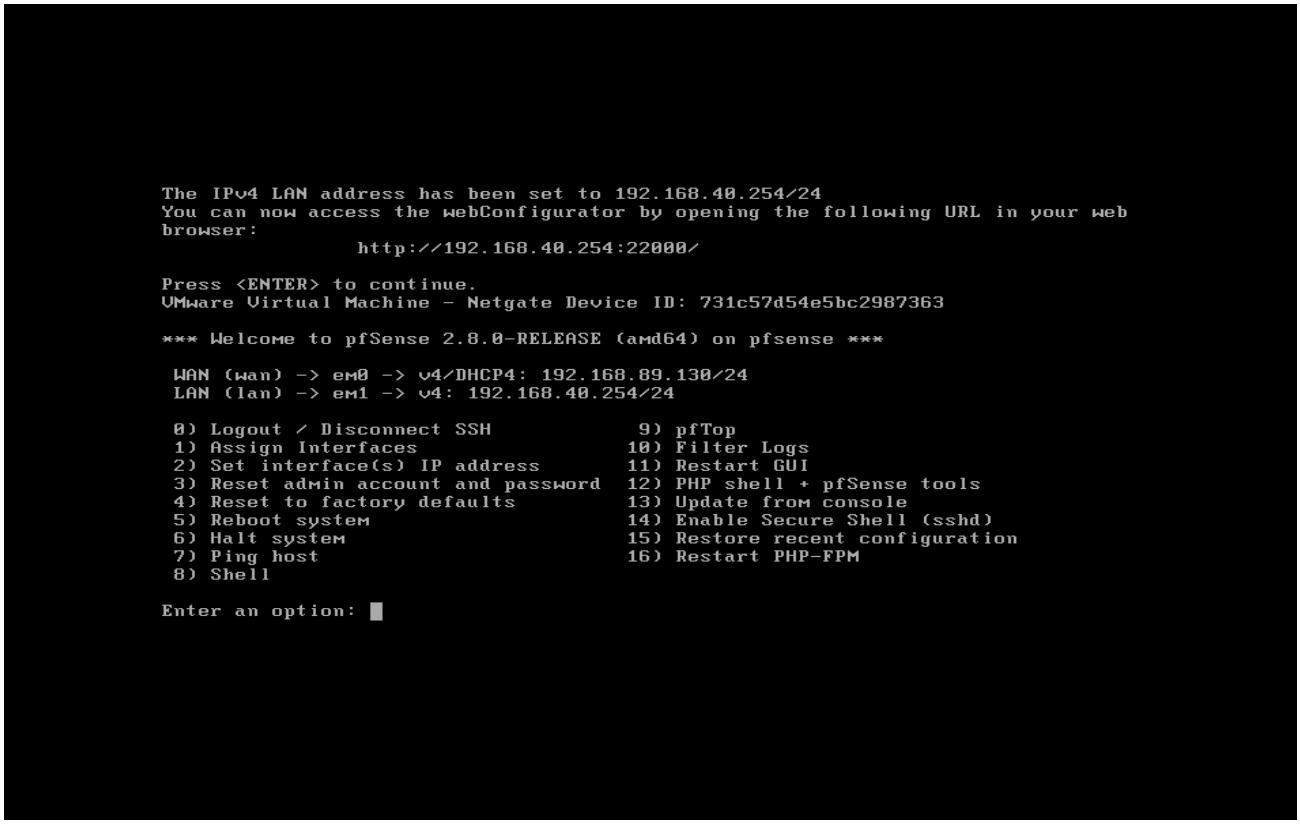
As páginas seguintes preservam as evidências e capturas do trabalho técnico original, com remoção da identidade visual institucional nas margens. O texto do enunciado foi substituído por esta apresentação independente.





Configurações do pfsense :

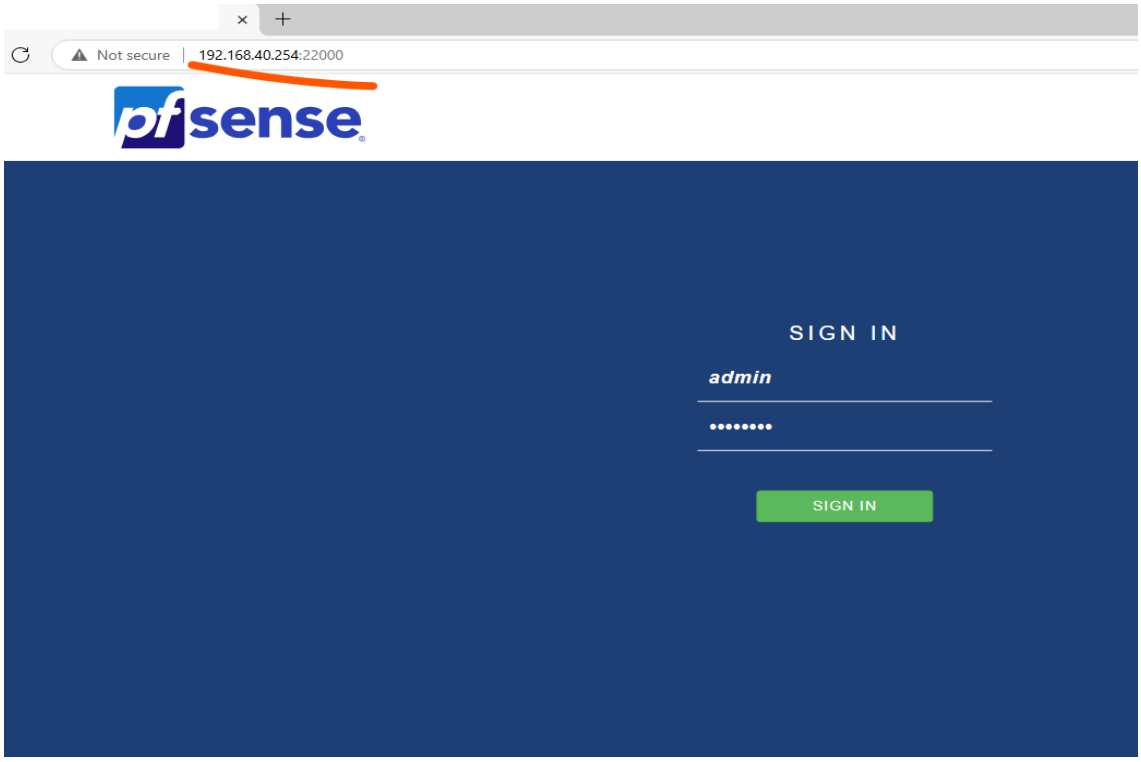
N -192.168.40.254



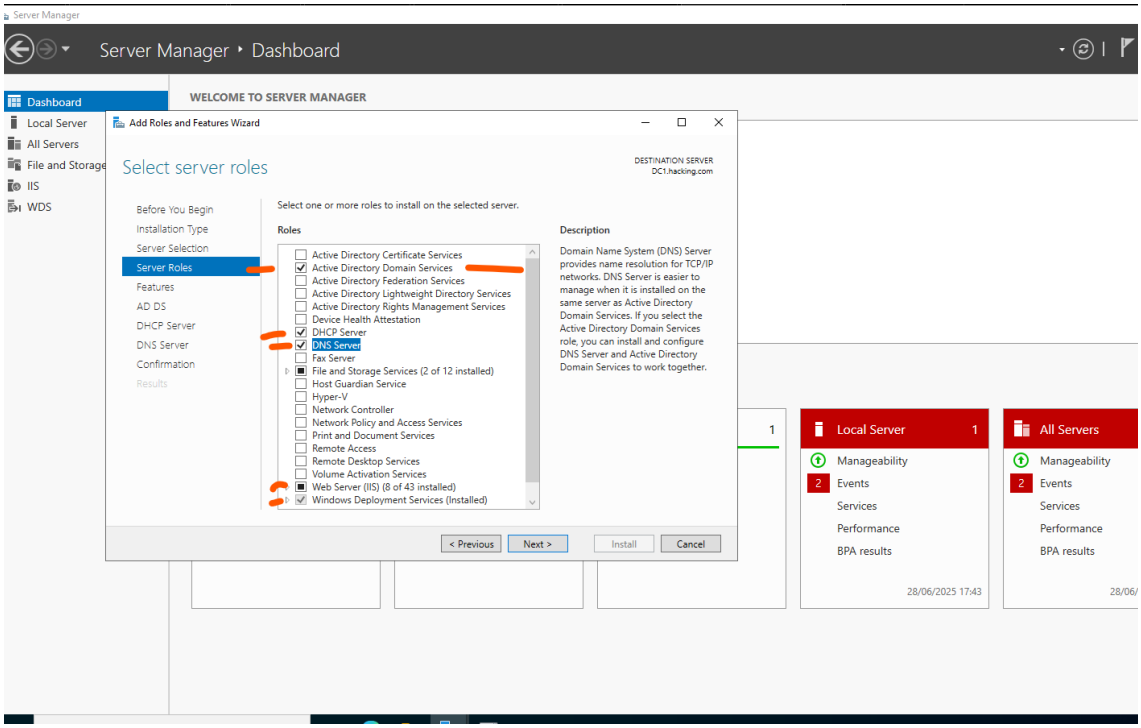
Acesso a pfsense , via web , endereço 192.168.40.254/24

User :admin

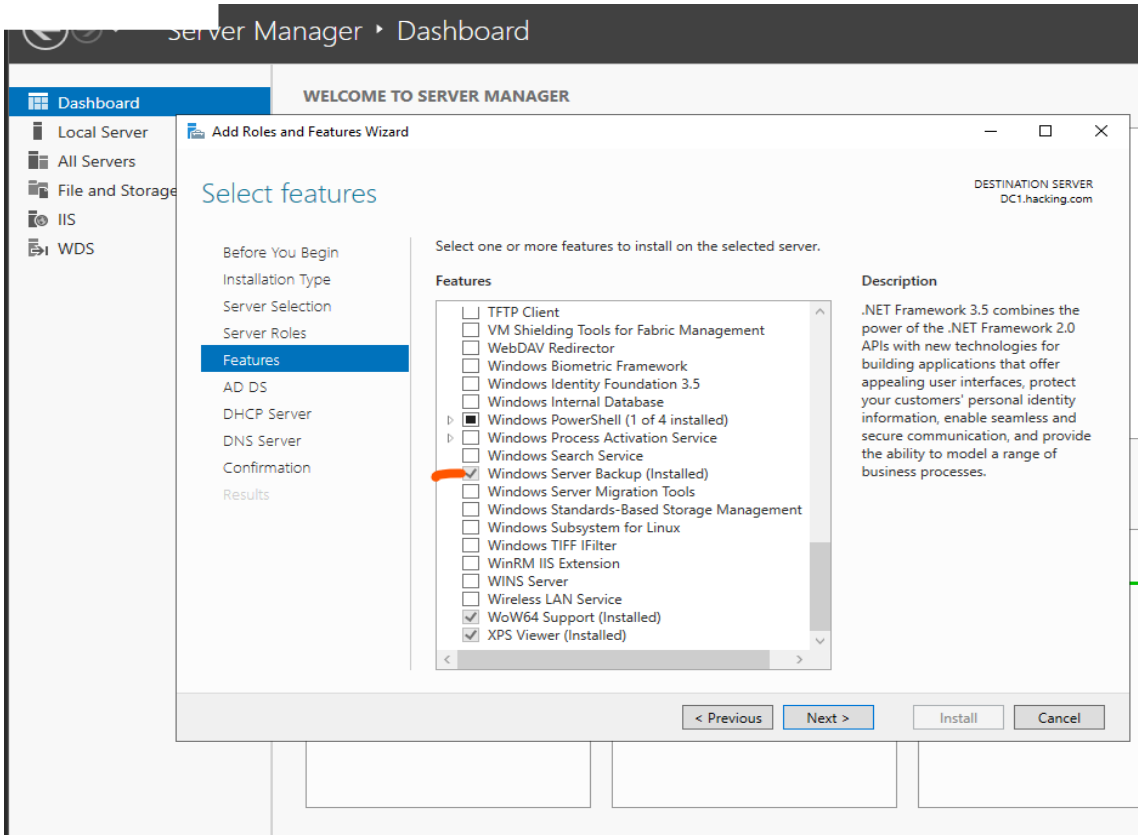
Passwd :

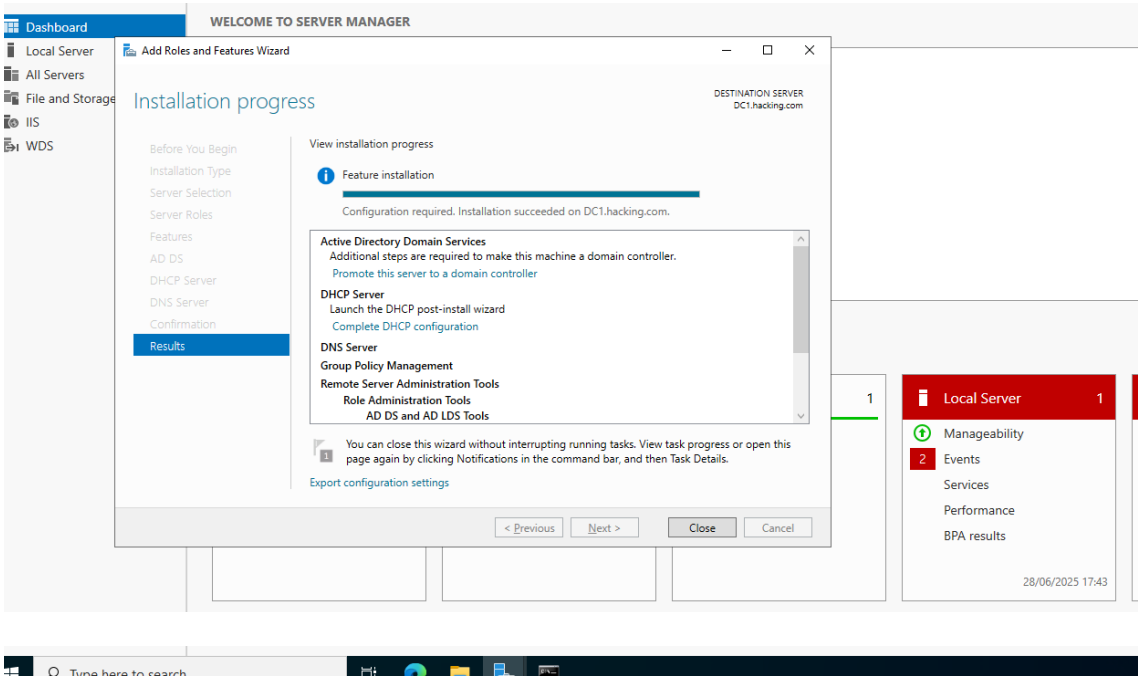


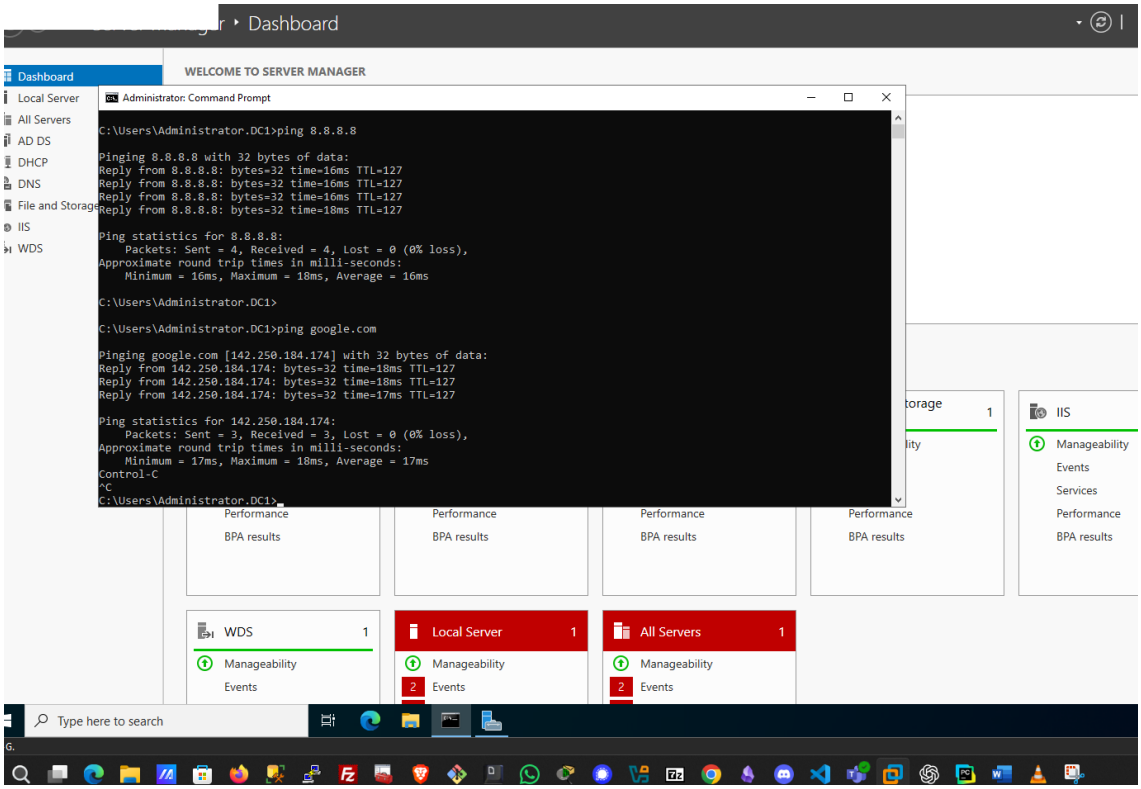
Instalação Active directory , DHCPserver , DNS server , WEB server (iis) Windows Deployment Services



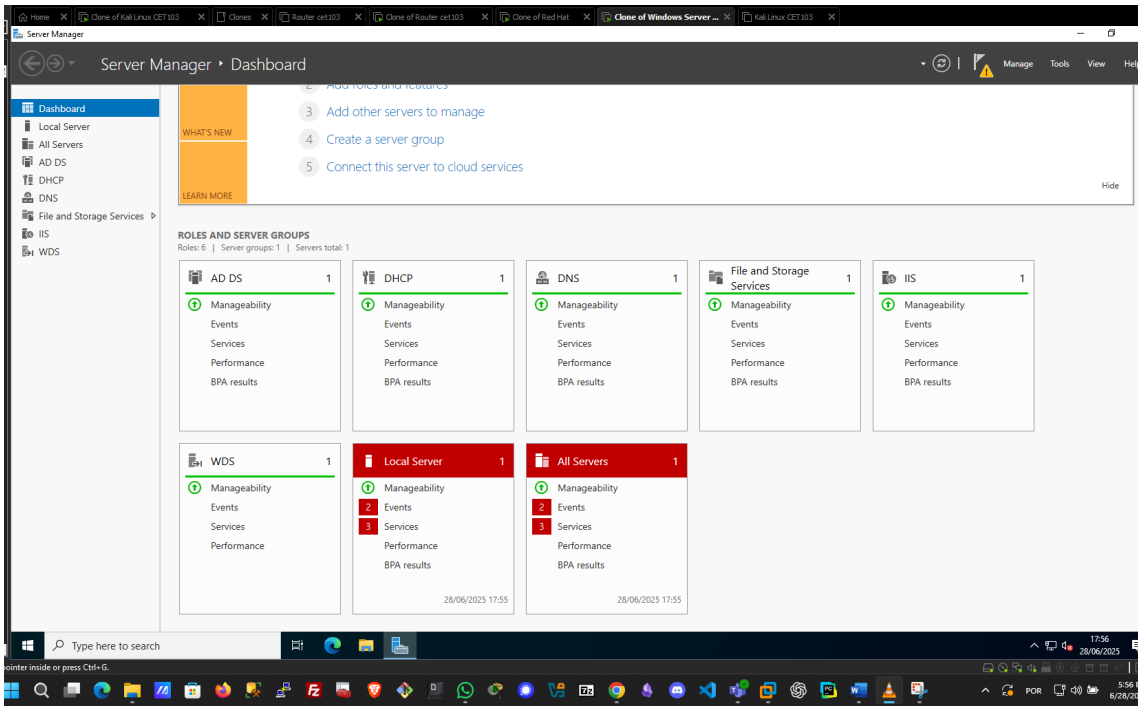
Windows Server Backup


Server Manager ▸ Dashboard
Dashboard
Local Server
All Servers
File and Storage
IIS
WDS
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Select features
DESTINATION SERVER
DC1.hacking.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD DS
DHCP Server
DNS Server
Confirmation
Results
Select one or more features to install on the selected server.
Features
TFTP Client
VM Shielding Tools for Fabric Management
WebDAV Redirector
Windows Biometric Framework
Windows Identity Foundation 3.5
Windows Internal Database
Windows PowerShell (1 of 4 installed)
Windows Process Activation Service
Windows Search Service
Windows Server Backup (Installed)
Windows Server Migration Tools
Windows Standards-Based Storage Management
Windows Subsystem for Linux
Windows TIFF IFilter
WinRM IIS Extension
WINS Server
Wireless LAN Service
WoW64 Support (Installed)
XPS Viewer (Installed)
Description
.NET Framework 3.5 combines the power of the .NET Framework 2.0 APIs with new technologies for building applications that offer appealing user interfaces, protect your customers' personal identity information, enable seamless and secure communication, and provide the ability to model a range of business processes.
< Previous
Next >
Install
Cancel

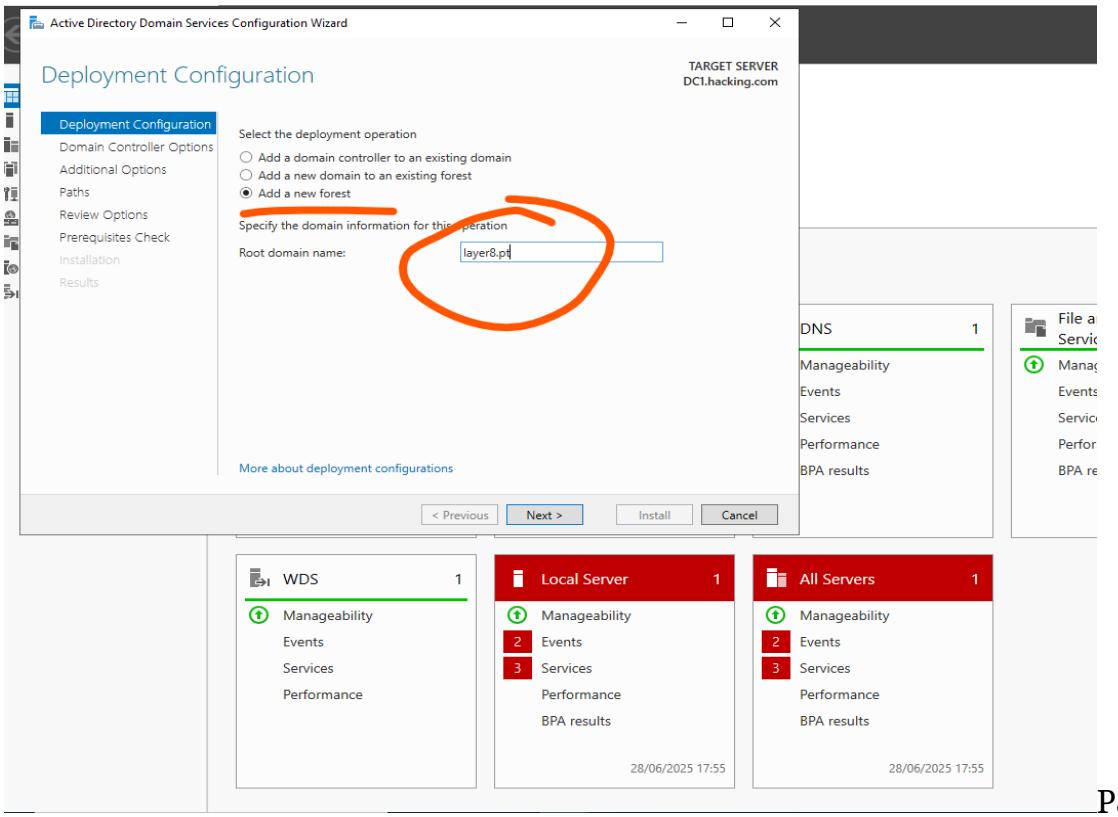

Dashboard
Local Server
All Servers
File and Storage
IIS
WDS
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Installation progress
DESTINATION SERVER
DC1.hacking.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD DS
DHCP Server
DNS Server
Confirmation
Results
View installation progress
Feature installation
Configuration required. Installation succeeded on DC1.hacking.com.
Active Directory Domain Services
Additional steps are required to make this machine a domain controller.
Promote this server to a domain controller
DHCP Server
Launch the DHCP post-install wizard
Complete DHCP configuration
DNS Server
Group Policy Management
Remote Server Administration Tools
Role Administration Tools
AD DS and AD LDS Tools
You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.
Export configuration settings
< Previous
Next >
Close
Cancel
Local Server
1
Manageability
2
Events
Services
Performance
BPA results
28/06/2025 17:43



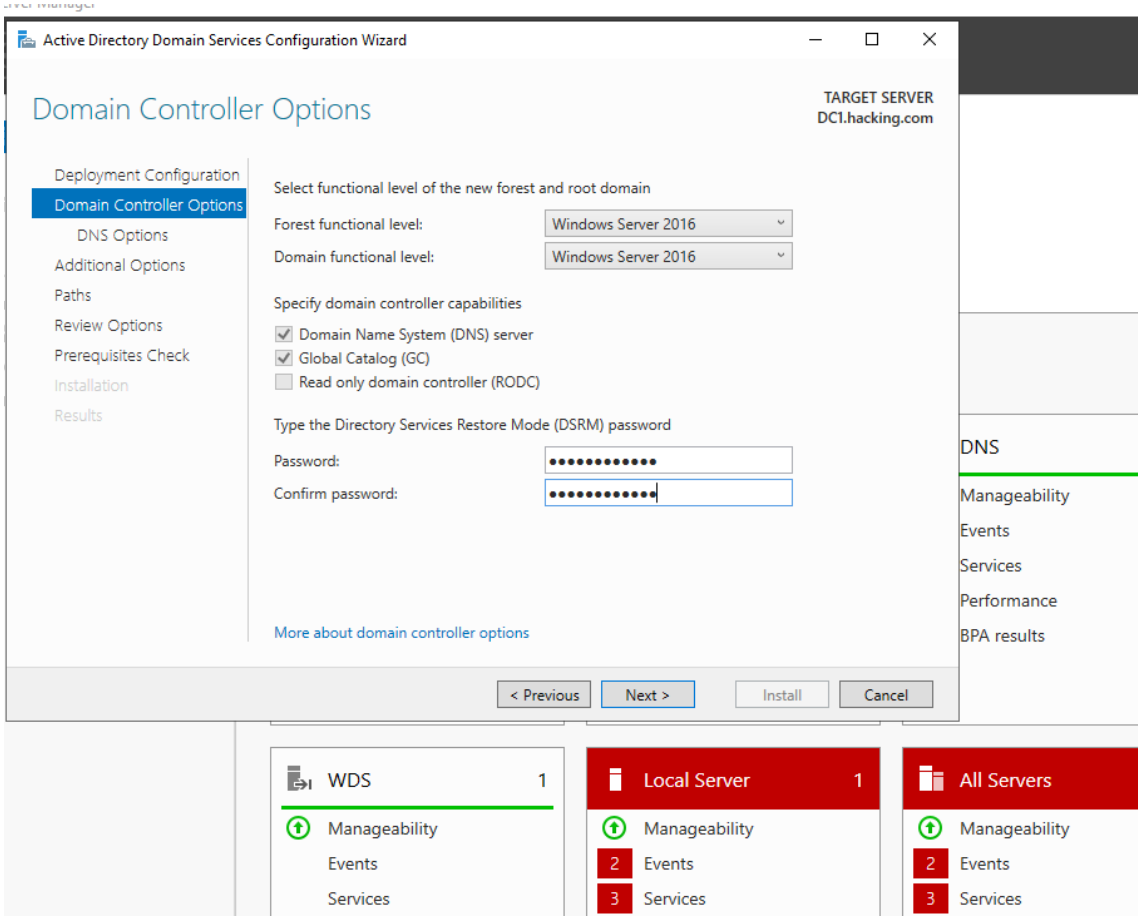
Resultado da Instalação (Serviços):



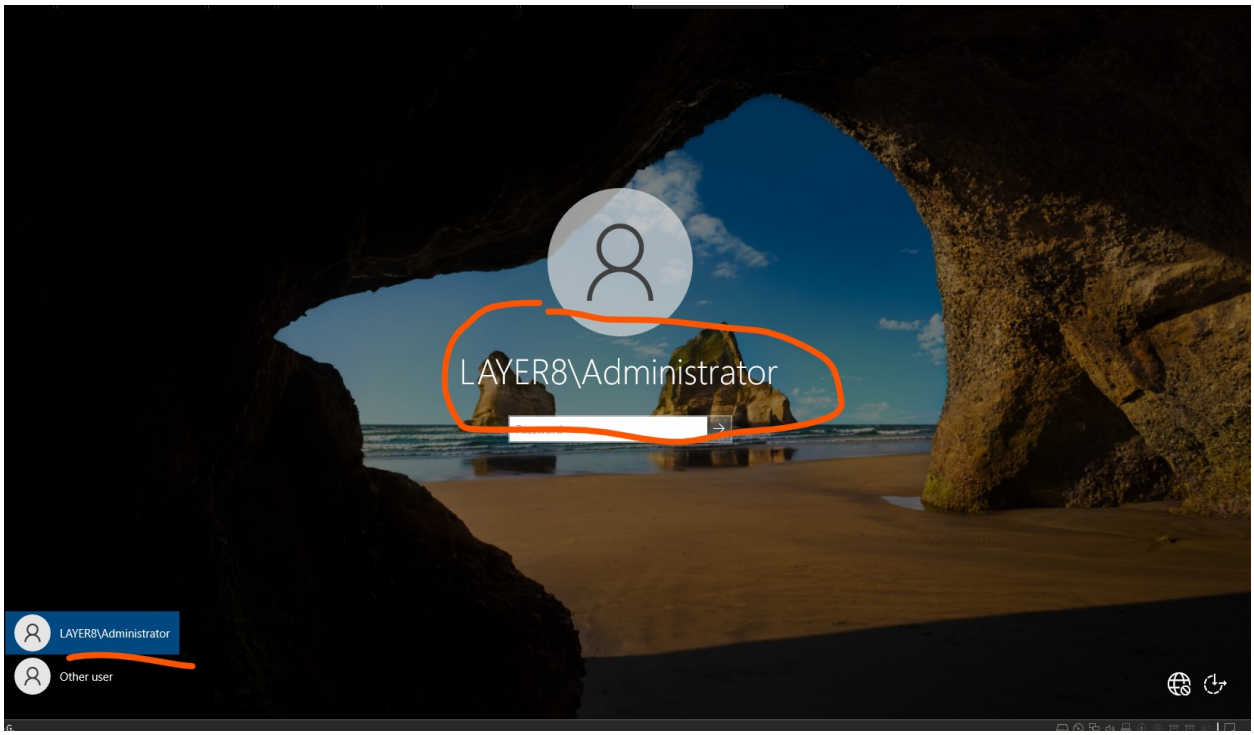
Adicionar um domínio (new forest) -**layer8.pt**


Active Directory Domain Services Configuration Wizard
Deployment Configuration
TARGET SERVER
DC1.hacking.com
Deployment Configuration
Domain Controller Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Select the deployment operation
Add a domain controller to an existing domain
Add a new domain to an existing forest
Add a new forest
Specify the domain information for this operation
Root domain name:
layer8.pt
More about deployment configurations
< Previous
Next >
Install
Cancel
DNS 1
Manageability
Events
Services
Performance
BPA results
WDS 1
Manageability
Events
Services
Performance
Local Server 1
Manageability
Events
Services
Performance
BPA results
28/06/2025 17:55
All Servers 1
Manageability
Events
Services
Performance
BPA results
28/06/2025 17:55

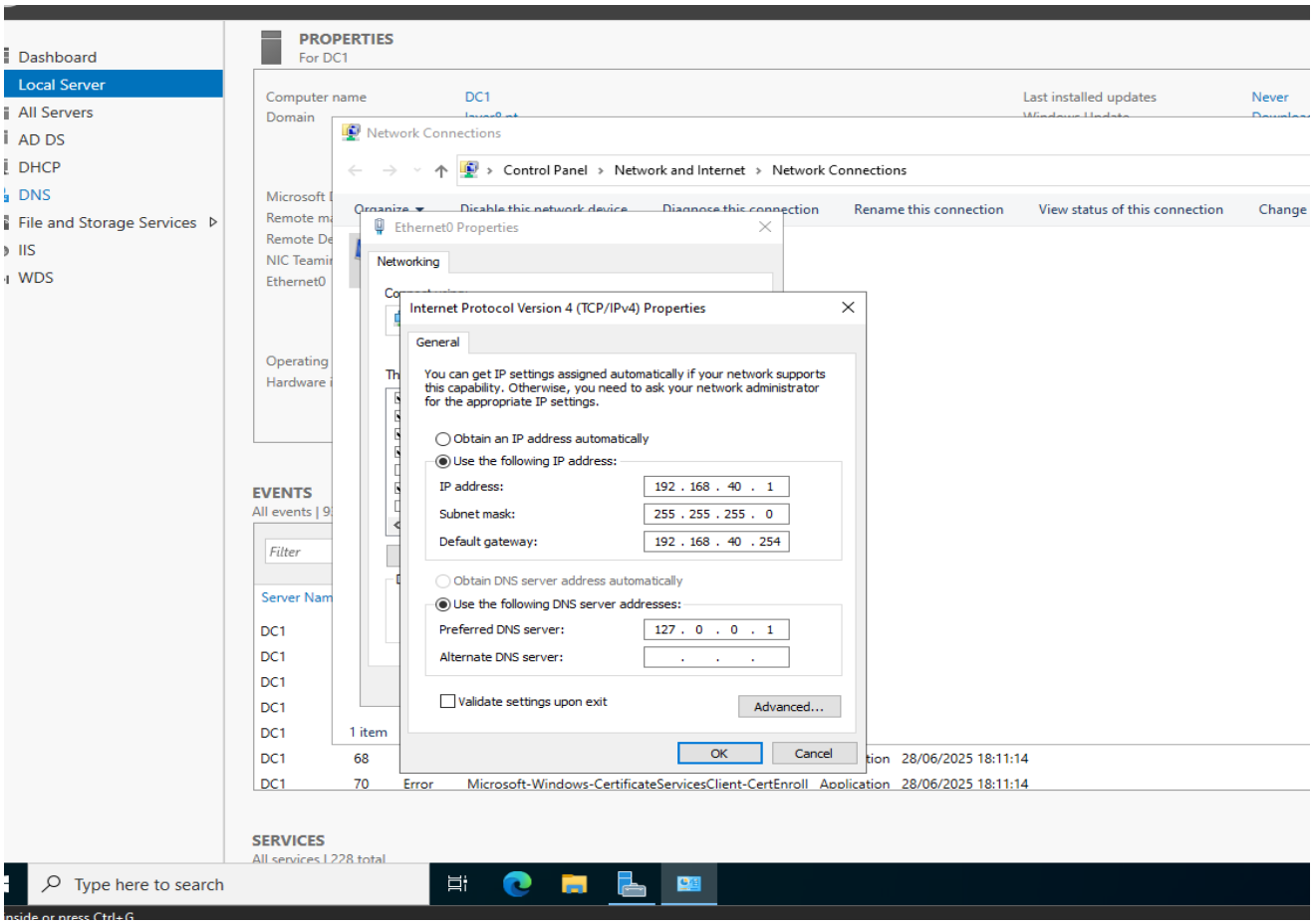
Password :


Active Directory Domain Services Configuration Wizard
Domain Controller Options
TARGET SERVER
DC1.hacking.com
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Select functional level of the new forest and root domain
Forest functional level:
Windows Server 2016
Domain functional level:
Windows Server 2016
Specify domain controller capabilities
Domain Name System (DNS) server
Global Catalog (GC)
Read only domain controller (RODC)
Type the Directory Services Restore Mode (DSRM) password
Password:
Confirm password:
More about domain controller options
< Previous
Next >
Install
Cancel
DNS
Manageability
Events
Services
Performance
BPA results
WDS 1
Manageability
Events
Services
Local Server 1
Manageability
Events
Services
All Servers
Manageability
Events
Services

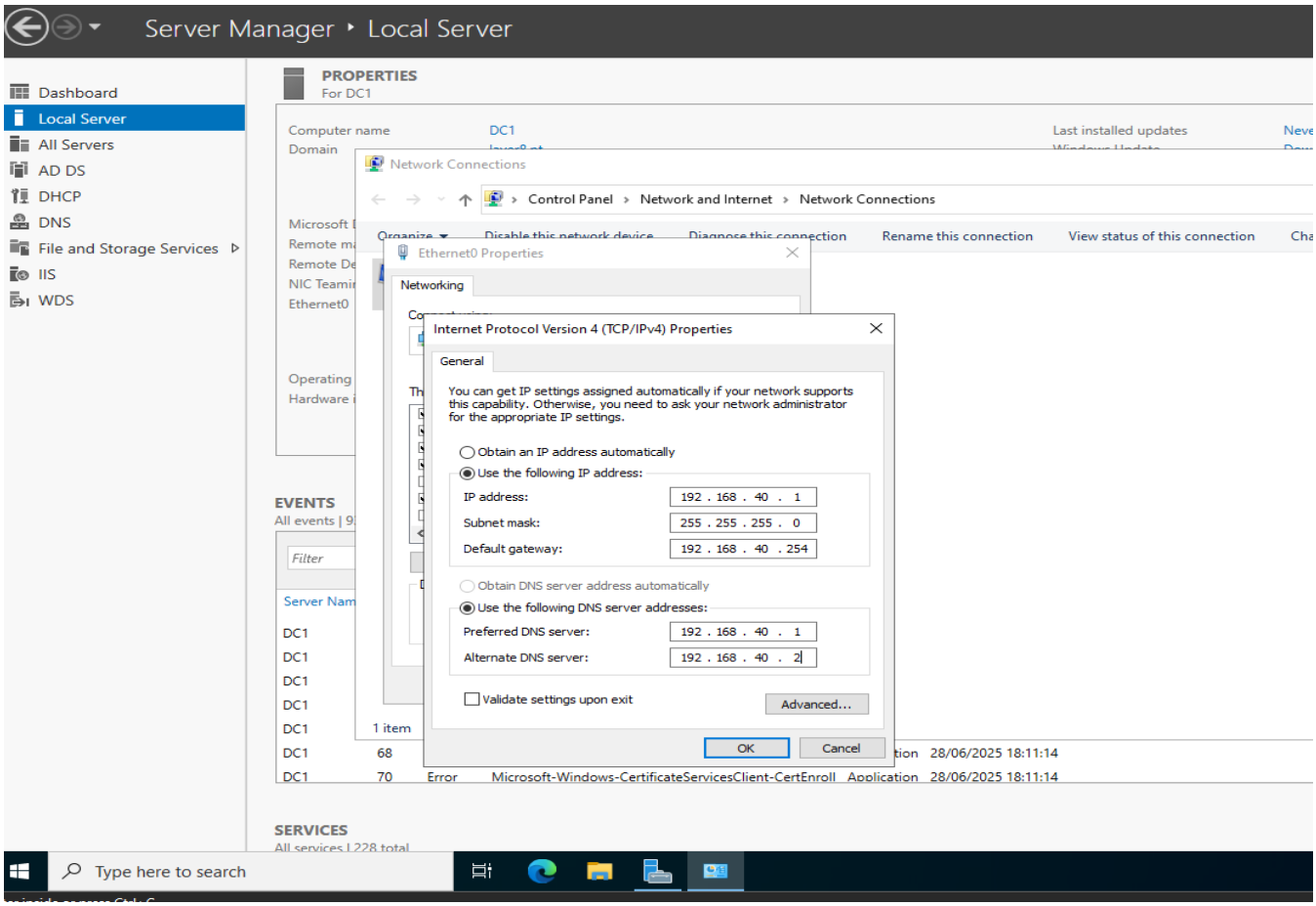
e abertura de sessão WindServer -**LAYER8\Administrator**



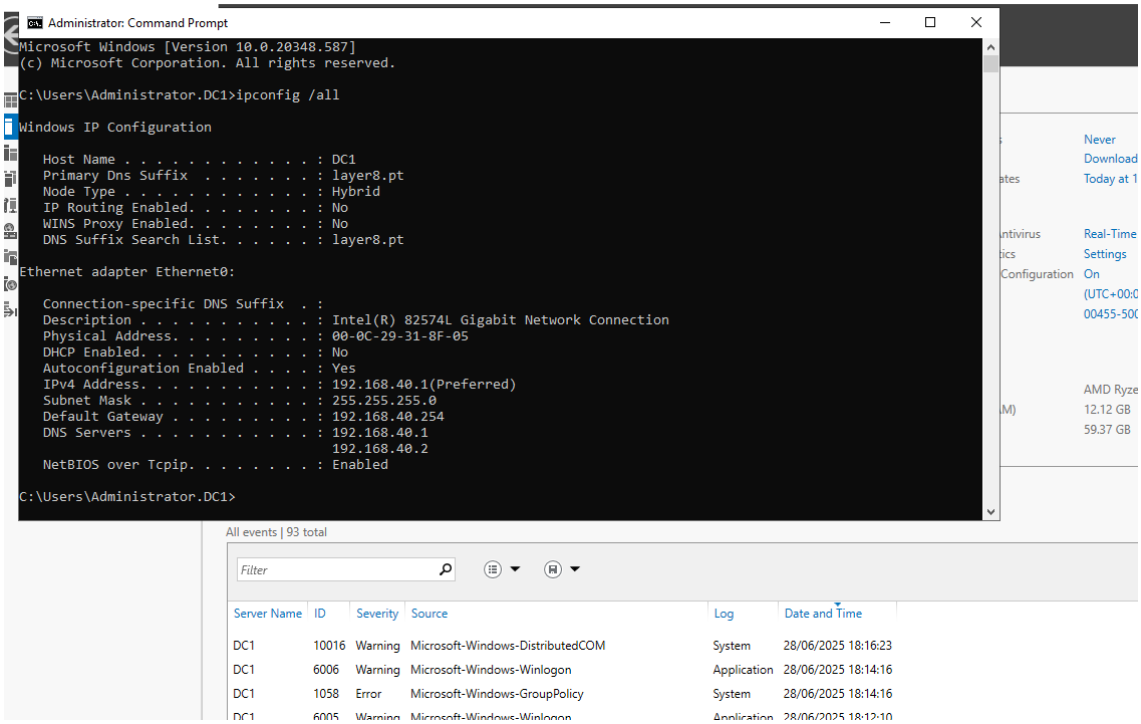
Confirmação que houve alteração automática apenas para DNS primário , proceder a alterações ....



terações :Prefere DNS server :192.168.40.1 , Alternate DNS Server:192.168.40.2

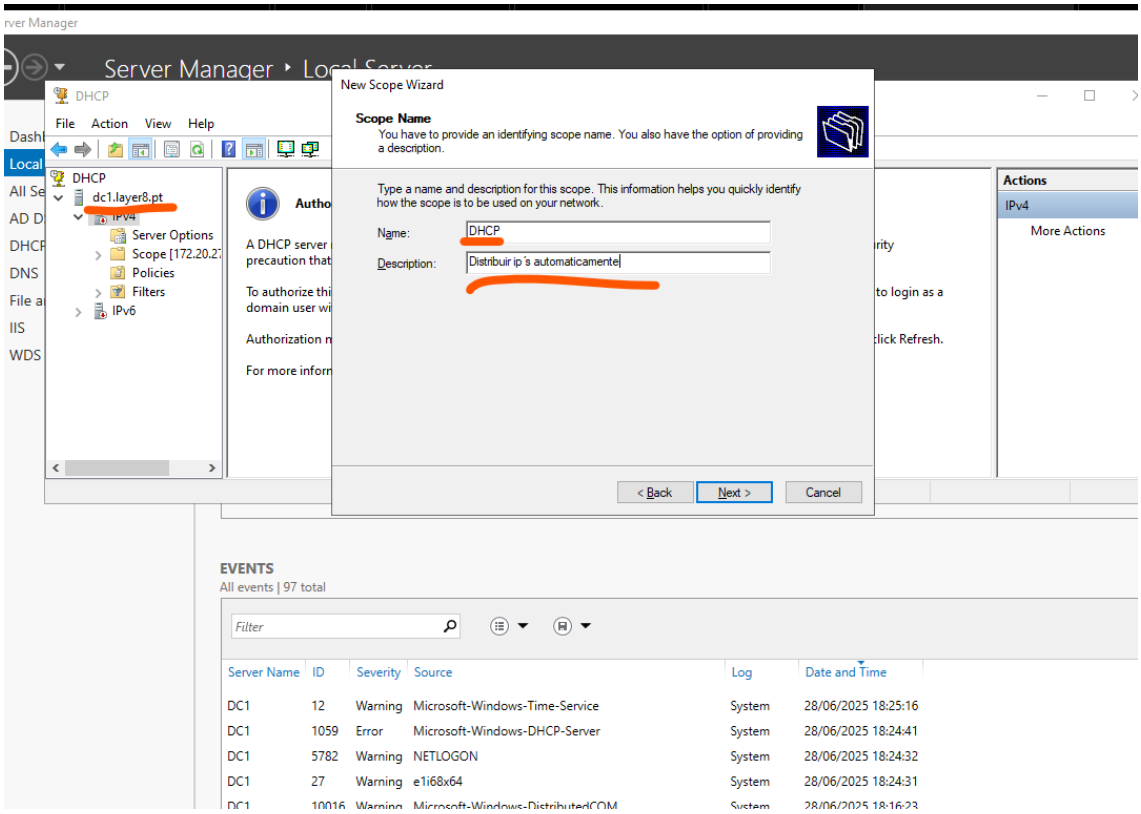


Confirmação de DNS: “192.168.40.1 , 192.168.40.2

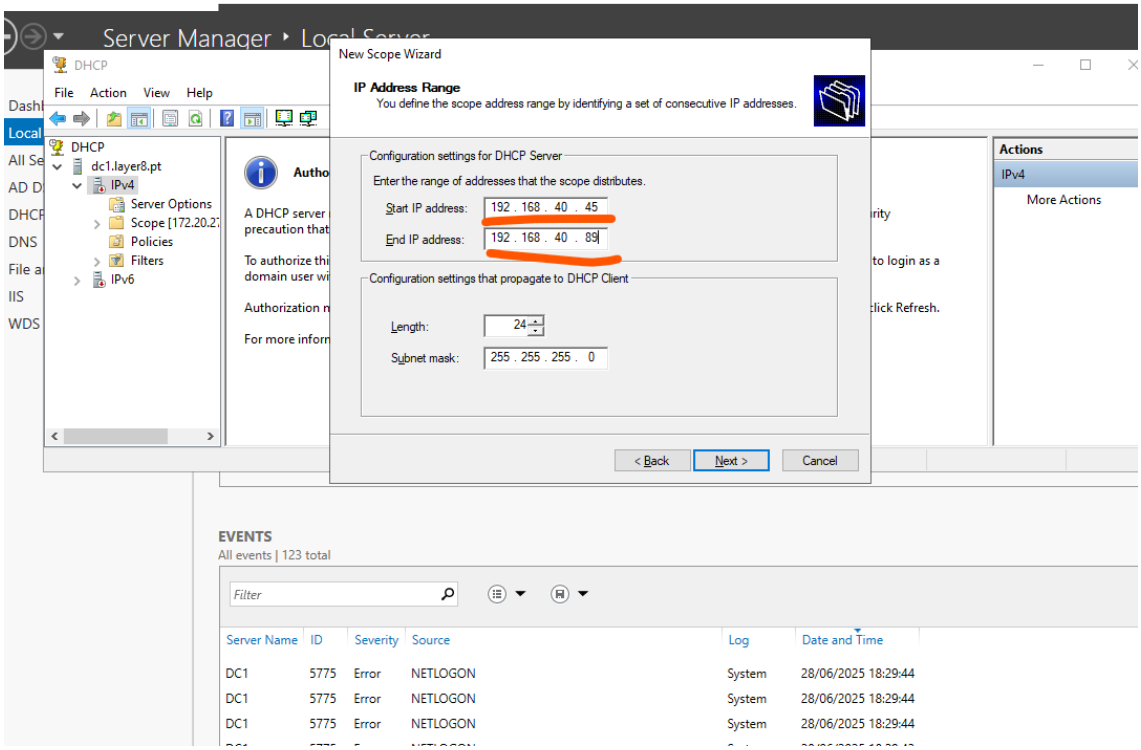


Configuração de DHCP :tools -DHCP

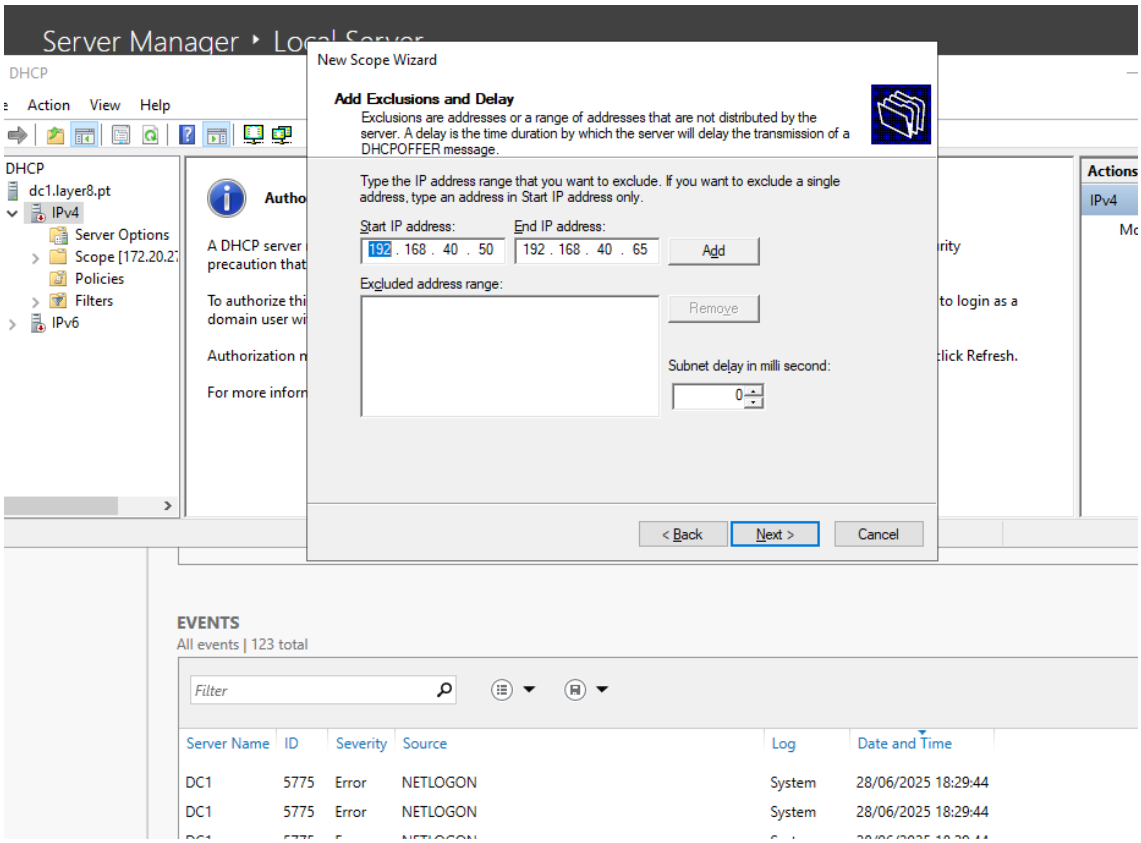
Distribuir ip´s automaticamente



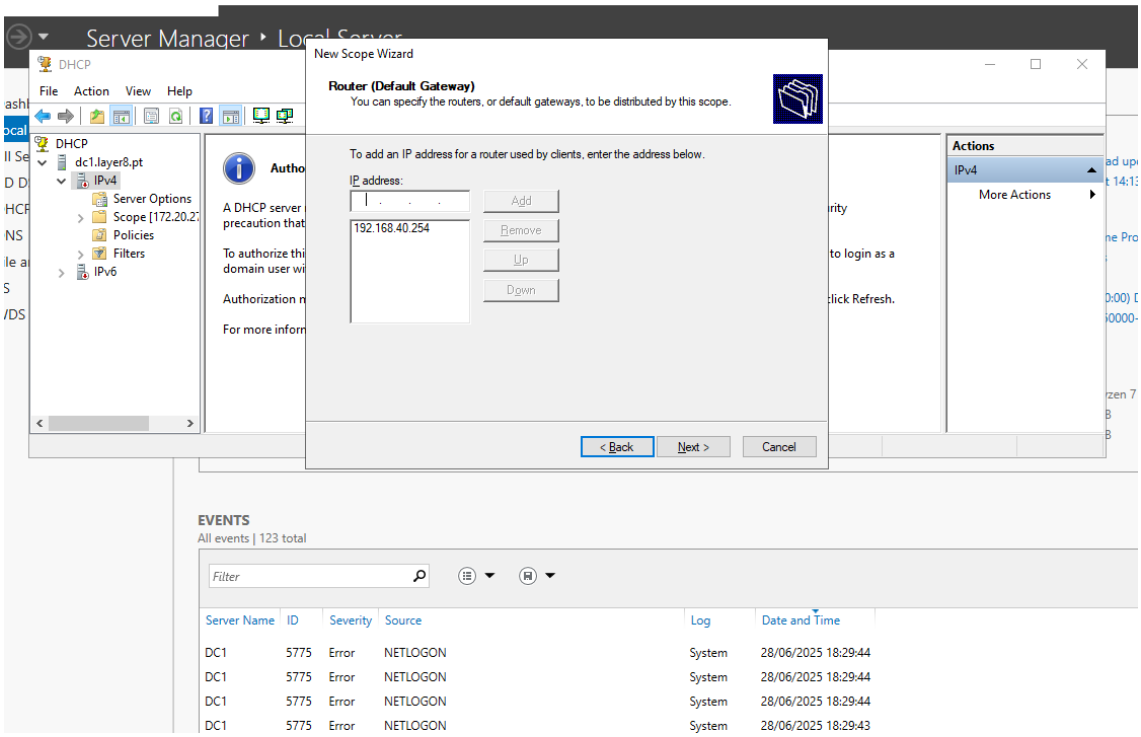
DHCP range – 192.168.40.45 – 192.168.40.89



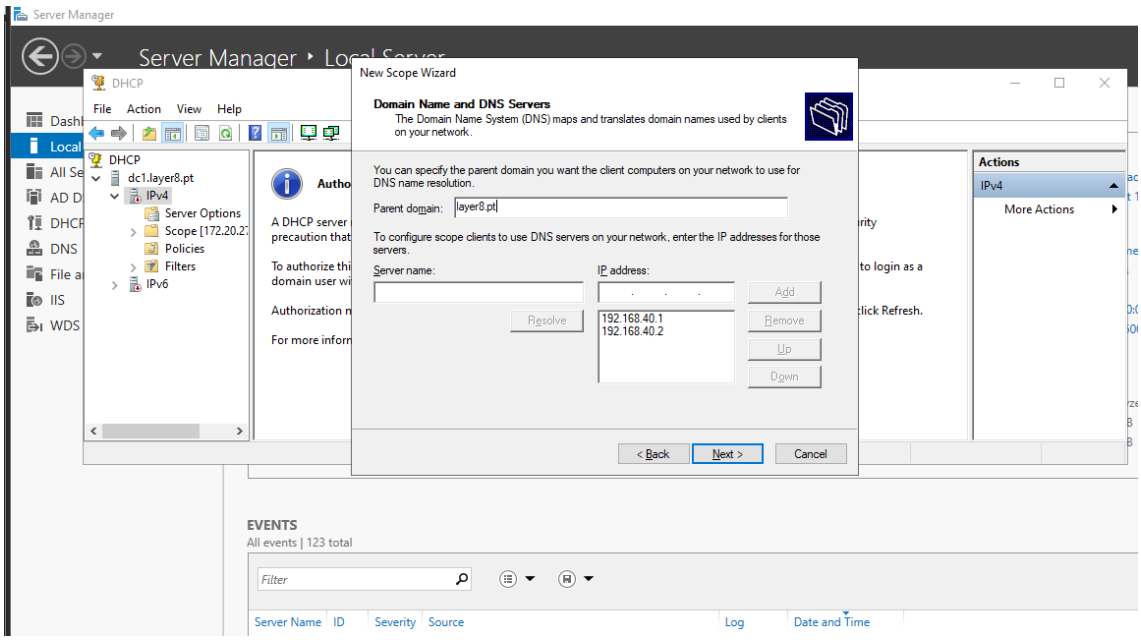
Ip´s exluidos 192.168.40.50 – 192.168.40.65



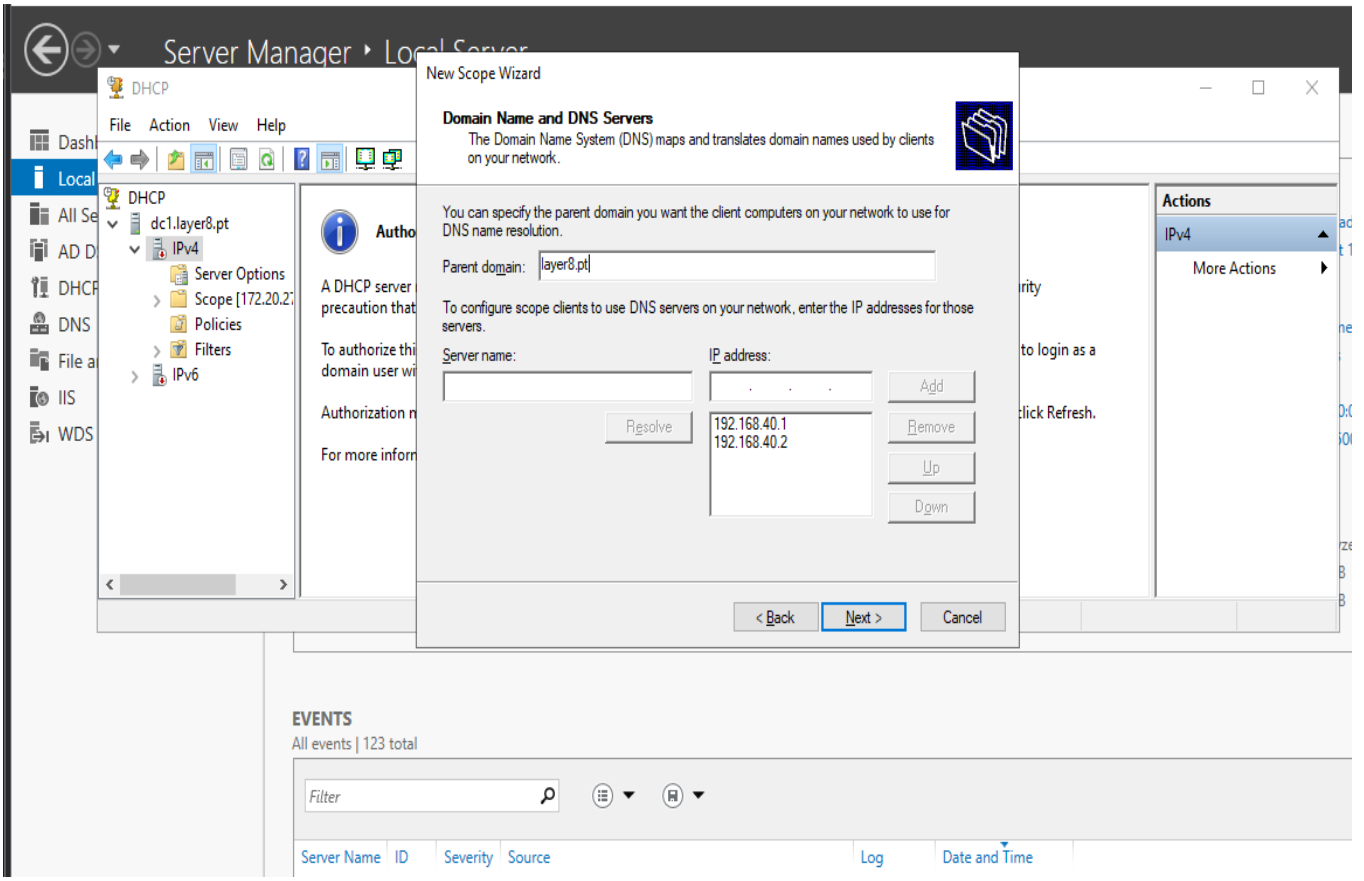
Ip default gateway -router pfsense 192.168.40.254



2 DNS servers 1 e 2 ,domain **layer8.pt -192.168.40.1 , 192.168.40.2**

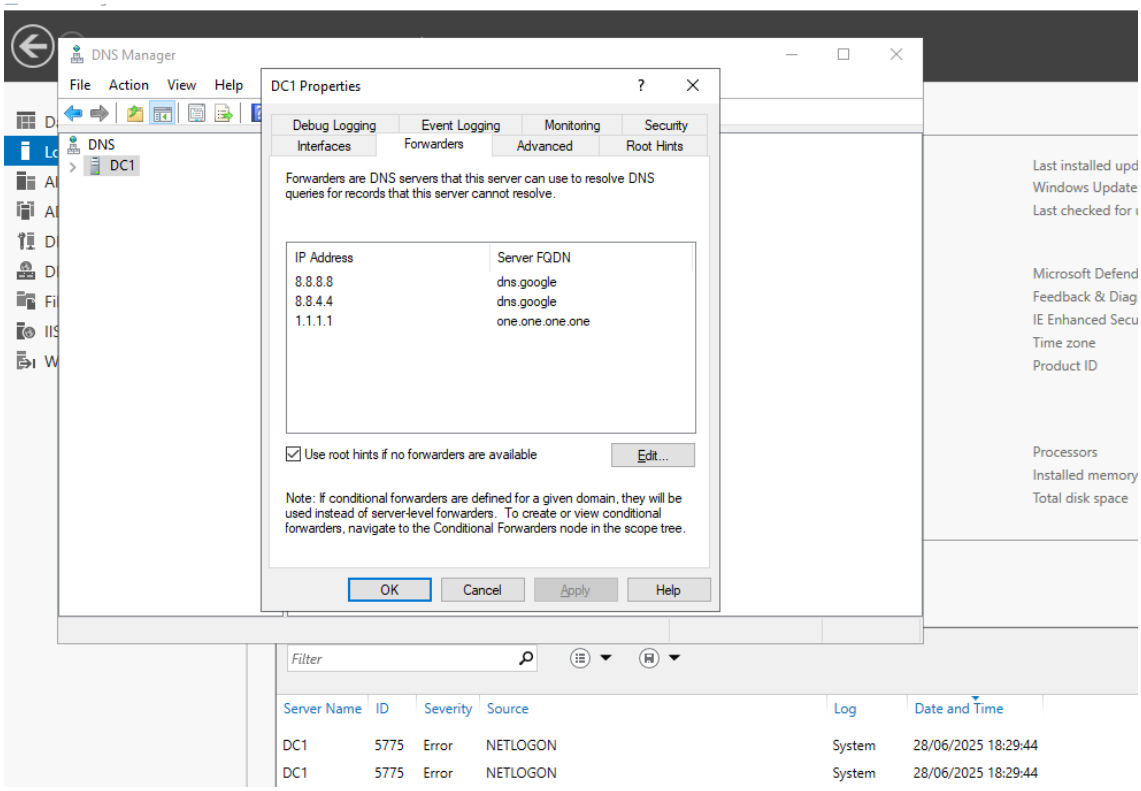


Verificação de dhcp ipv4 “Scope”

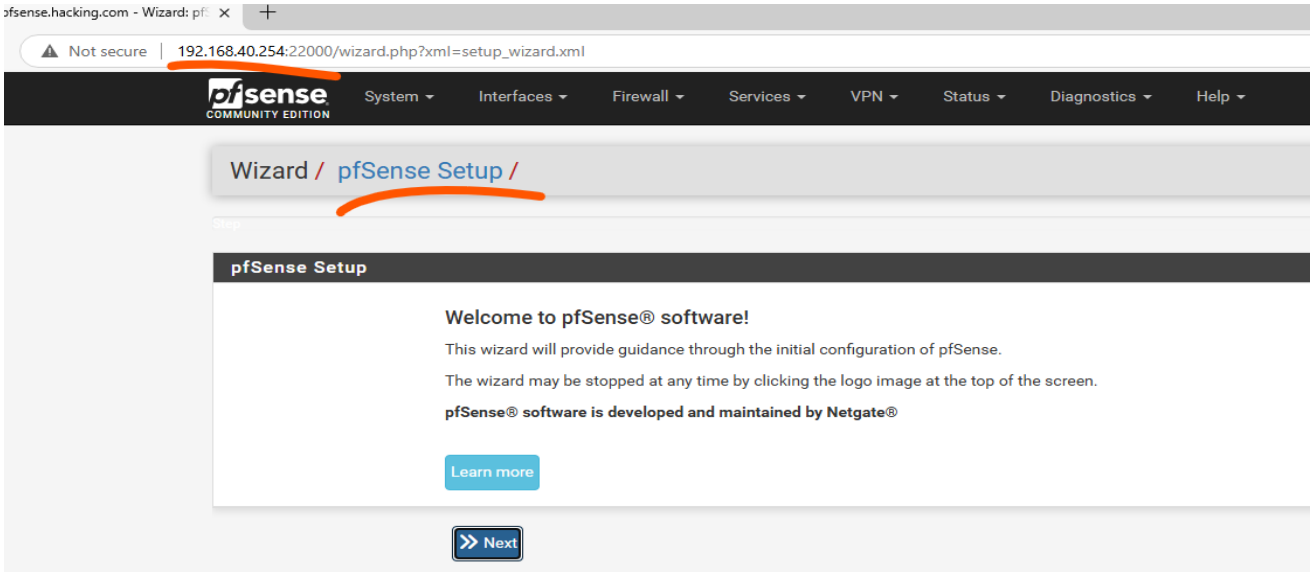


Configuração de Dns :tools -Dns -DC-forwarders

8.8.8.8 dns.google , 8.8.4.4 dns.google , 1.1.1.1 one.one.one.one



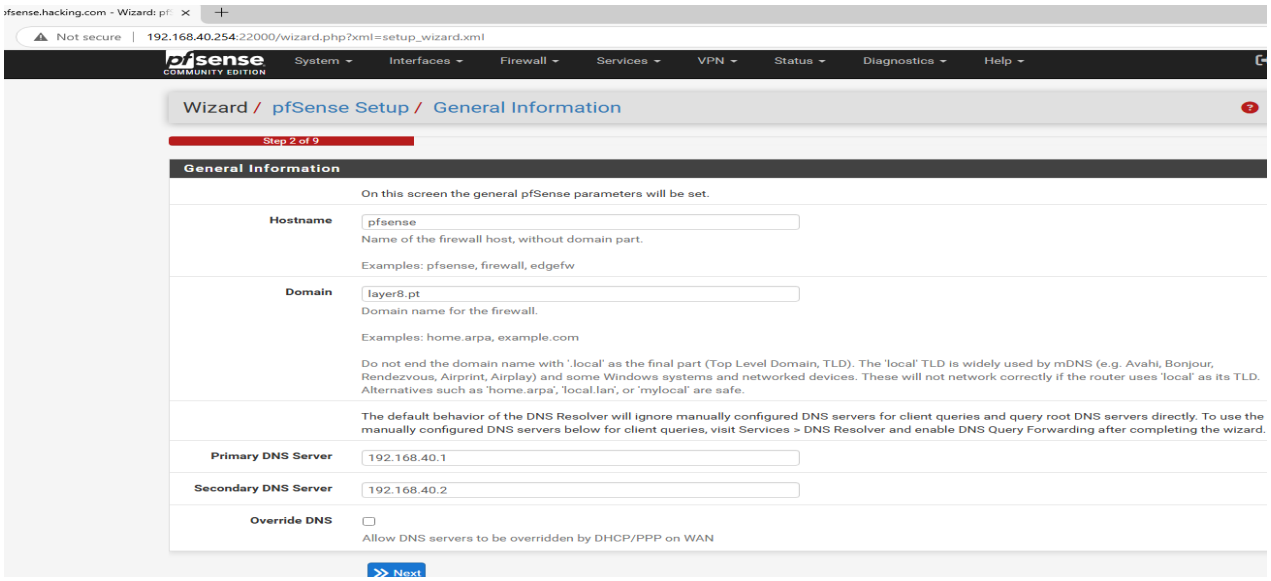
Configuração de pfsense :



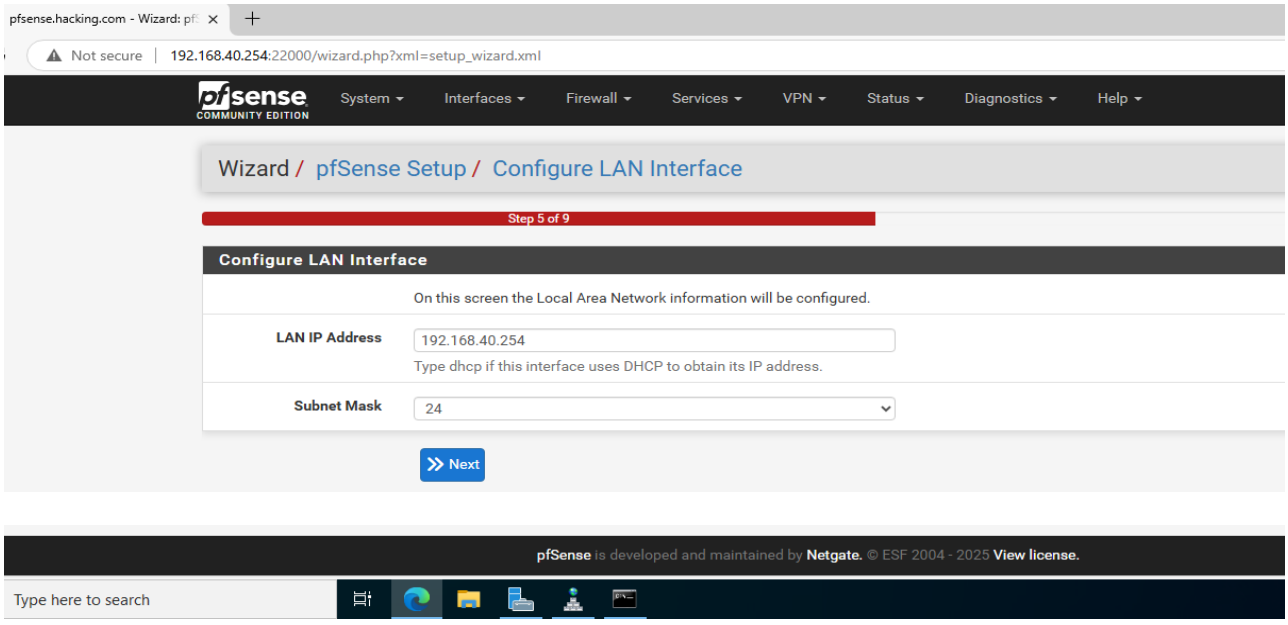
Wizard/pfSense Setup /General Information

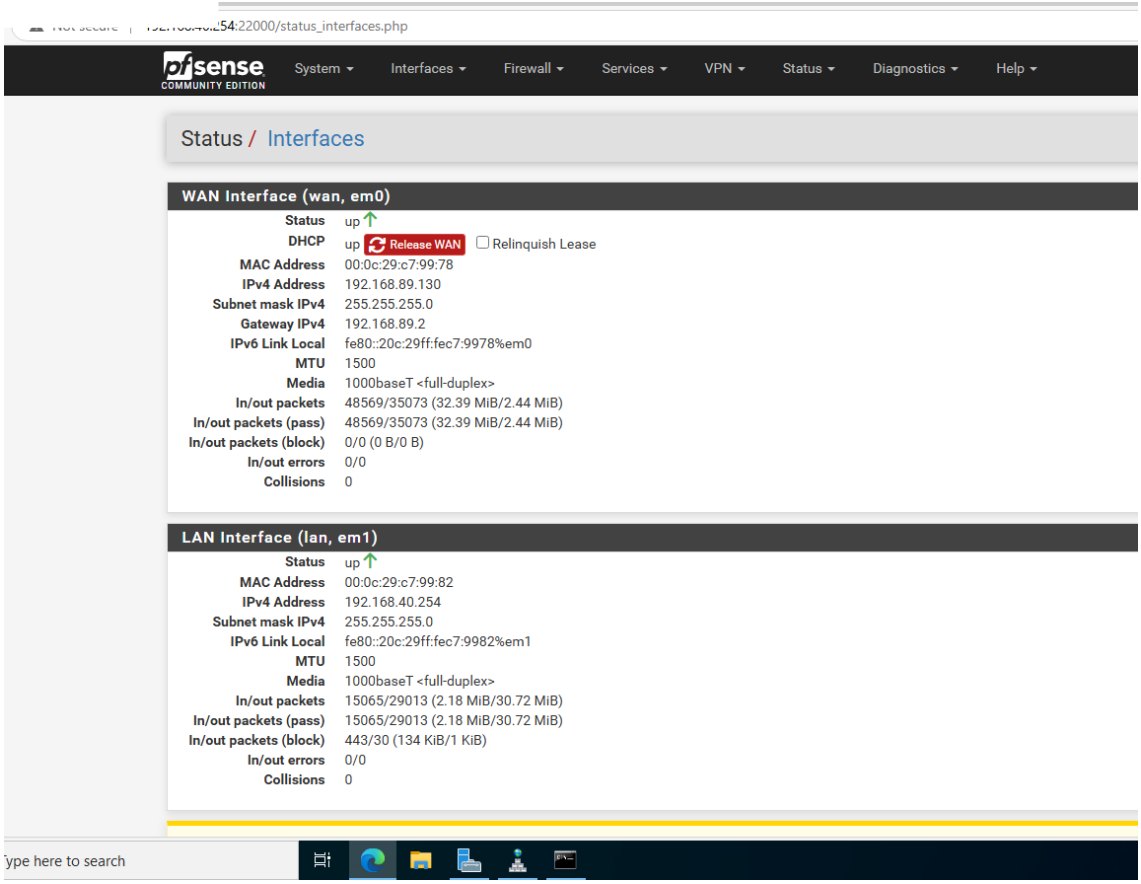
Hostname :pfsense

8.pt ;Primary DNS Server : 192.168.40.1 ;Secondary Server :192.168.40.2



Wizard/pfSense Setup /ConfigureLAN Interface -Configuração do router -192.168.40.254

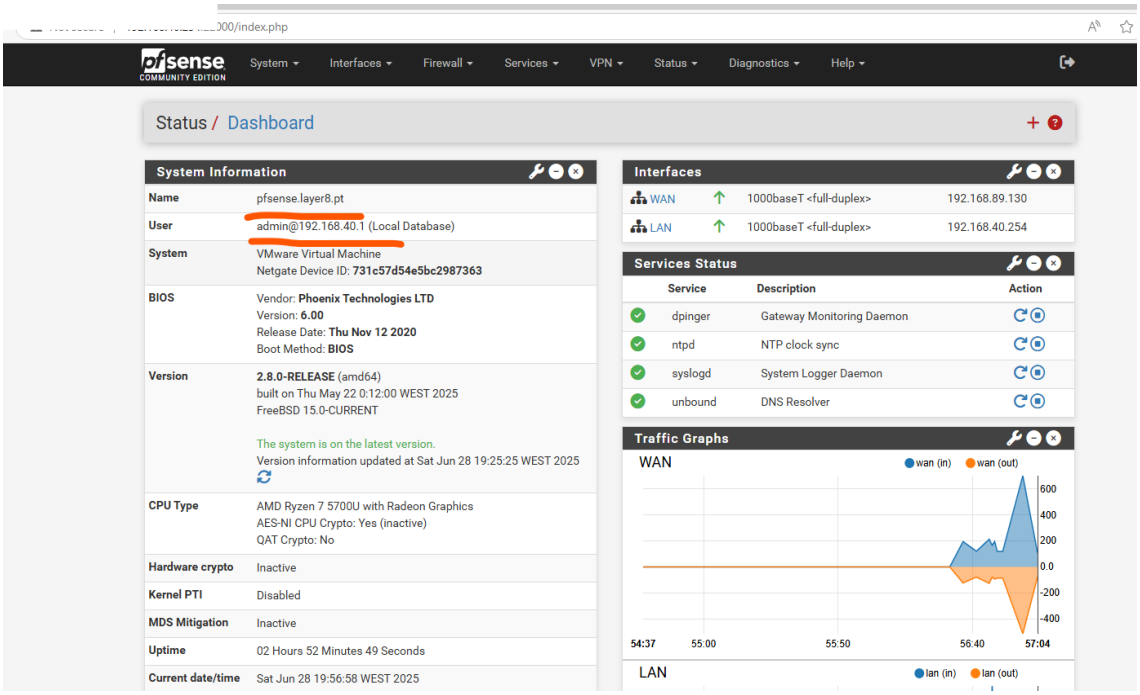




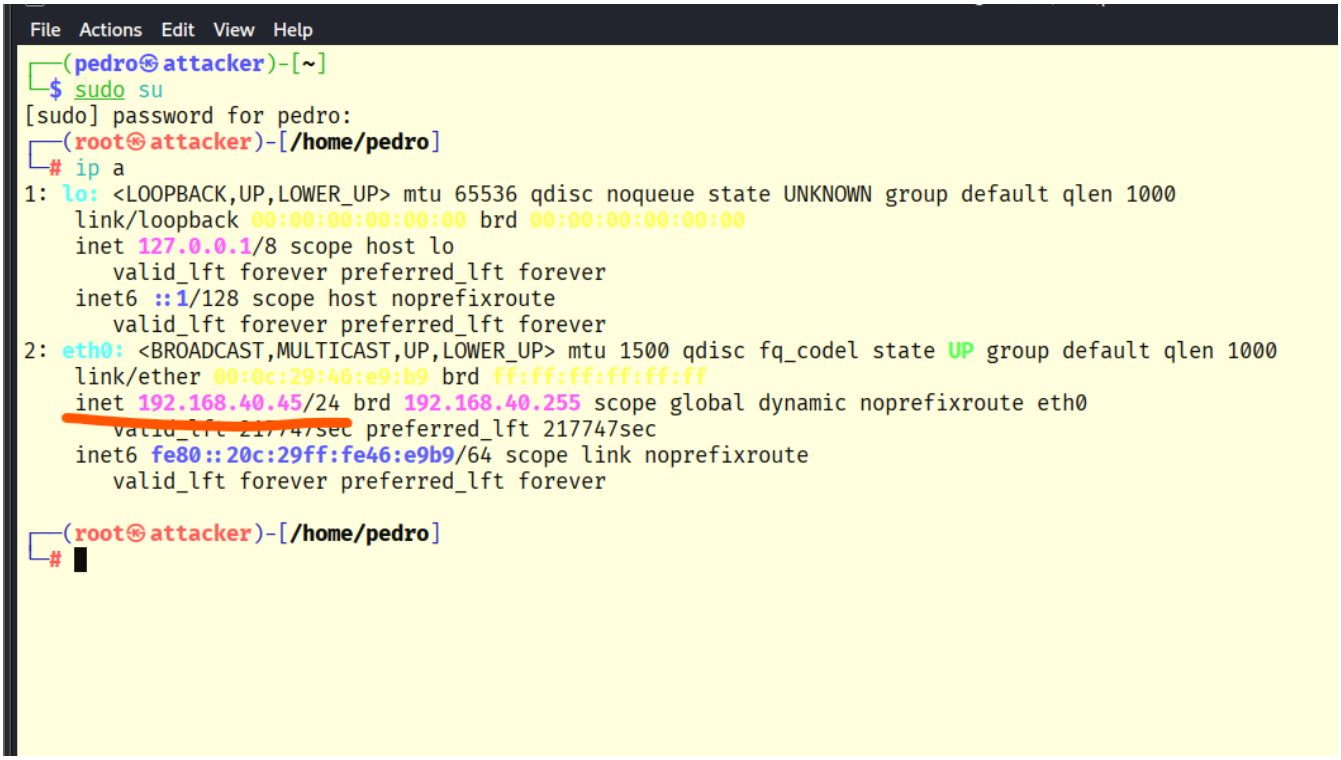
Verificação de configuração final :

Name :pfsense.layer8.pt

User:admin@192.168.40.1

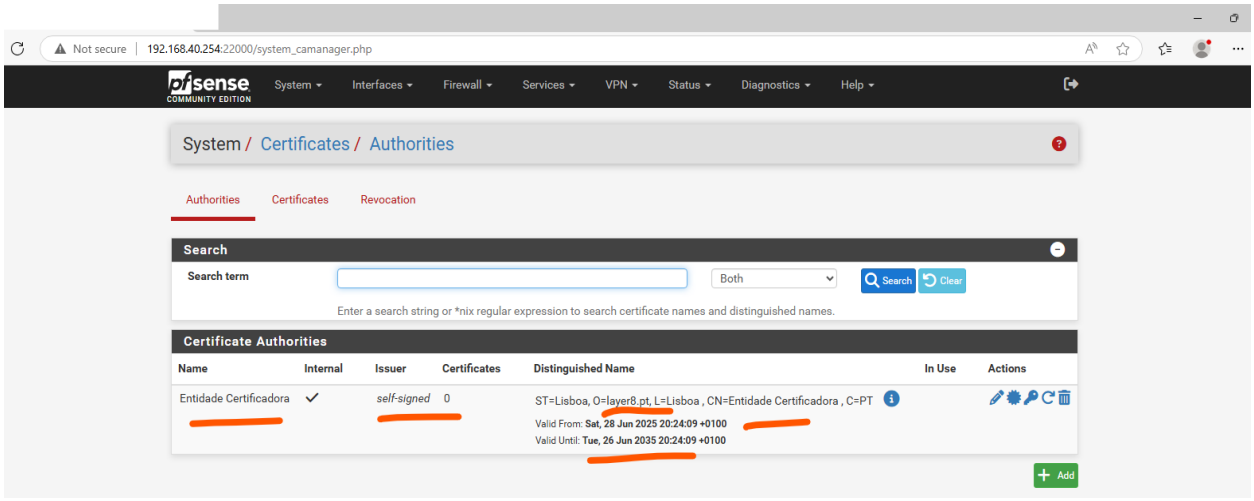


Após colocação do kali em LAn , ip **192.168.40.45**,confirmação que este já panaha endereço do range dhcp (192.168.10.45-192.168.40.89) -

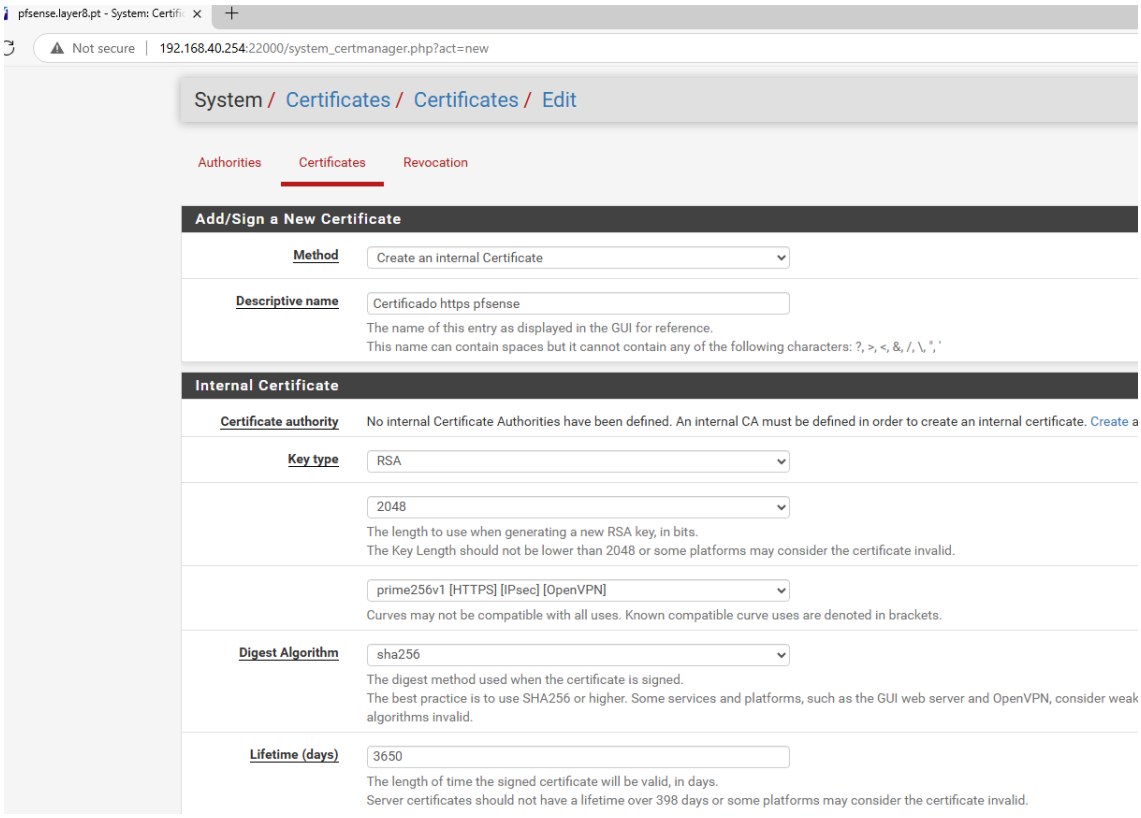


Criar uma Entidade Certificadora no pfsense :

Confirmação :Certificado Authority



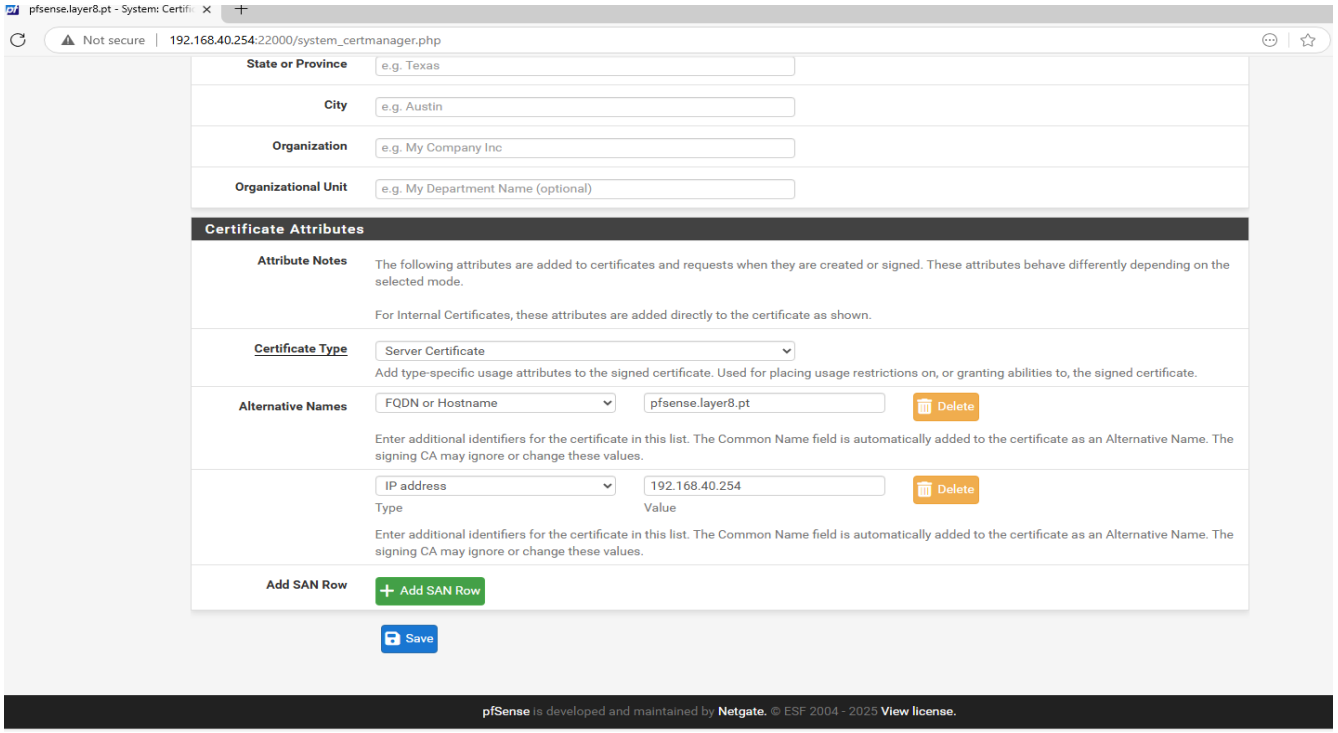
Criação de certificado Interno :



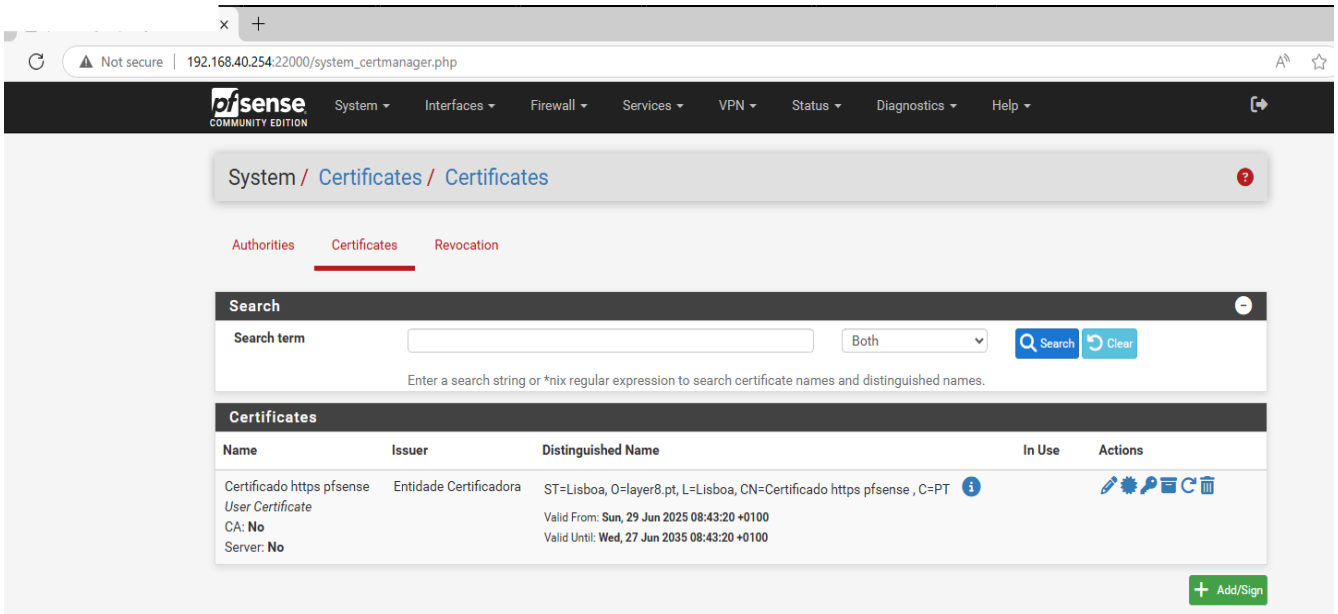
:ServerCertificate

Alternative Names :FQDN or Hostname pfsense.layer8.pt

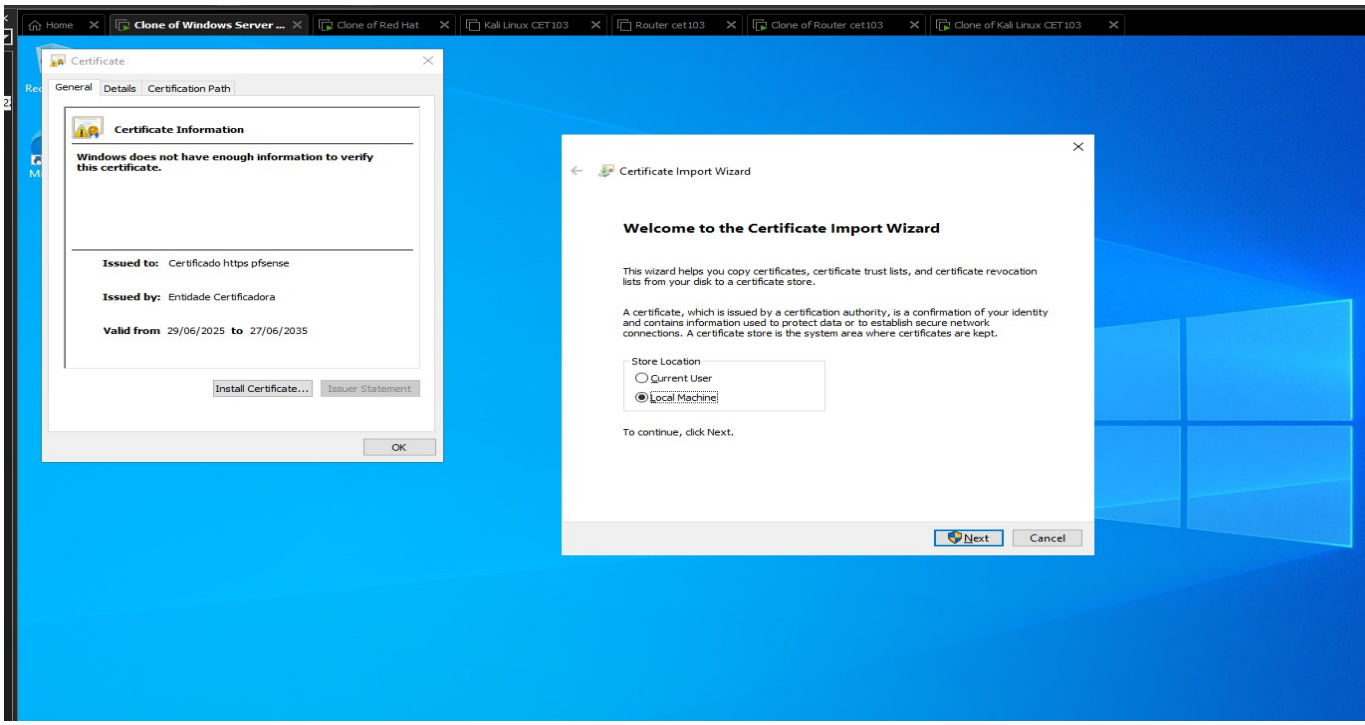
IP address 192.168.40.254



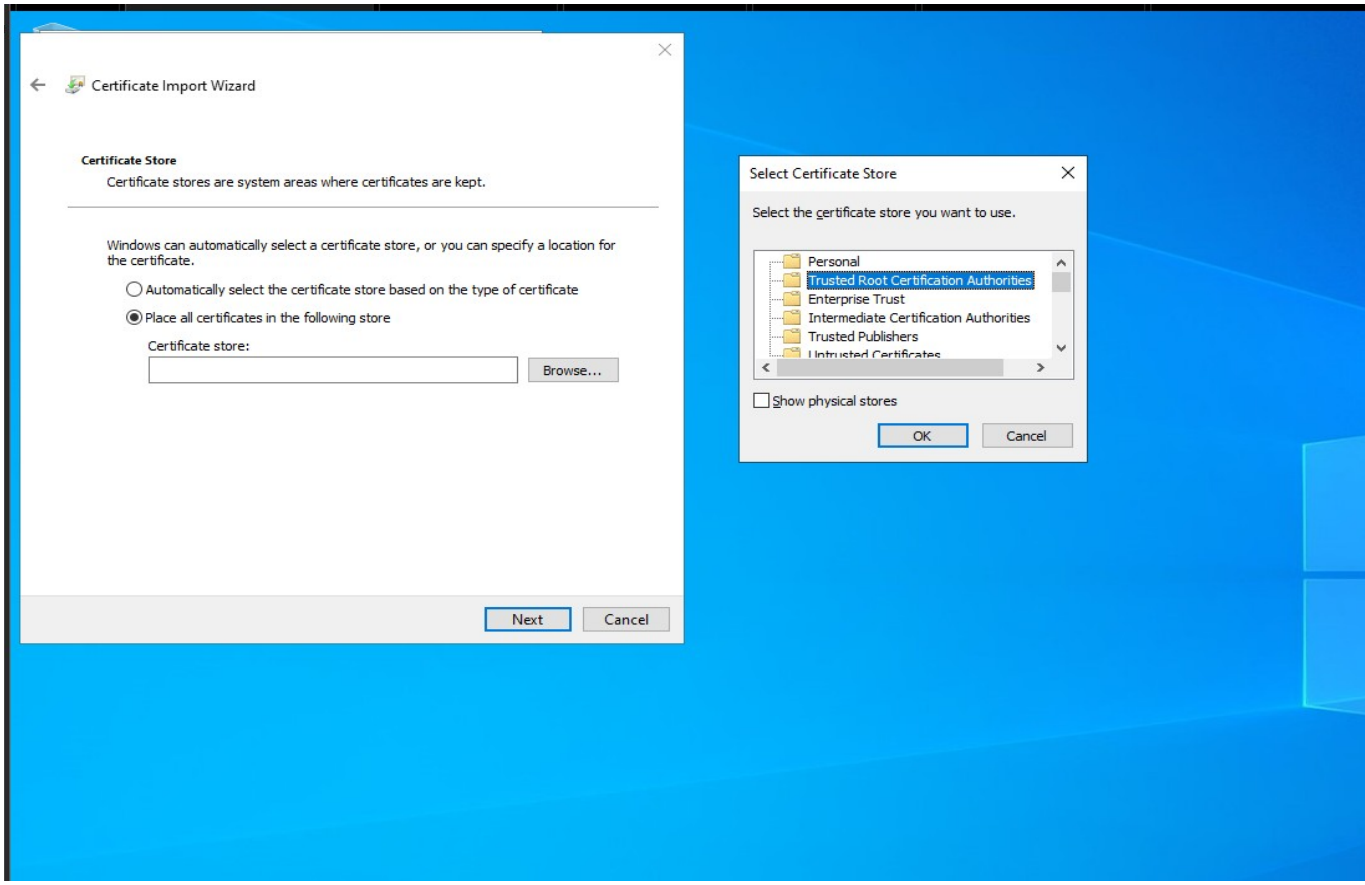
Verificação de certificado , export certificate (clicar em Aactions “star”)



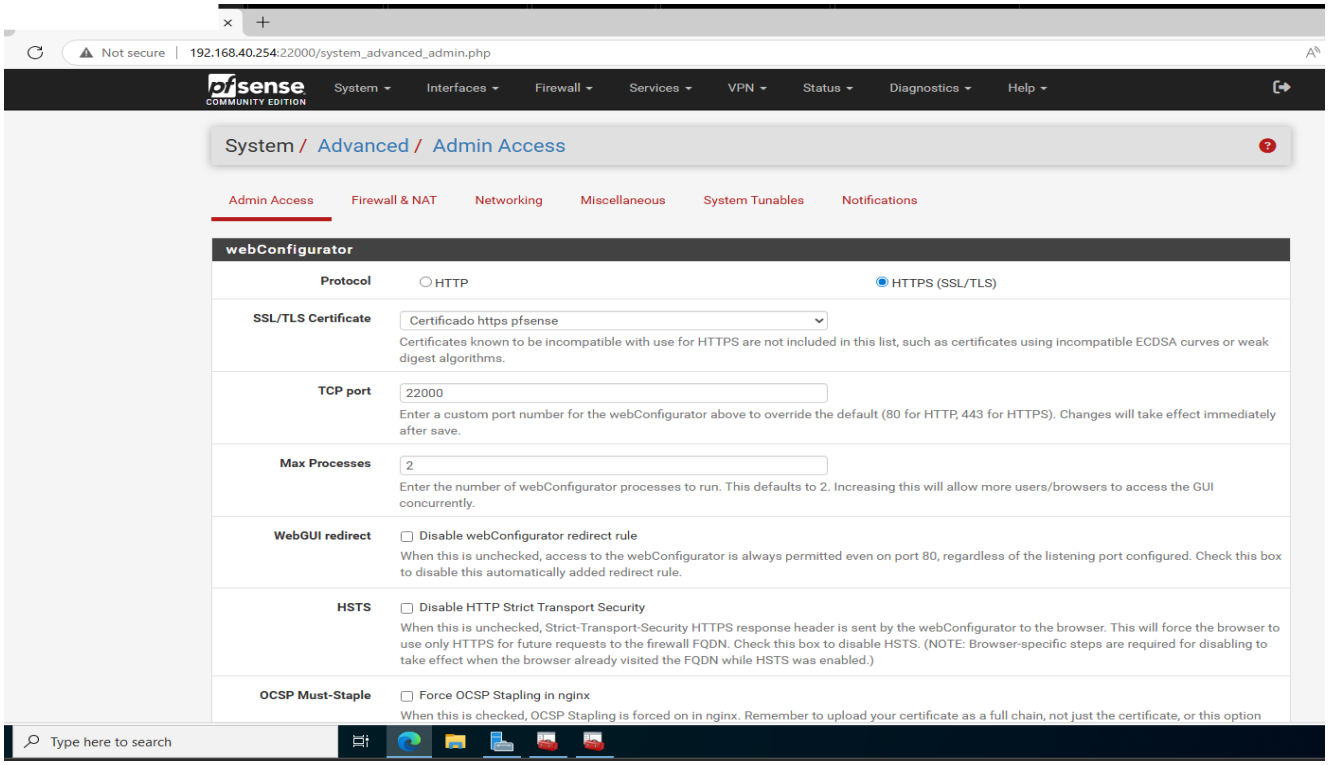
Instalação do certificado



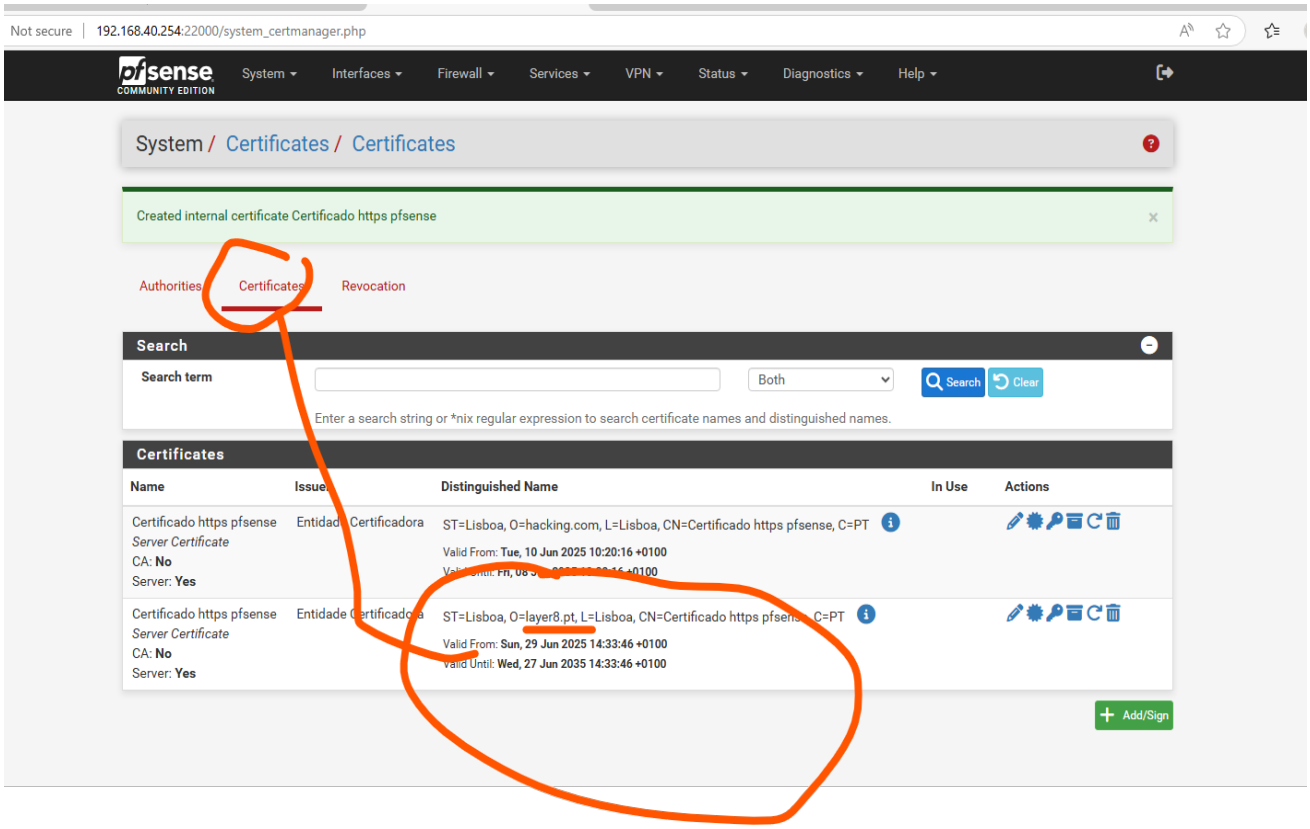
th :destino:



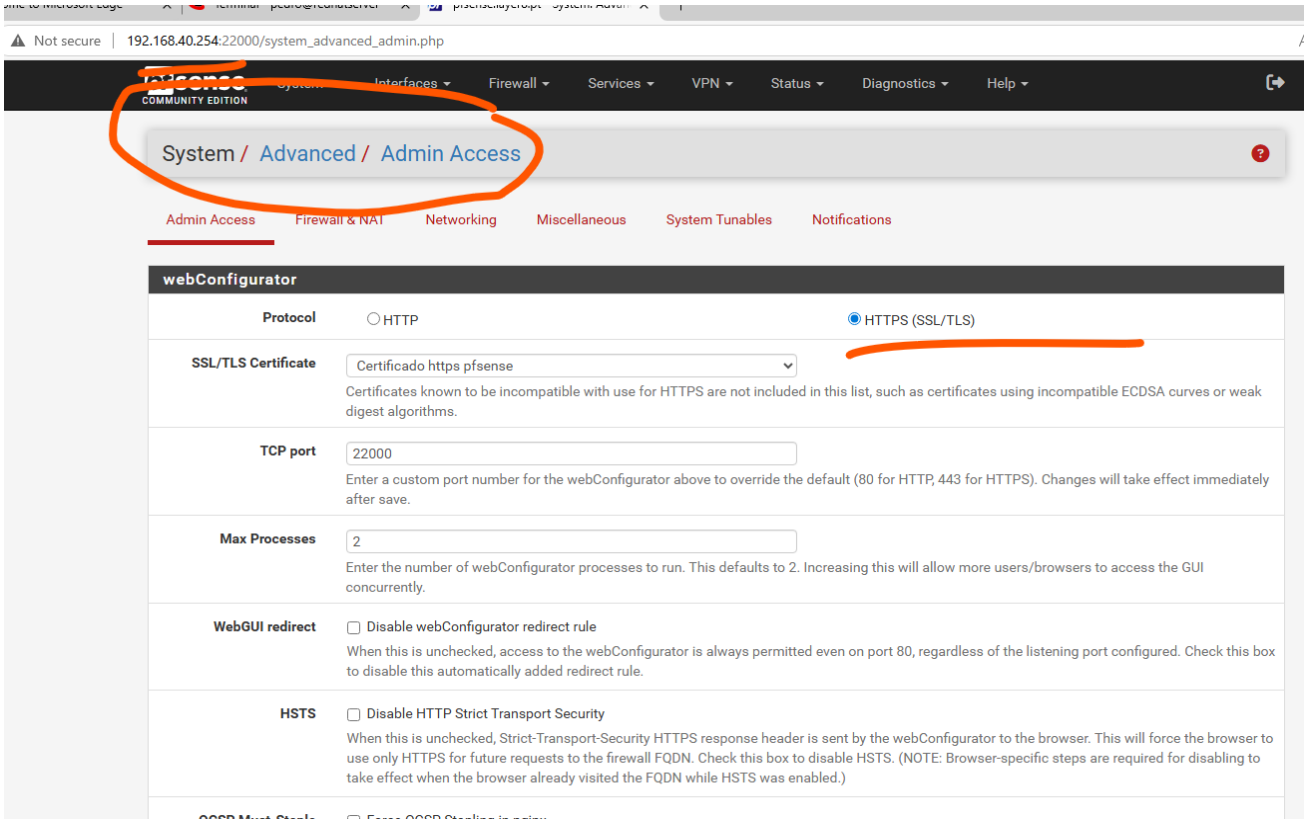
System /Advanced / Admin Access-Colocar web configuração em https -seguro



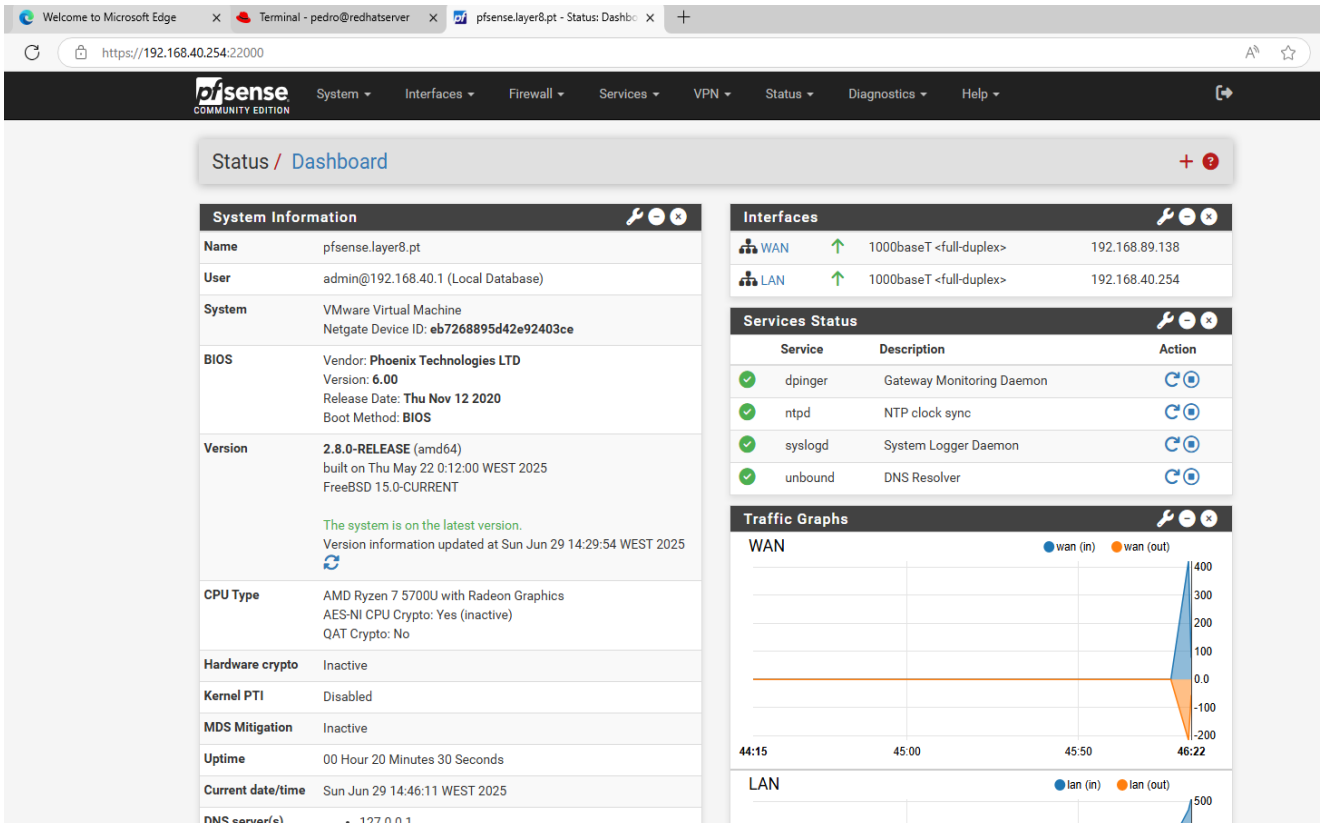
Instalar o certificado no mesmo pasta destino de :**Authorities certificate**



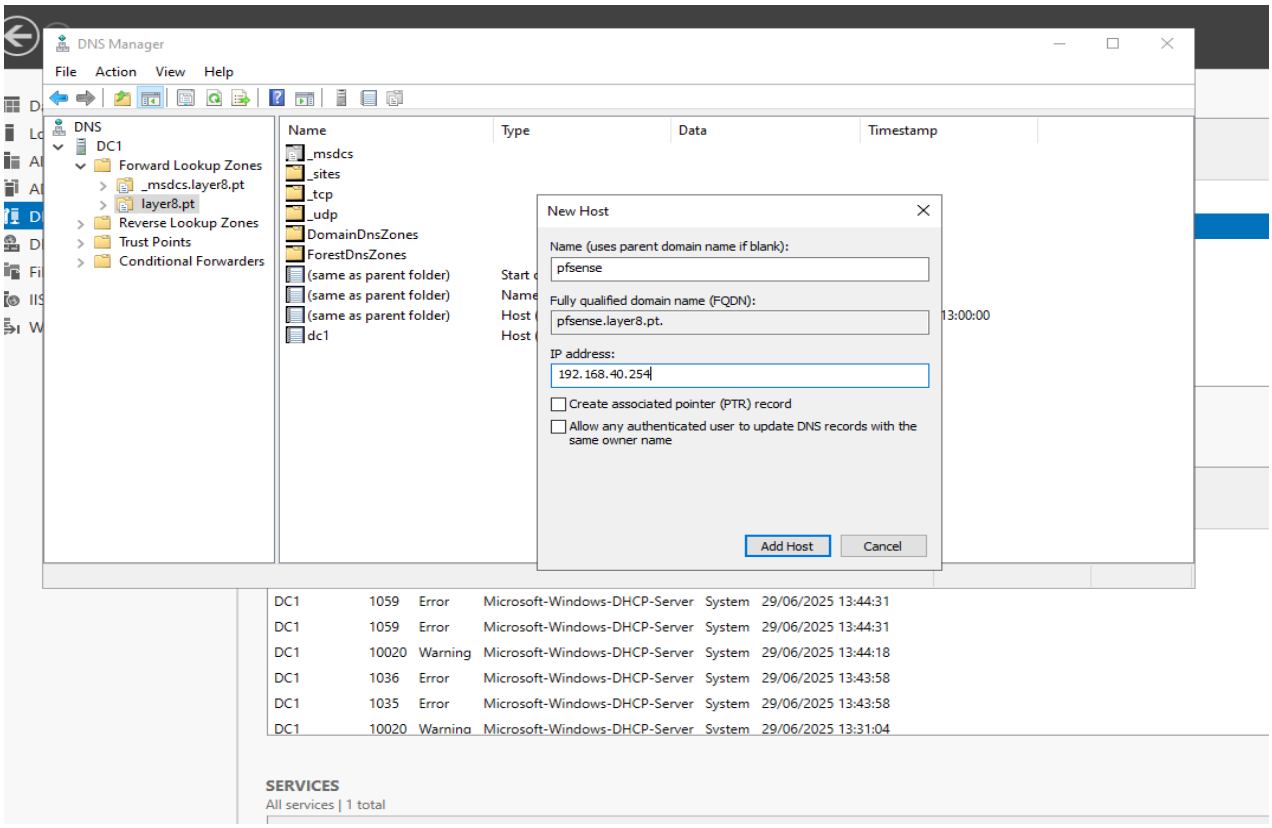
min Acess em https



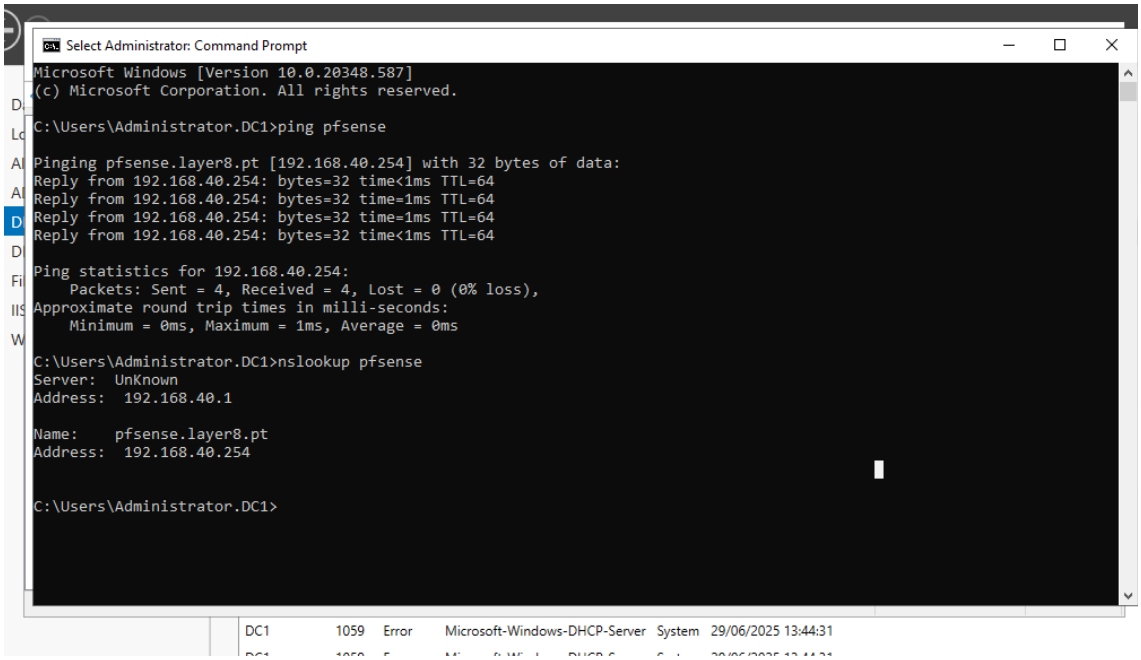
Confirmação :https://192.168.40.254:22000



novo host DNS



Confirmação de ping ao Domain name :



r | Secondary Server

OS: Red Hat Server 10.0

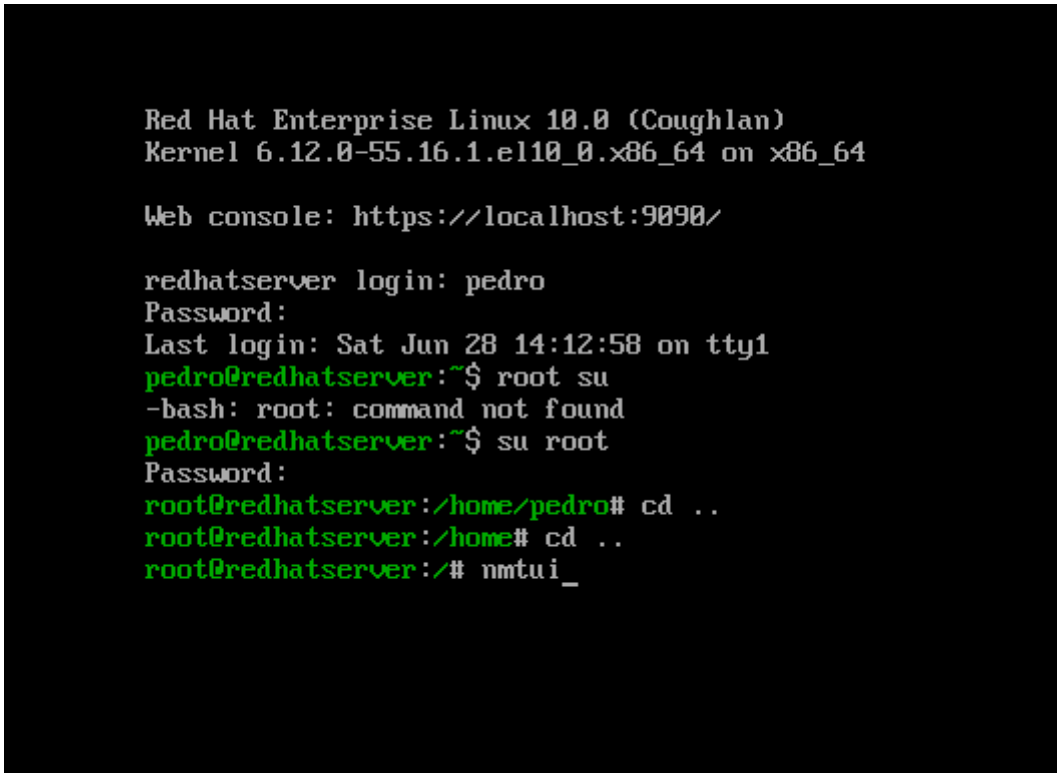
Hostname: redhatserver

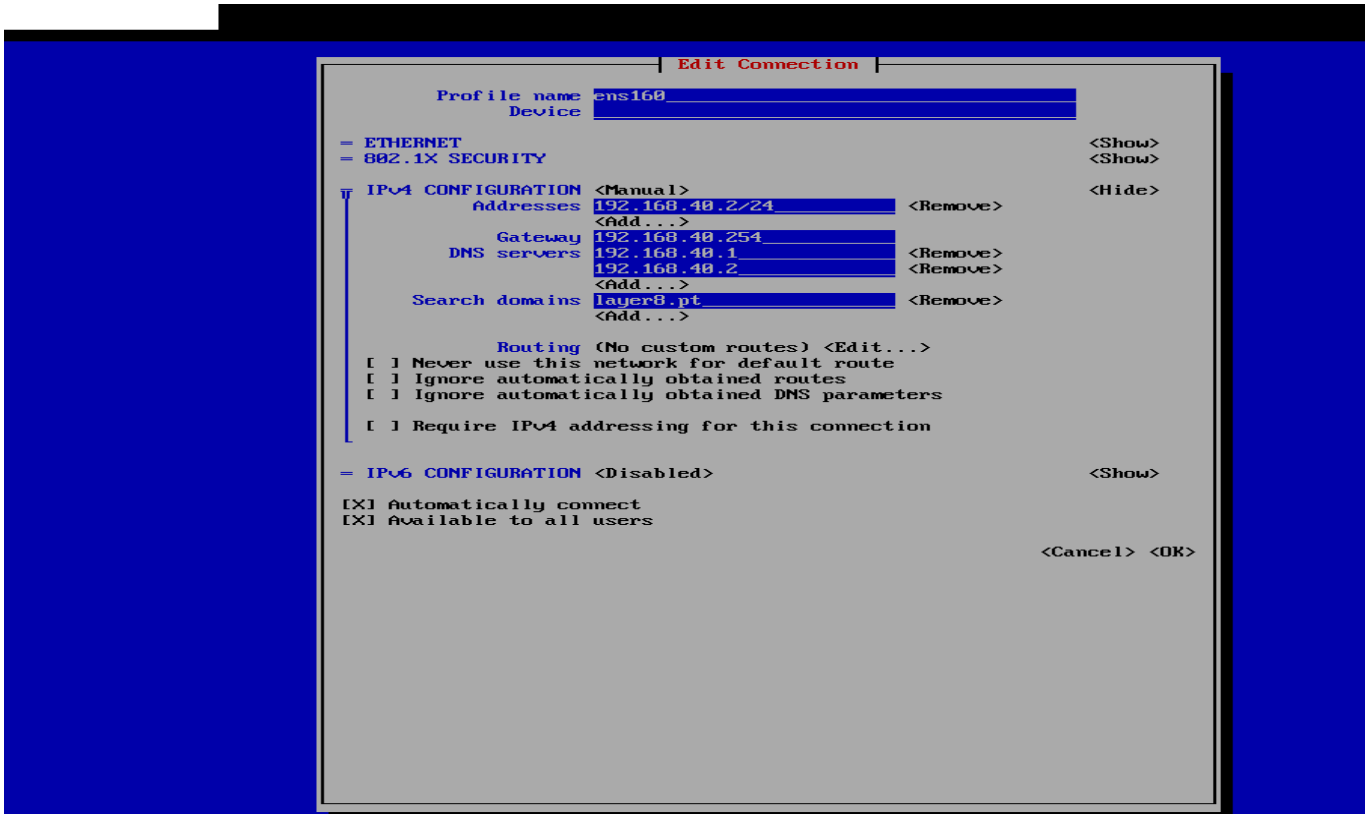
Interfaces IP: ens160 (192.168.40.2)

Serviços: DHCP | DNS | Failover | SSH

Descrição: Como servidor Secundário, é responsável por ser servidor de failover de DNS e DHCP do Windows Server. Deve criar um script em python para verificar se os serviços estão sempre ativos mesmo que o DC1 vá abaixo. Mudar a porta do serviço do SSH para uma à escolha.

Configuração de ligações no nmtui do RedHat



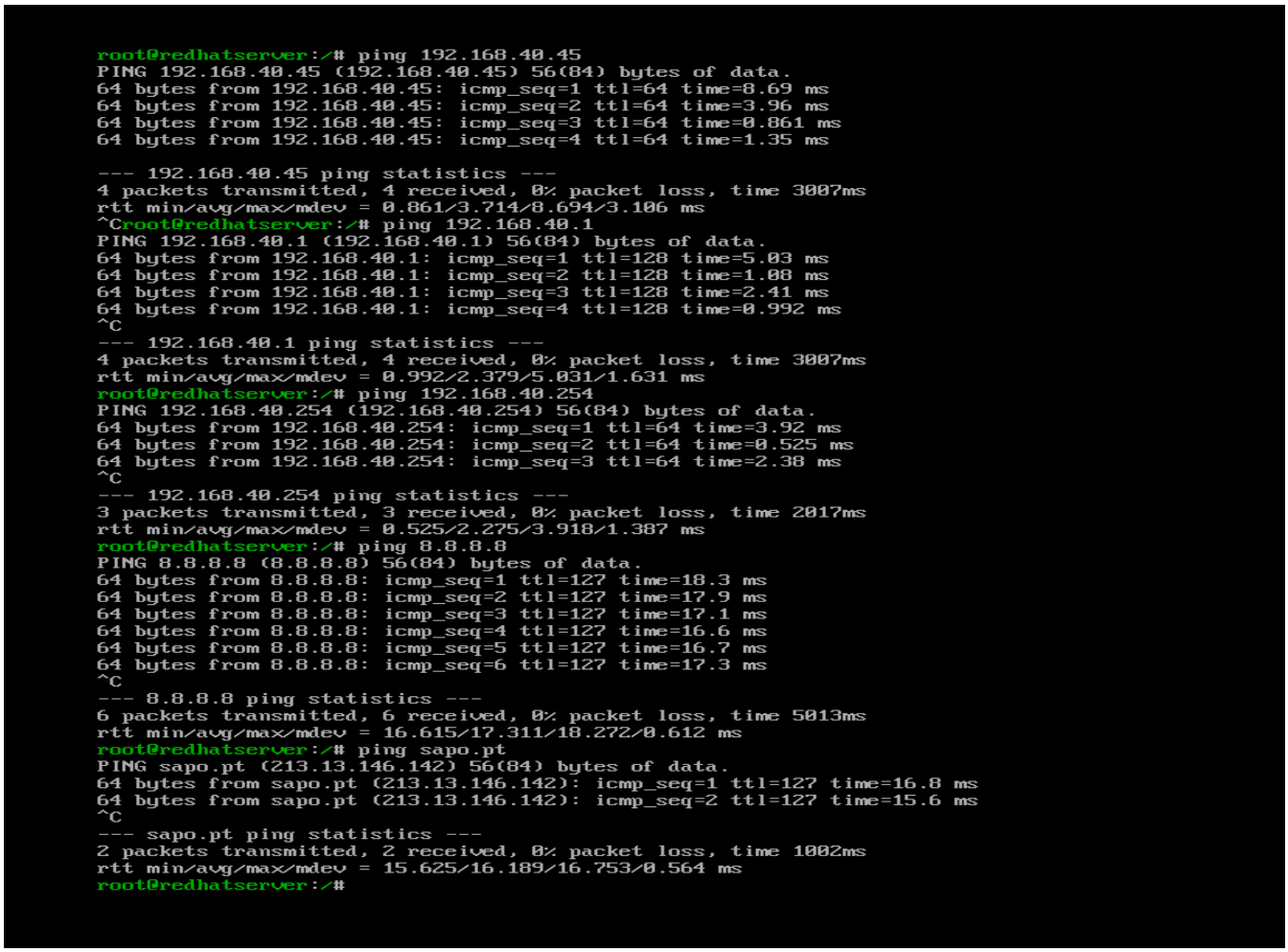


Confirmação de ip a redhat -192.168.40.2

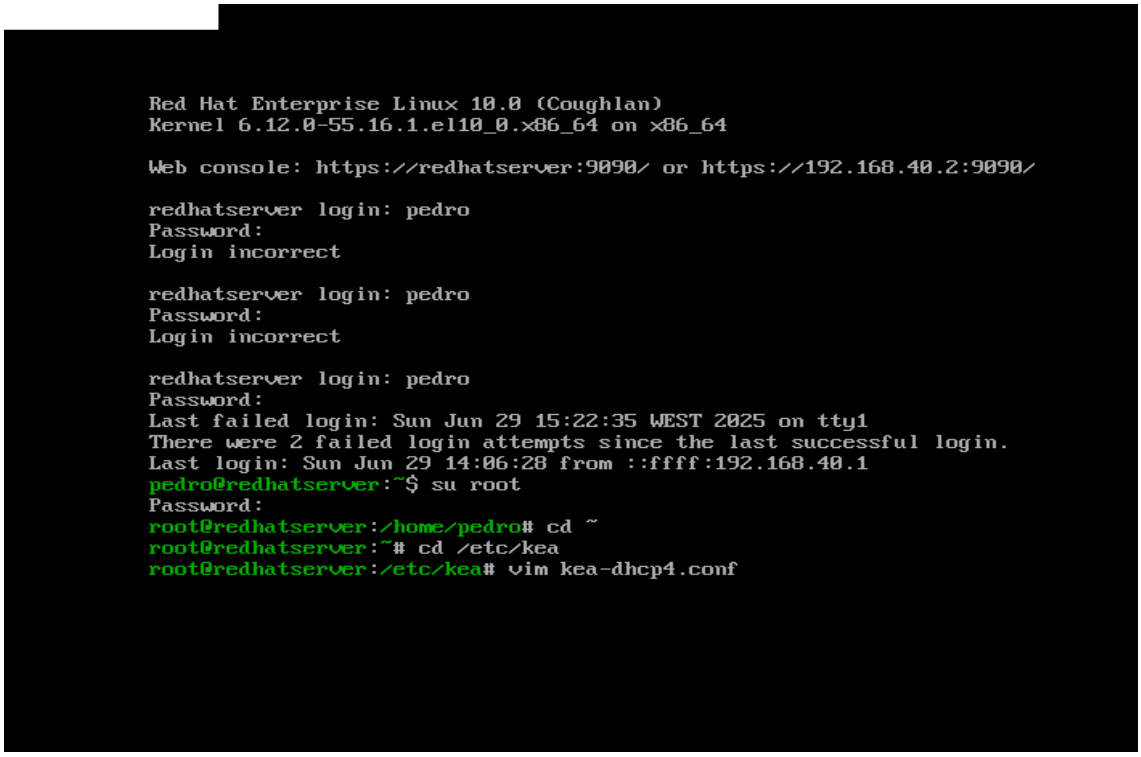
winserver ping192.168.40.1 , KaliLinux ping 192.168.40.45, Pfsense ping 192.168.10..254, google 8.8.8.8 ,ping sapo.pt

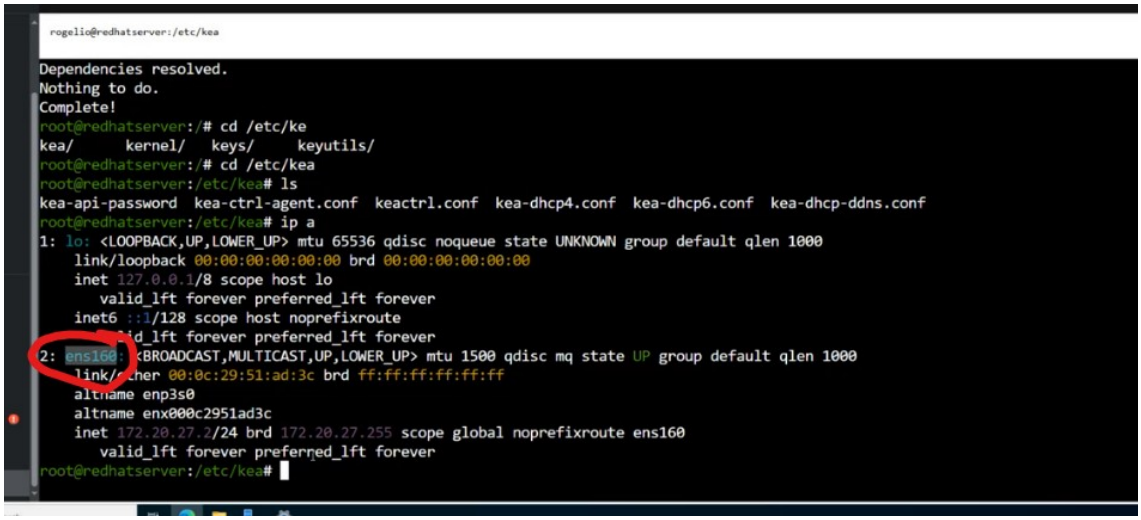
```
root@redhatserver:/# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:c9:93:39 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    altname enx000c29c99339
    inet 192.168.40.2/24 brd 192.168.40.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
root@redhatserver:/# ping 192.168.40.45
PING 192.168.40.45 (192.168.40.45) 56(84) bytes of data.
64 bytes from 192.168.40.45: icmp_seq=1 ttl=64 time=8.69 ms
64 bytes from 192.168.40.45: icmp_seq=2 ttl=64 time=3.96 ms
64 bytes from 192.168.40.45: icmp_seq=3 ttl=64 time=0.861 ms
64 bytes from 192.168.40.45: icmp_seq=4 ttl=64 time=1.35 ms

--- 192.168.40.45 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 0.861/3.714/8.694/3.106 ms
^Croot@redhatserver:/# ping 192.168.40.1
PING 192.168.40.1 (192.168.40.1) 56(84) bytes of data.
64 bytes from 192.168.40.1: icmp_seq=1 ttl=128 time=5.03 ms
64 bytes from 192.168.40.1: icmp_seq=2 ttl=128 time=1.08 ms
64 bytes from 192.168.40.1: icmp_seq=3 ttl=128 time=2.41 ms
64 bytes from 192.168.40.1: icmp_seq=4 ttl=128 time=0.992 ms
^C
--- 192.168.40.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 0.992/2.379/5.031/1.631 ms
root@redhatserver:/#
```



Vim /etc/kea/kea-dhcp4.conf





root@redhatserver:/etc/kea# cat kea-dhcp4.conf

```
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "ens160" ]
    },
  "lease-database": {
  "type": "memfile",
```

```
"name": "/var/lib/kea/kea-leases4.csv",
 "lfc-interval": 3600
 },
 "renew-timer": 15840,
 "rebind-timer": 27720,
 "valid-lifetime": 31680,
    "option-data": [
{
 "name": "domain-name-servers",
        "data": "192.168.40.1, 192.168.40.2"
},
 {
"name": "domain-search",
 "data": "layer8.pt"
 }
 ],
"subnet4":
{"id": 1,"subnet": "192.168.40.0/24",
"pools": [

         { "pool": "192.168.40.45 - 192.168.40.89"}
       ],
"option-data": [
 {
```

ers",

"data": "192.168.40.254"

}

]

}

]

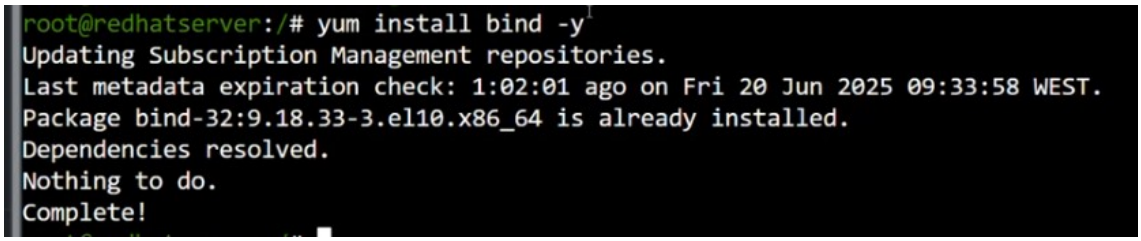
}

}

Instalação de Serviço DNS

O **BIND (Berkeley Internet Name Domain)** é o software mais utilizado no mundo para implementar servidores **DNS (Domain Name System)** em sistemas Linux/Unix.

Instalação do software “bind”



Edição do serviço “named”

Edição do ficheiro “named.conf”

root@redhatserver:/etc# vim named.conf

root@redhatserver:/etc# cat named.conf

//

// named.conf

// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS

// server as a caching only nameserver (as a localhost DNS resolver only).

//

// See /usr/share/doc/bind*/sample/ for example named configuration files.

//

options {

listen-on port 53 { 127.0.0.1; **192.168.40.2; };**

#listen-on-v6 port 53 { ::1; };

directory "/var/named";

dump-file "/var/named/data/cache_dump.db";

statistics-file "/var/named/data/named_stats.txt";

memstatistics-file "/var/named/data/named_mem_stats.txt";

secroots-file "/var/named/data/named.secroots";

recursing-file "/var/named/data/named.recursing";

allow-query { localhost; **192.168.40.0/24;** };

forwarders **{ 8.8.8.8; 8.8.4.4; 1.1.1.1;** };

/*

- If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.

- If you are building a RECURSIVE (caching) DNS server, you need to enable

recursion.

- If your recursive DNS server has a public IP address, you MUST enable access

```
                o limit queries to your legitimate users. Failing to do so will
          cause your server to become part of large scale DNS amplification
          attacks. Implementing BCP38 within your network would greatly
          reduce such attack surface
      */
      recursion yes;

      dnssec-validation yes;

      managed-keys-directory "/var/named/dynamic";
      geoip-directory "/usr/share/GeoIP";

      pid-file "/run/named/named.pid";
      session-keyfile "/run/named/session.key";

      /* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
      include "/etc/crypto-policies/back-ends/bind.config";
};
logging {
      channel default_debug {
            file "data/named.run";
            severity dynamic;
      };
};
```

```
        type hint;
        file "named.ca";
};


zone "hacking.com" IN {
        type master;
        file "forward.layer8.pt";
        allow-query {192.168.40.0/24; };
};
zone "40.168.192.in-addr.arpa" IN {
        type master;
        file "reverse.layer8.pt";
        allow-query { 192.168.40.0/24; };
};
include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
```

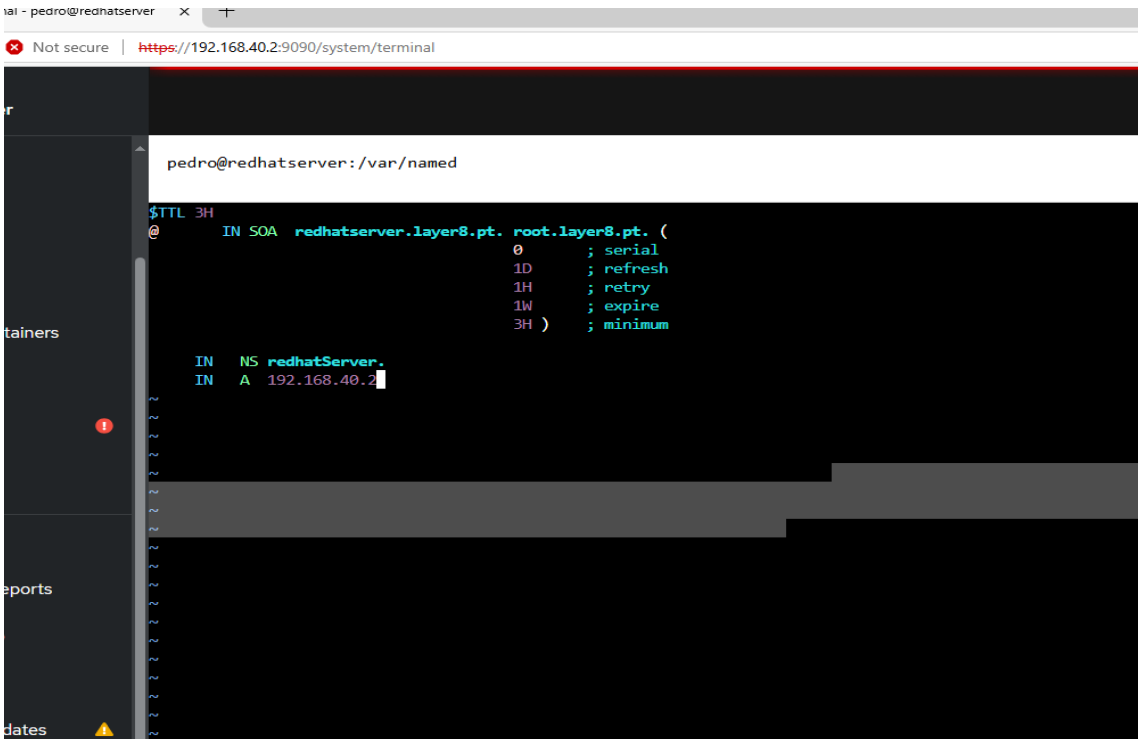
copia dos ficheiros named.empty para foward.layer8.pt e para reverse.layer8.pt

```
root@redhatserver:/etc# cd /var/named/
root@redhatserver:/var/named# cp named.empty forward.layer8.pt
root@redhatserver:/var/named# cp named.empty reverse.layer8.pt
root@redhatserver:/var/named# ll
total 32
drwxrwx---. 2 named named   49 Jun 29 13:26 data
drwxrwx---. 2 named named   60 Jun 20 16:57 dynamic
-rw-r-----. 1 root  named  210 Jun 20 15:16 forward.hacking.com
-rw-r-----. 1 root  root   152 Jun 29 17:11 forward.layer8.pt
-rw-r-----. 1 root  named 3314 Feb 13 00:00 named.ca
-rw-r-----. 1 root  named  152 Feb 13 00:00 named.empty
-rw-r-----. 1 root  named  152 Feb 13 00:00 named.localhost
-rw-r-----. 1 root  named  168 Feb 13 00:00 named.loopback
-rw-r-----. 1 root  named  238 Jun 20 15:17 reverse.hacking.com
-rw-r-----. 1 root  root   152 Jun 29 17:11 reverse.layer8.pt
drwxrwx---. 2 named named    6 Feb 13 00:00 slaves
root@redhatserver:/var/named#
```

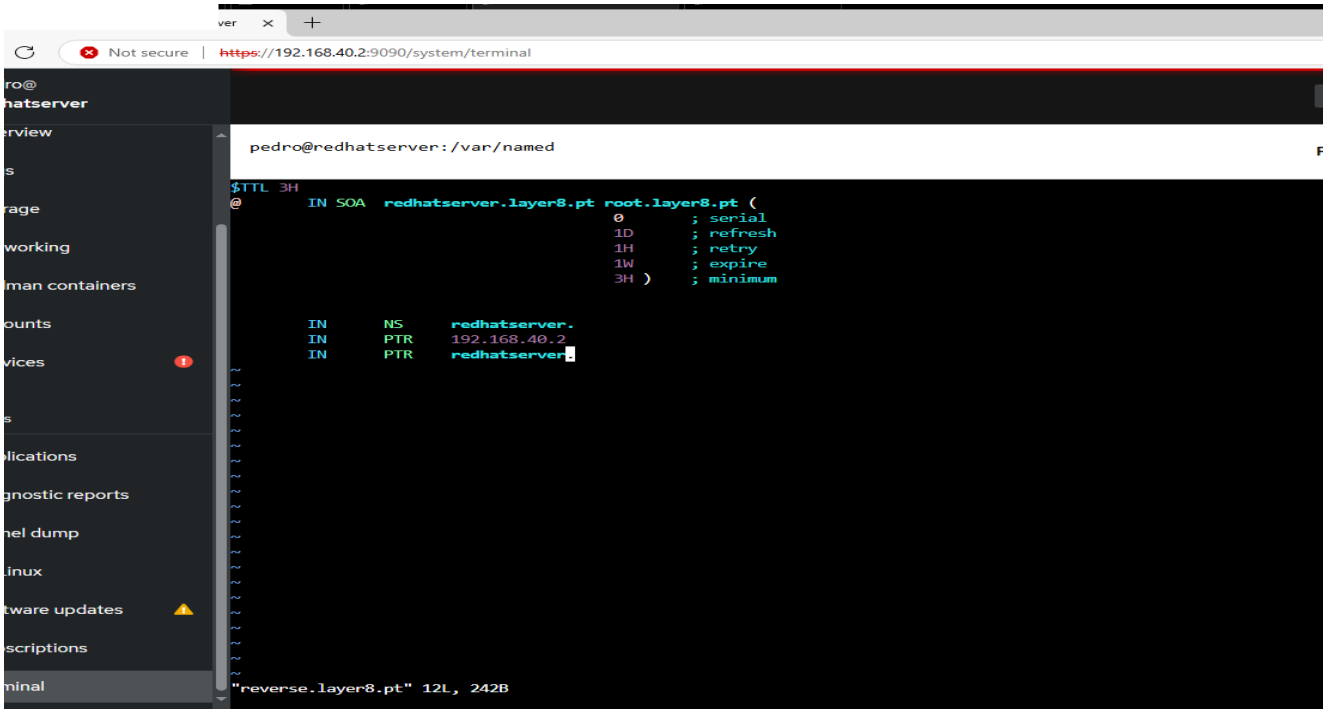
Alteração do owners :

```
root@redhatserver:/var/named# chown root:named forward.layer8.pt
root@redhatserver:/var/named# chown root:named reverse.layer8.pt
```

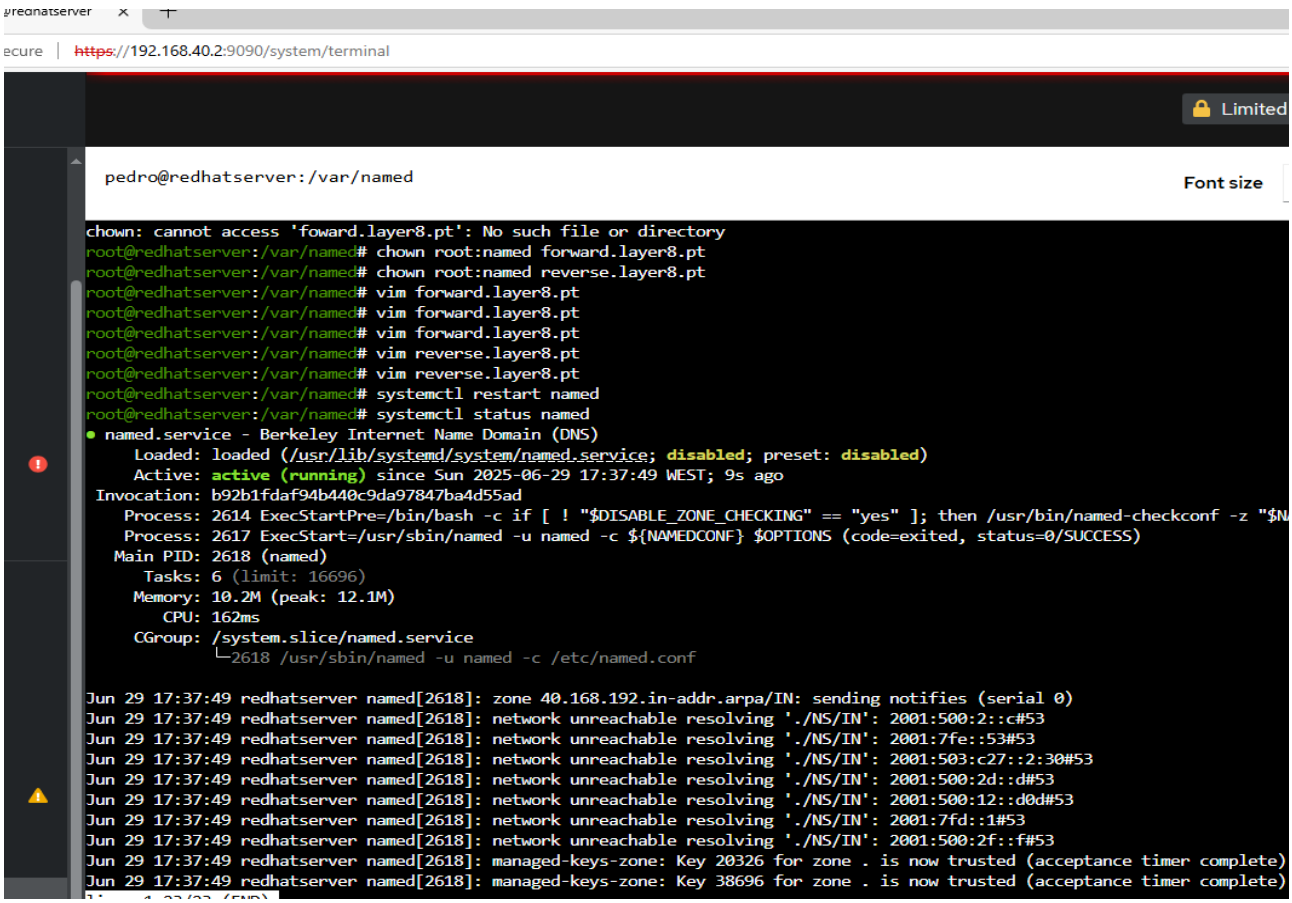
root@redhatserver:/var/named# vim forward.layer8.pt




```
root@redhatserver:/var/named# vim reverse.layer8.pt
```



Start named , status



e **ping layer8.pt**

```
root@redhatserver:/var/named# cd
root@redhatserver:~# ping layer8.pt
PING layer8.pt (192.168.40.1) 56(84) bytes of data.
64 bytes from 192.168.40.1: icmp_seq=1 ttl=128 time=0.765 ms
64 bytes from 192.168.40.1: icmp_seq=2 ttl=128 time=3.88 ms
64 bytes from 192.168.40.1: icmp_seq=3 ttl=128 time=0.933 ms
64 bytes from 192.168.40.1: icmp_seq=4 ttl=128 time=0.772 ms
64 bytes from 192.168.40.1: icmp_seq=5 ttl=128 time=0.761 ms
64 bytes from 192.168.40.1: icmp_seq=6 ttl=128 time=1.20 ms
^C
--- layer8.pt ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5010ms
rtt min/avg/max/mdev = 0.761/1.385/3.880/1.126 ms
root@redhatserver:~#
```

Implementação de Script Python- Failover

Explicação prática : Como não poem existir dois servidores WS e Redhat na mesma zona a responder em dhcp e dns com ip´s diferentes **, é criado como solução Script em Python que constantemente pinga do redhat para WS inativando os serviços DHCP e DNS do próprio -(redhat)** , Os Serviços Dhcp e Dns do Redhat serão ativados quando os mesmos do WS deixarem de funcionar(por erro ou avaria).

Testes de ping do KaliLinux-> redhat

Desativar -no redhat de forma permanente a firewall para a porta -53

Prótocolos: tcp, udp, dns

```
root@redhatserver:/home# firewall-cmd --permanent --add-port=53/tcp
success
root@redhatserver:/home# firewall-cmd --permanent --add-port=53/udp
success
root@redhatserver:/home# firewall-cmd --permanent --add-service=dns
success
root@redhatserver:/home#
```

```
success
root@redhatserver:/home# firewall-cmd --reload
success
root@redhatserver:/home# firewall-cmd --list-ports
53/tcp 53/udp
root@redhatserver:/home# firewall-cmd --list-services
cockpit dhcpv6-client dns ssh
root@redhatserver:/home# _
```

Converter o ficheiro para executável (chmod-permissões)

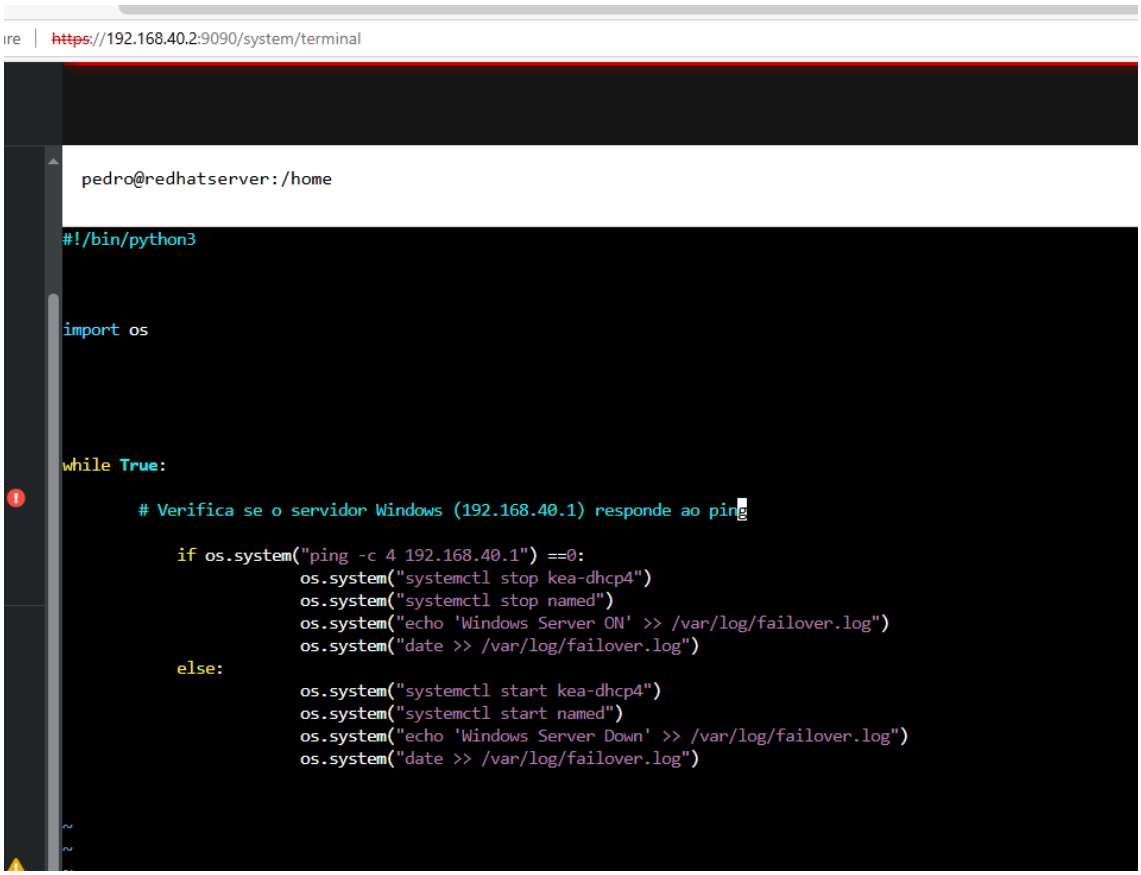
```
root@redhatserver:/home# chmod +x script.py
```

heiro , 2 formas possíveis :

./script.py ou python3 script.py



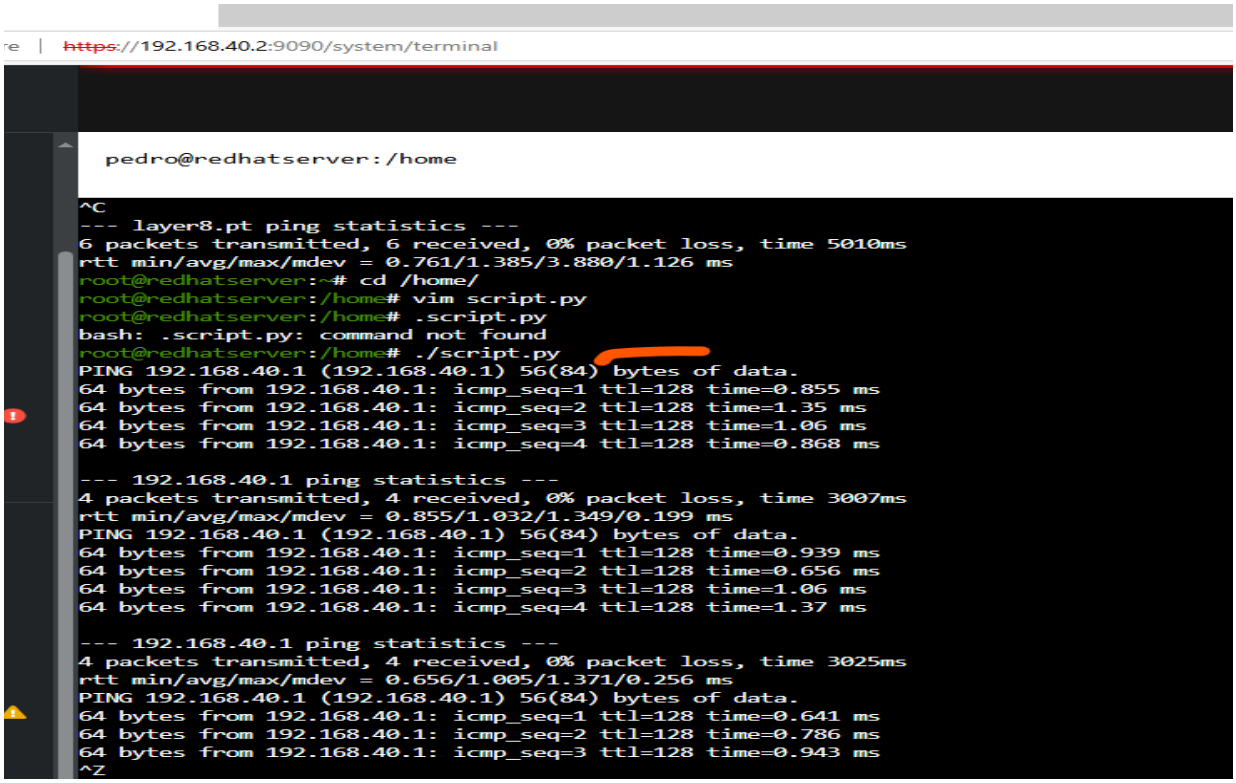
Editar ficheiro (comando nano ou vim)



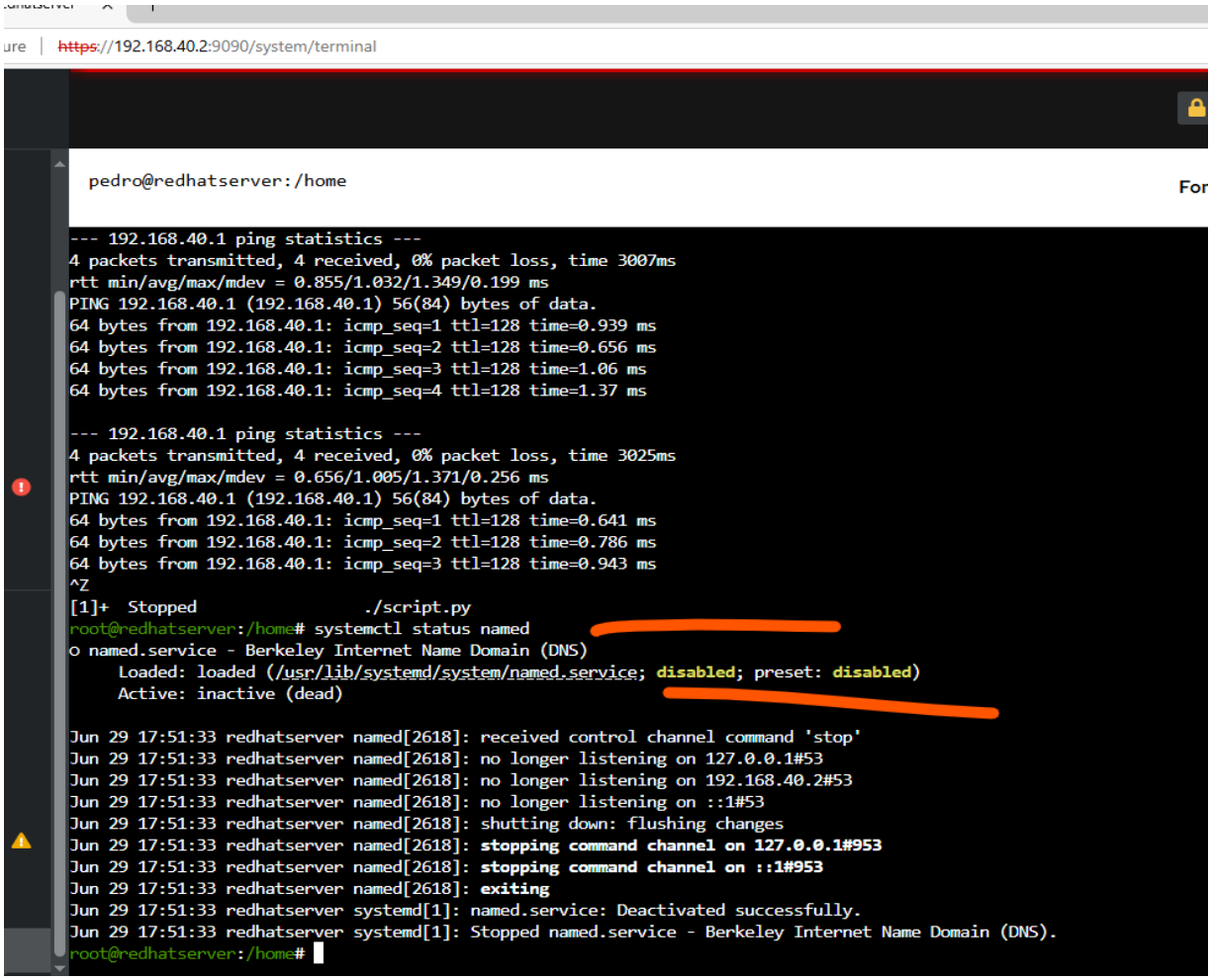
Testar Script



Apara parar Script comando: crt +z

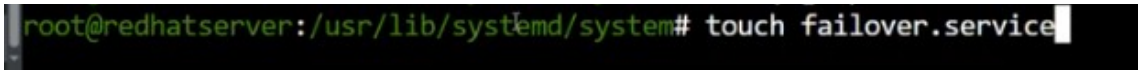


Verificação de status named :"disable"

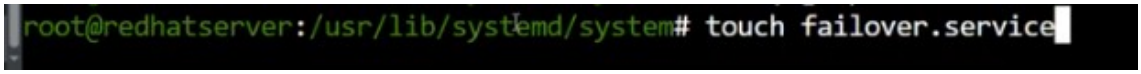


over ,é um mecanismo automático ou manual que garante a continuidade de um serviço mesmo que o sistema principal falhe.

Criação do serviço failover:



Edição do serviço failover:





Descrição detalhada do ficheiro failover.service

[Unit]

Description=Failover entre Windows Server & Red Hat Server

After=network.target

- **[Unit]** — Identificação do serviço
  - Description: Dá uma descrição legível do serviço — neste caso, indica que se trata de um mecanismo de failover entre Windows e Red Hat Server.
  - After=network.target: Este serviço só será iniciado após a rede estar disponível (importante, já que ele depende de ping).

[Service]

on3 /home/script.py

- [Service] — Comportamento do serviço
  - ExecStart: Define o comando que será executado quando o serviço for iniciado.
    - Aqui: executa o script Python script.py que está no /home/.
    - Esse script é o que faz o monitoramento por ping e ativa/desativa os serviços kea-dhcp4 e named.

[Install]

WantedBy=multi-user.target

- [Install] — Como o serviço será ativado
  - WantedBy=multi-user.target: Faz com que este serviço seja iniciado automaticamente quando o sistema atinge o nível de execução multiusuário (modo texto completo, normal em servidores**).**

Este ficheiro transforma o teu script Python de failover num serviço de sistema Linux (systemd), permitindo:

Funcionalidade	Resultado
systemctl start failover	**Inicia o script de failover**
systemctl enable failover	**Ativa o serviço para iniciar automaticamente**
systemctl status failover	**Verifica se está ativo e a correr**
journalctl -u failover	**Vê os logs do serviço**

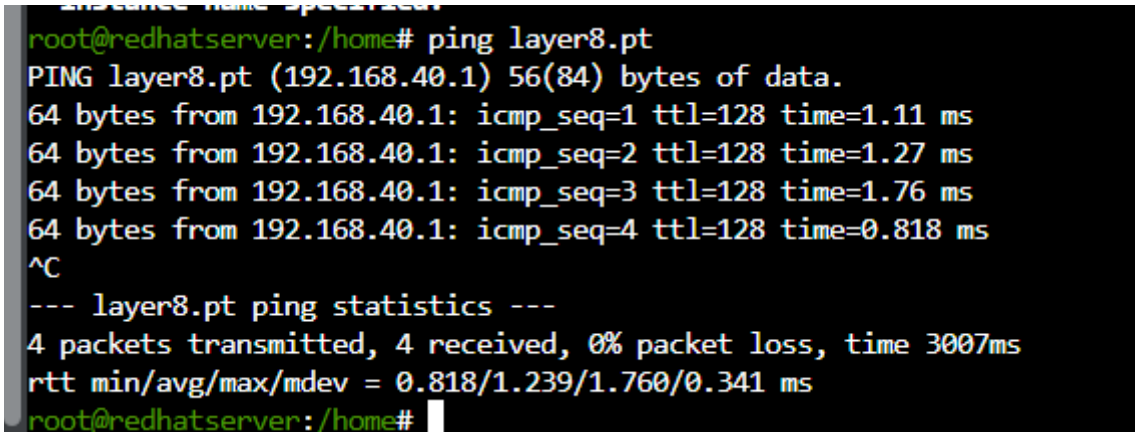
Ativação do Serviço failover

Comando→ systemctl enable failover.service

```
root@redhatserver:/home# systemctl enable failover.service
The unit files have no installation config (WantedBy=, RequiredBy=, UpheldBy=,
Also=, or Alias= settings in the [Install] section, and DefaultInstance= for
template units). This means they are not meant to be enabled or disabled using systemctl.

Possible reasons for having these kinds of units are:
• A unit may be statically enabled by being symlinked from another unit's
  .wants/, .requires/, or .upholds/ directory.
• A unit's purpose may be to act as a helper for some other unit which has
  a requirement dependency on it.
• A unit may be started when needed via activation (socket, path, timer,
  D-Bus, udev, scripted systemctl call, ...).
• In case of template units, the unit is meant to be enabled with some
  instance name specified.
```

Testar : ping layer8.pt

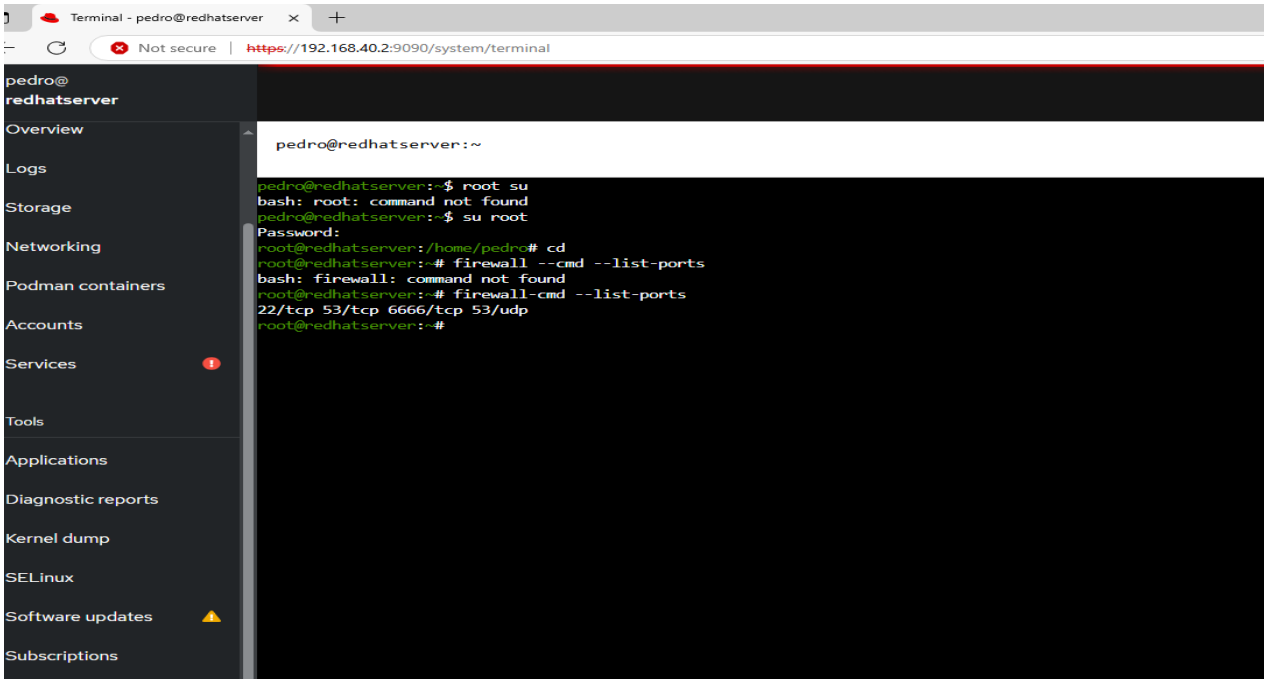


Desligar a placa de rede do WinServer

Verificar que faz a troca de ip automático -192.168.40.2

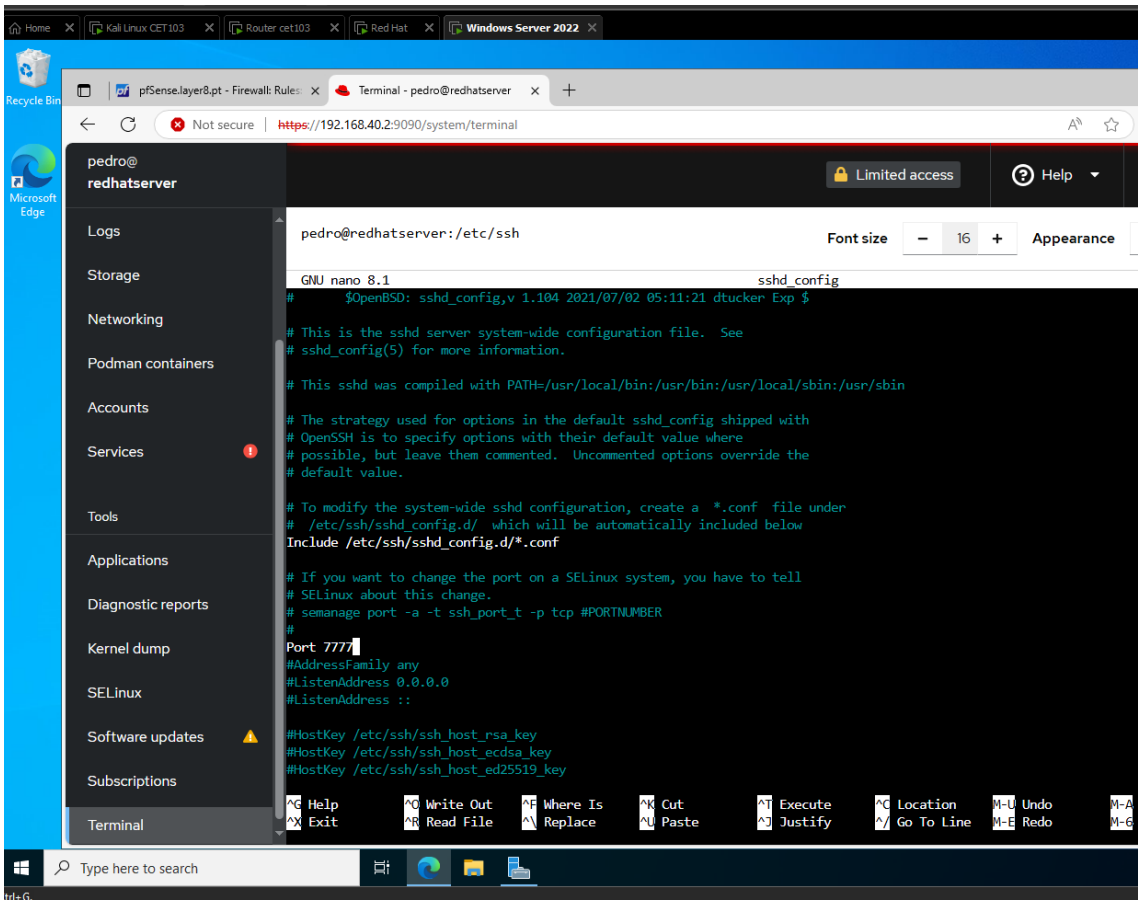
>>>>>>Erro

Confirmação se a porta 22 está ativa para a firewall



Redirecionar no ssh a “port per default -22 “ para “ port -7777“ - número aleatório”

erver:/etc/ssh# **nano sshd_config**



linhas alteradas

21 **Port 7777**

40 **PermitRootLogin no**

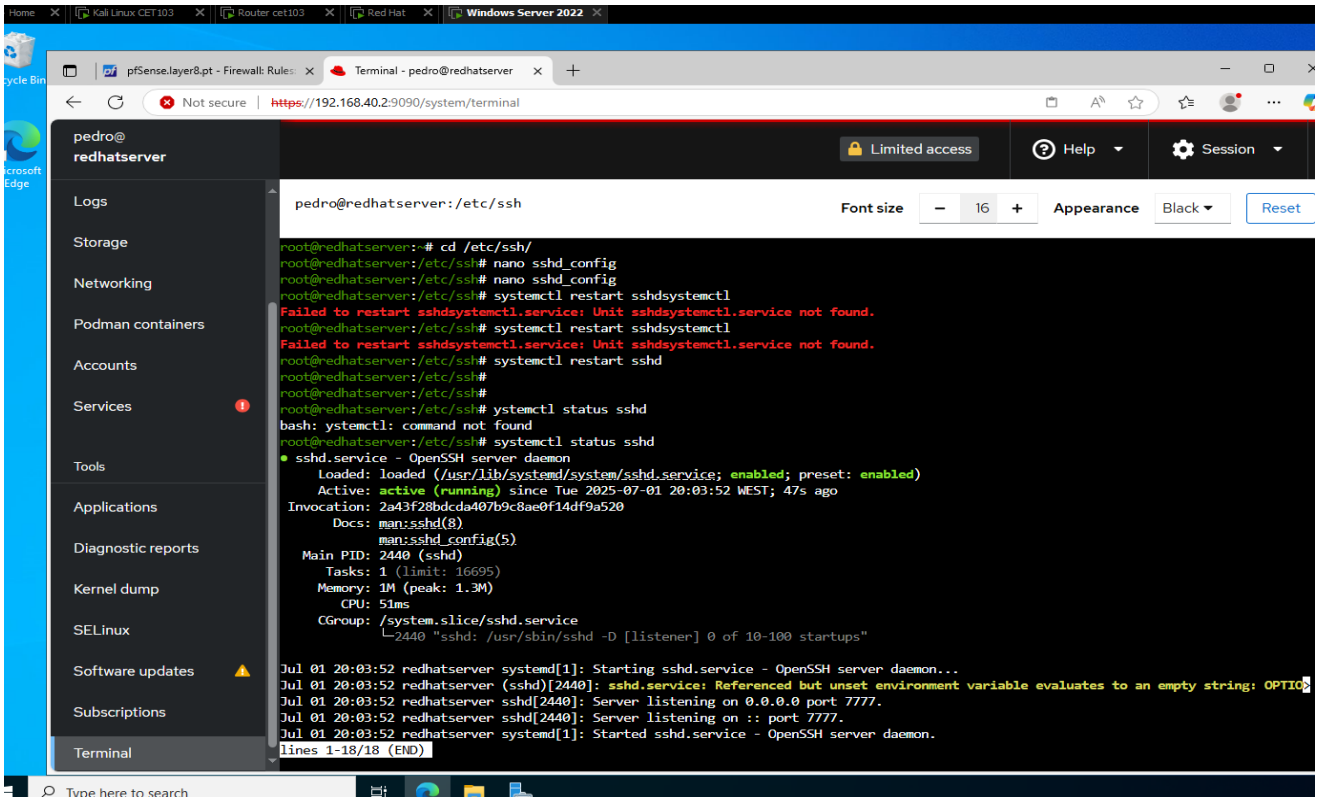
42 **MaxAuthTries 10000**

43 **MaxSessions 10**

Reniciar o serviço ssh

root@redhatserver:/etc/ssh# **systemctl restart sshd**

root@redhatserver:/etc/ssh# **systemctl status sshd**



ssh-access.target

- Um .target no systemd é um tipo especial de unidade que agrupa serviços.
- Serve para organizar o arranque ou ativação conjunta de vários serviços relacionados.
- ssh-access.target pode ser usado, por exemplo, para representar um estado ou grupo de serviços relacionados ao acesso via SSH.

root@redhatserver:/usr/lib/systemd/system# **cat ssh-access.target**

SPDX-License-Identifier: LGPL-2.1-or-later

#

This file is part of systemd.

#

systemd is free software; you can redistribute it and/or modify it

under the terms of the GNU Lesser General Public License as published by

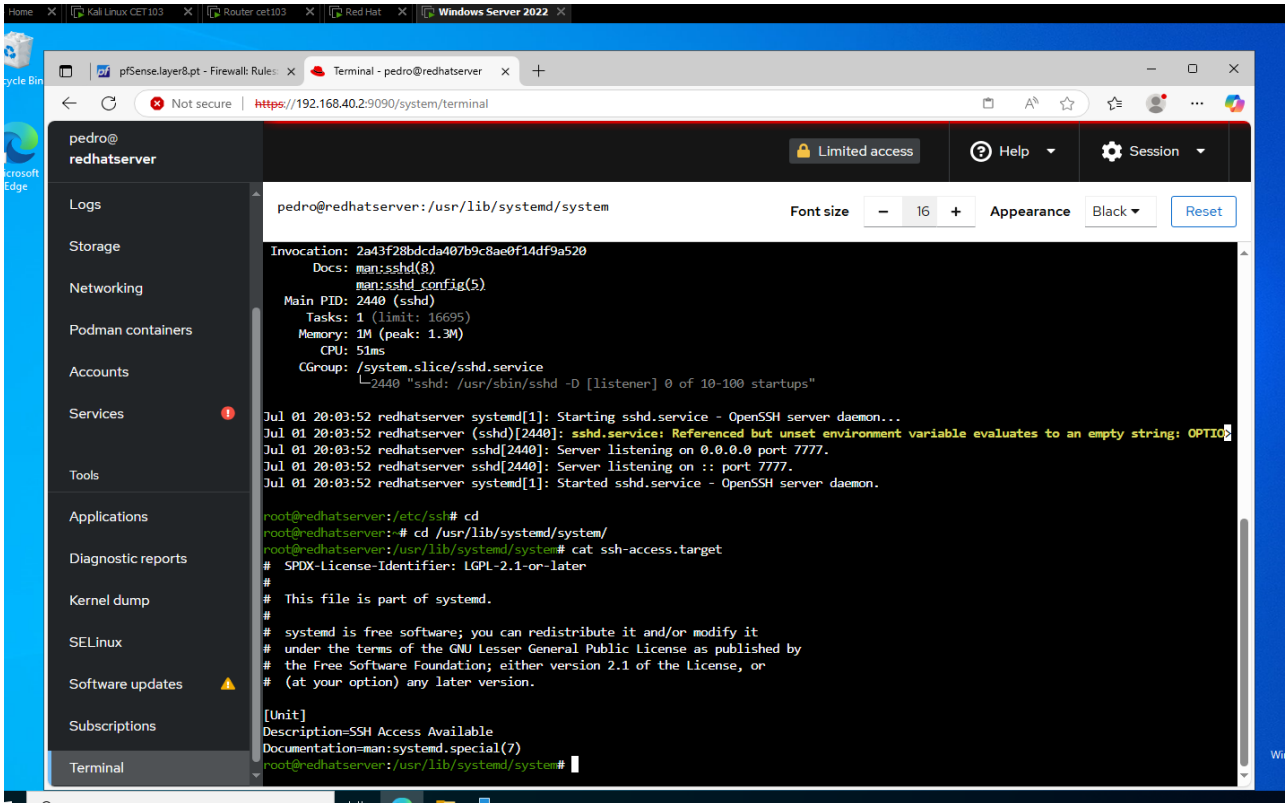
the Free Software Foundation; either version 2.1 of the License, or

any later version.

[Unit]

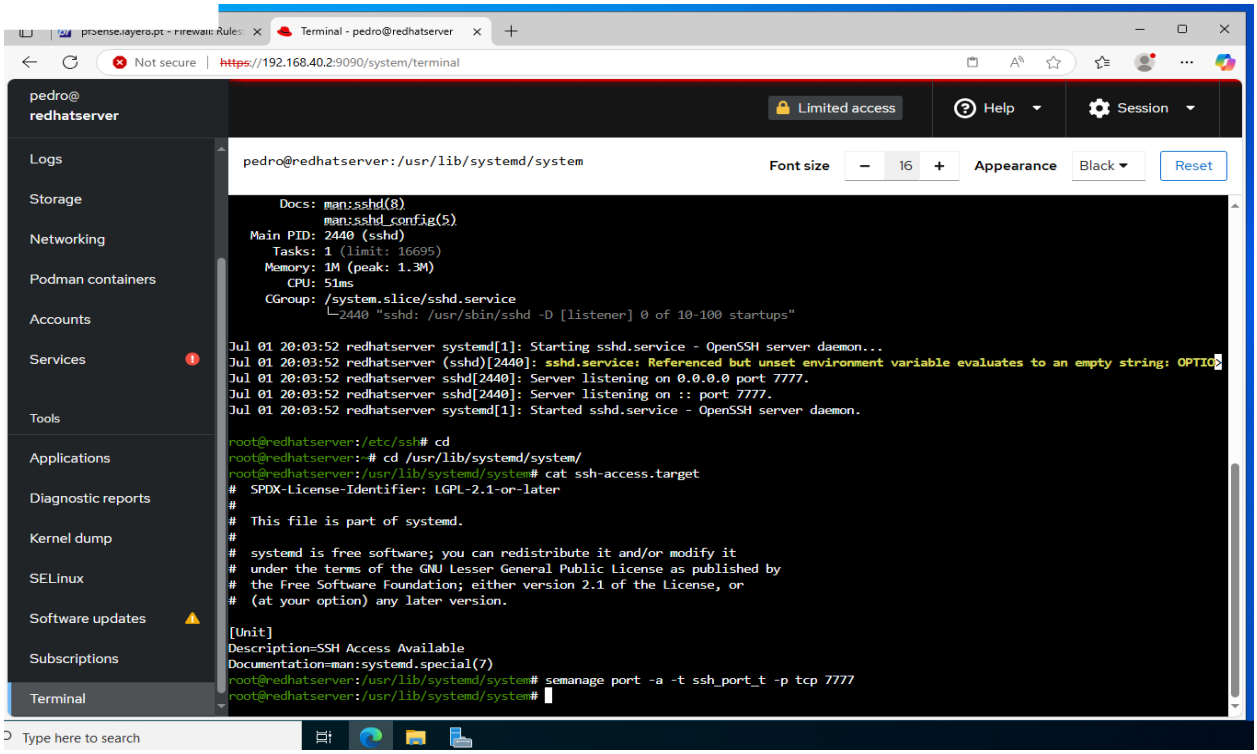
Description=SSH Access Available

Documentation=man:systemd.special(7)



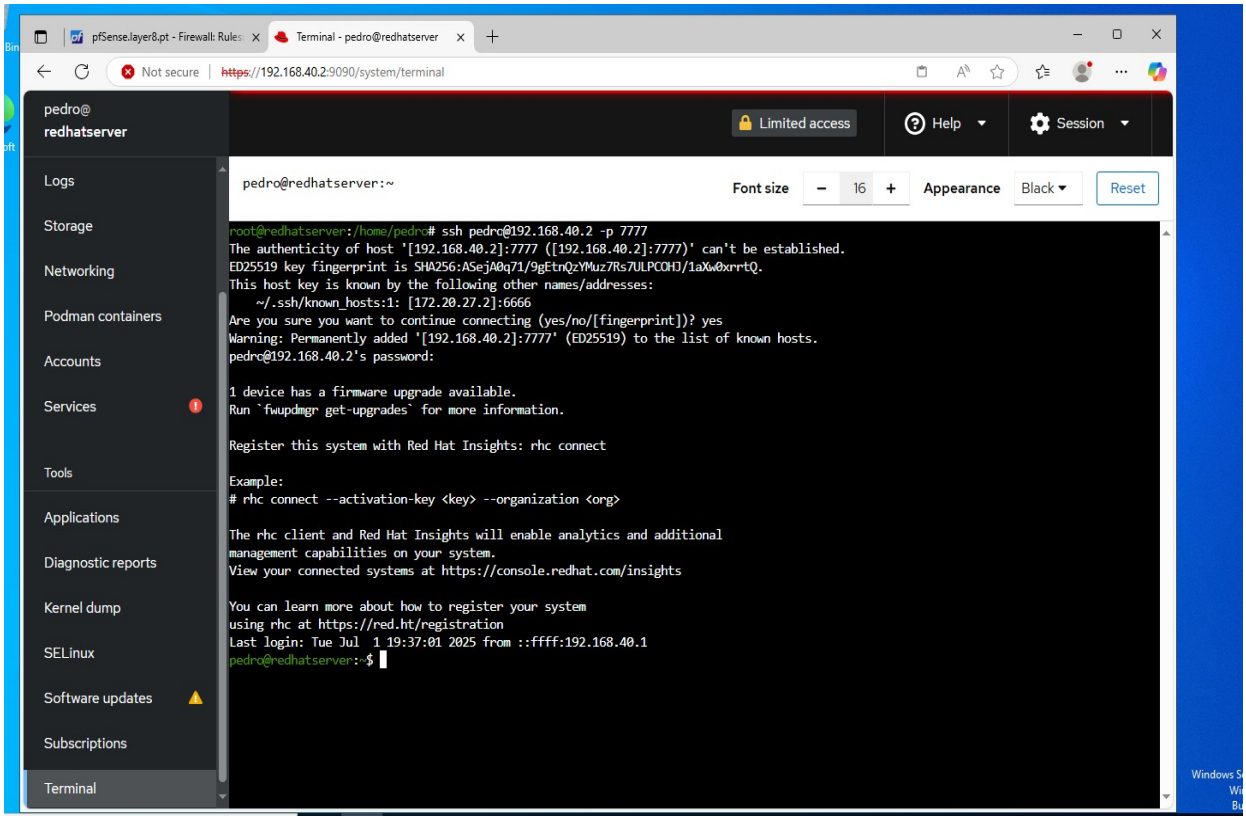
root@redhatserver:/usr/lib/systemd/system# **semanage port -a -t ssh_port_t -p tcp 7777**

comando é usado em sistemas Linux com SELinux (como Red Hat, CentOS, Rocky Linux) para permitir que o serviço SSH use uma porta diferente da padrão (22), neste caso, a porta 7777.



Acesso via ssh á porta 7777no redhat

root@redhatserver:/home/pedro# **ssh pedro@192.168.40.2 -p 7777**



á porta 6666no KaliLinux ao redhat

┌──(root㉿attacker)-[/usr/share/nmap/scripts]

└─# ssh pedro@192.168.89.139

Desativação da porta 7777 na firewall do RedHat

root@redhatserver:/home# **firewall-cmd --permanent --add-port=7777/tcp**

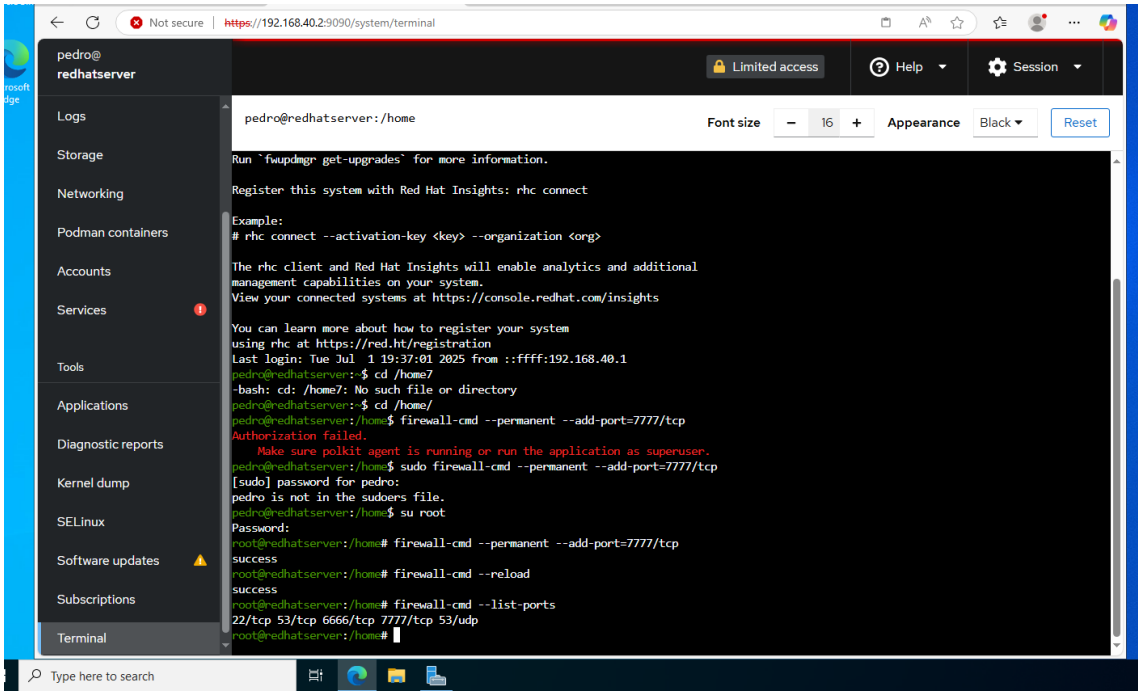
success

root@redhatserver:/home# **firewall-cmd --reload**

success

root@redhatserver:/home# **firewall-cmd --list-ports**

22/tcp 53/tcp 6666/tcp 53/udp



Verificação final :

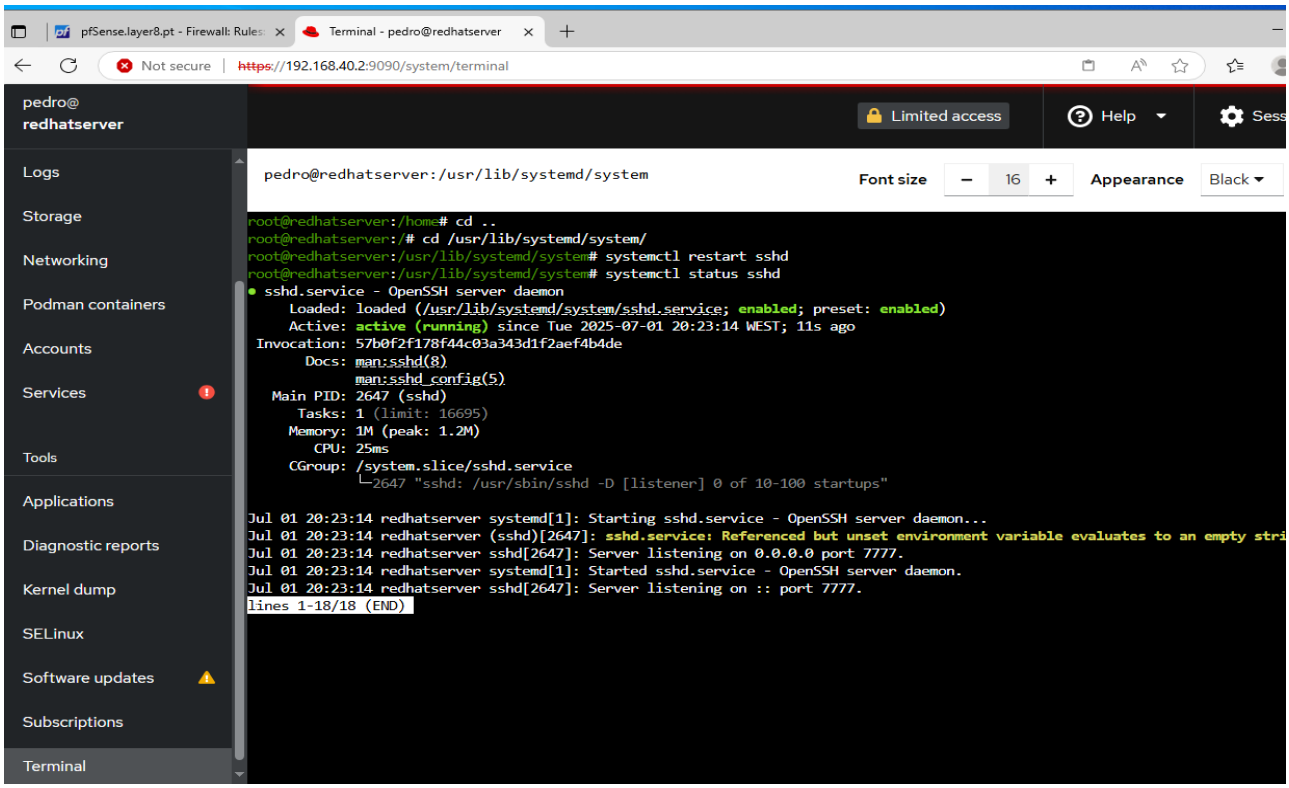
root@redhatserver:/usr/lib/systemd/system# **systemctl restart sshd**

root@redhatserver:/usr/lib/systemd/system# **systemctl status sshd**

● sshd.service - OpenSSH server daemon

Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)

Active: **active (running)** since Tue 2025-06-24 19:39:04 WEST; 13s ag



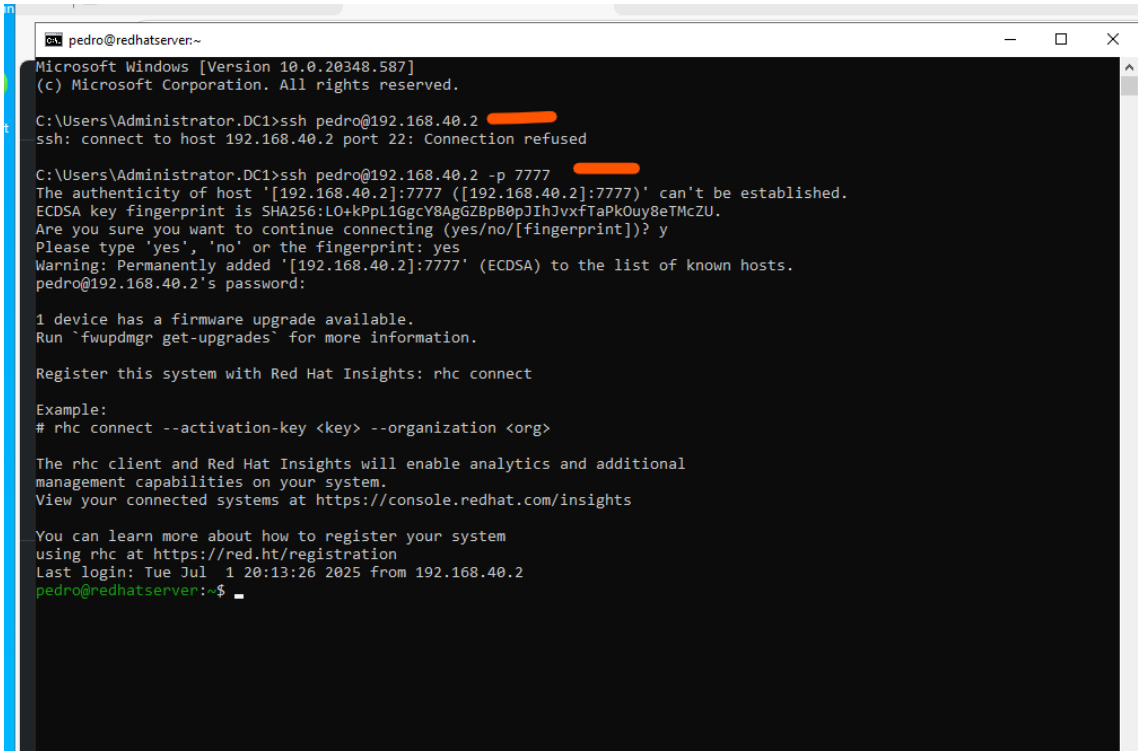
Confirmação no Windows Server de que o serviço ssh já não se encontra ativo

Comand prompt- cmd

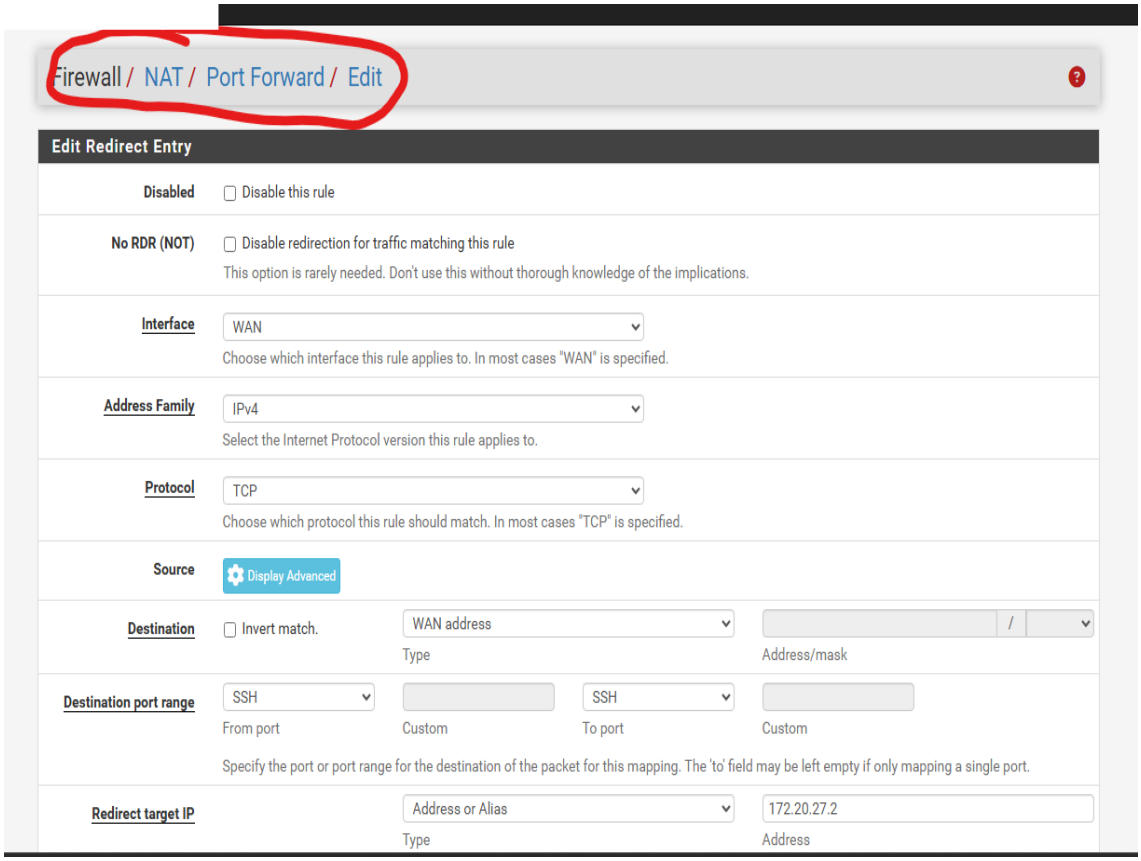
C:\Users\Administrator>C:\Users\Administrator>**ssh pedro@192.168.40.2**

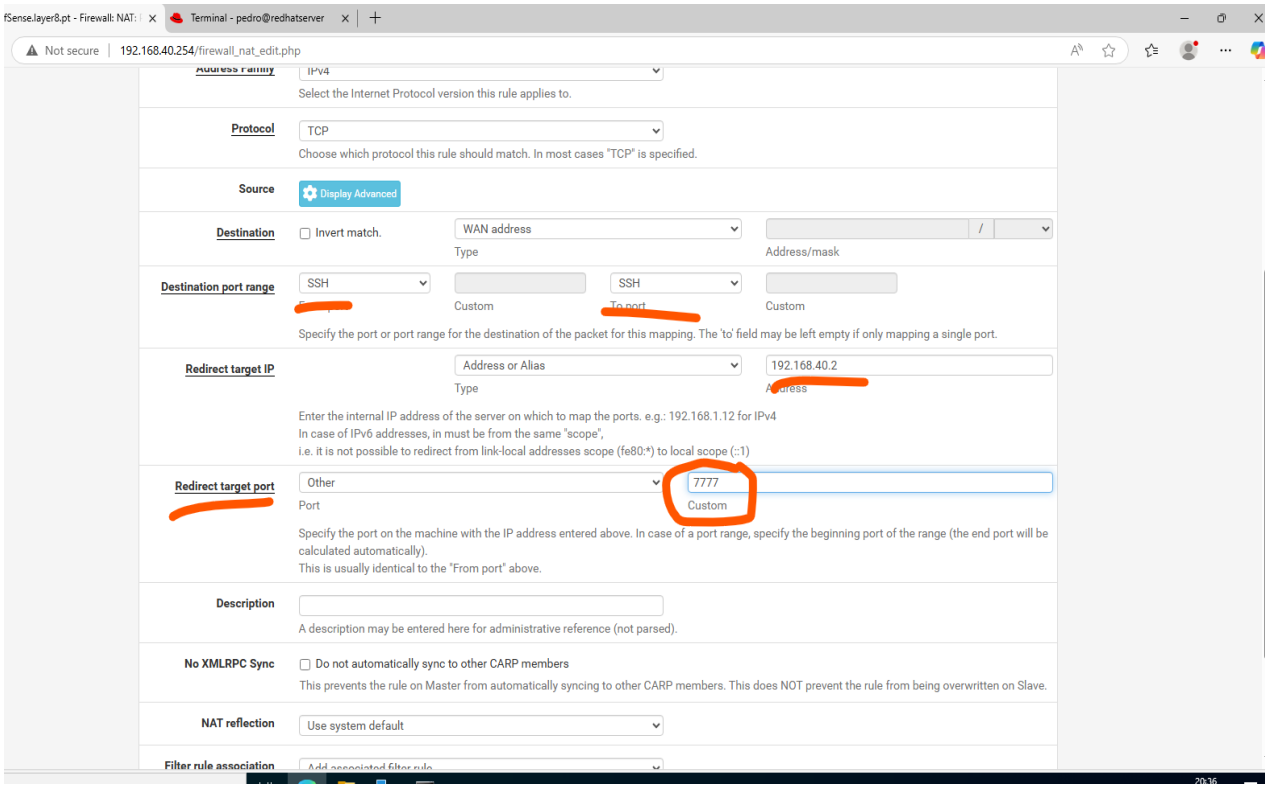
istrator' is not recognized as an internal or external command, operable program or batch file.

Confirmação de ligação em Windows via ssh na porta 7777

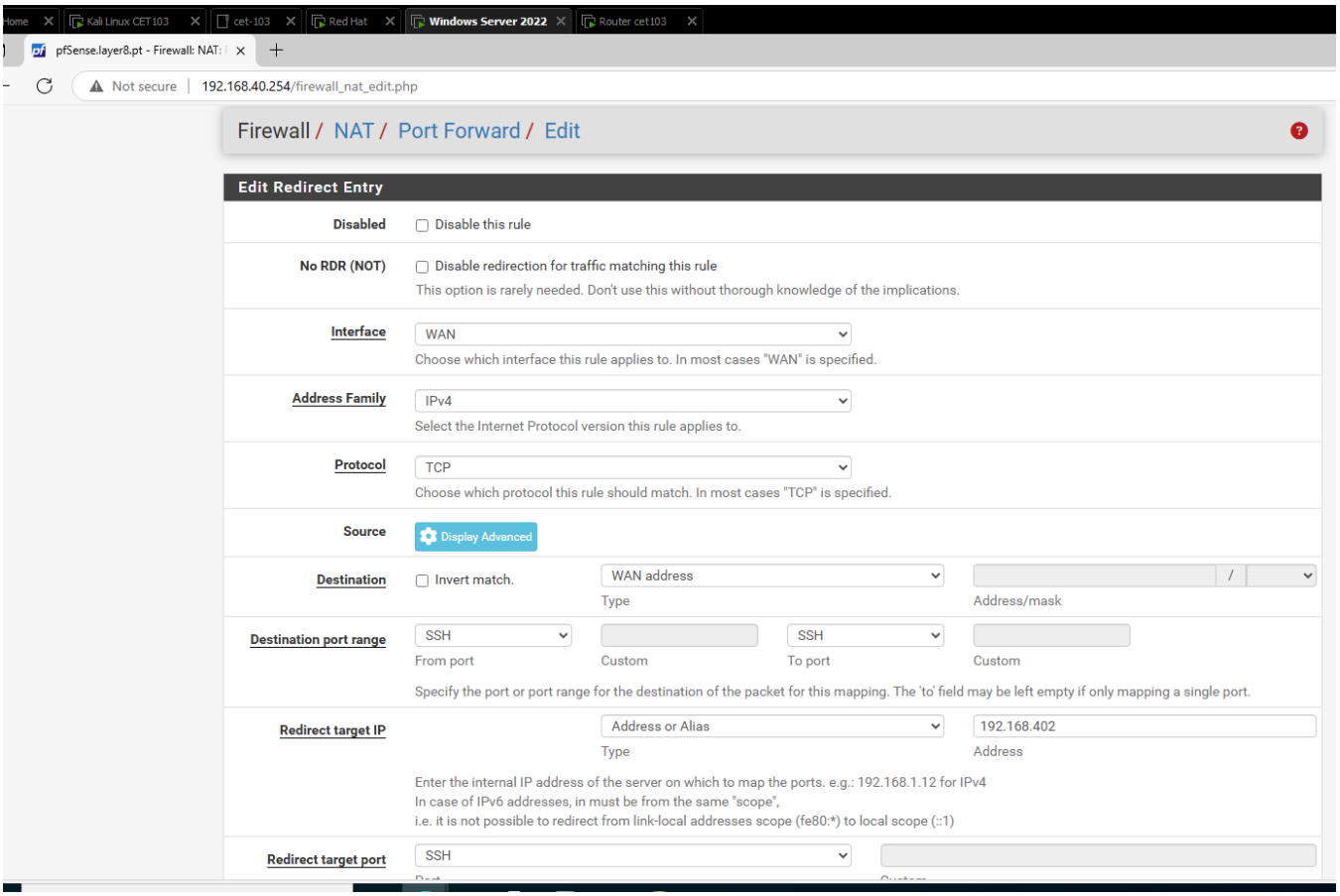


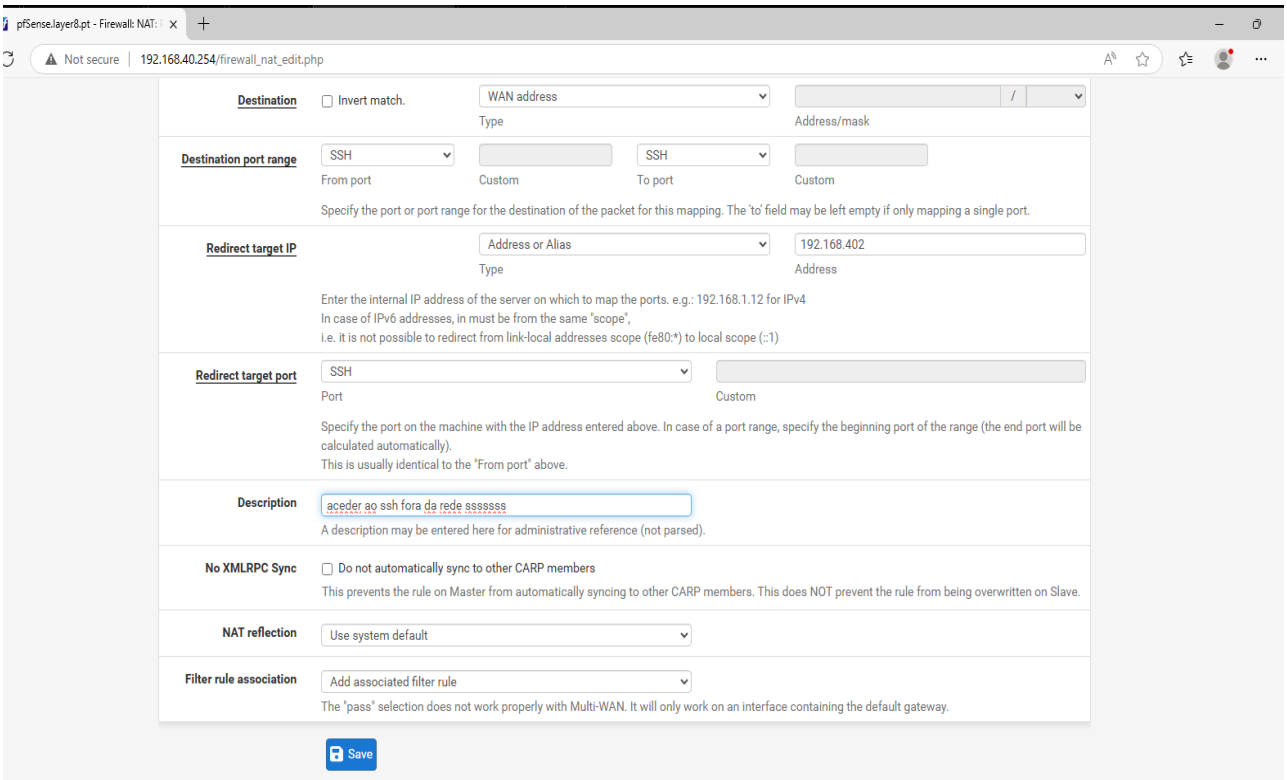
Redirecionar da porta - forward 22 ssh do routrer pfsense para a porta 7777

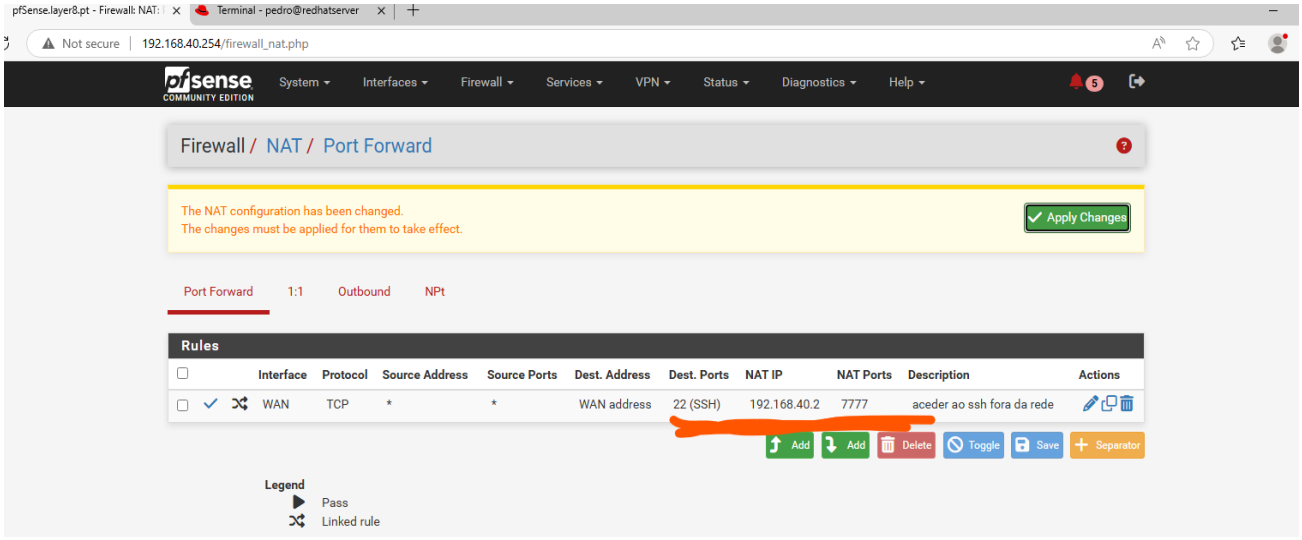

Firewall / NAT / Port Forward / Edit
Edit Redirect Entry
Disabled
Disable this rule
No RDR (NOT)
Disable redirection for traffic matching this rule
This option is rarely needed. Don't use this without thorough knowledge of the implications.
Interface
WAN
Choose which interface this rule applies to. In most cases "WAN" is specified.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which protocol this rule should match. In most cases "TCP" is specified.
Source
Display Advanced
Destination
Invert match.
WAN address
Type
Address/mask
Destination port range
SSH
From port
Custom
SSH
To port
Custom
Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.
Redirect target IP
Address or Alias
Type
172.20.27.2
Address


Terminal - pedro@redhatserver
Not secure
192.168.40.254/firewall_nat_edit.php
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which protocol this rule should match. In most cases "TCP" is specified.
Source
Display Advanced
Destination
Invert match.
WAN address
Type
Address/mask
Destination port range
SSH
Custom
SSH
To port
Custom
Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.
Redirect target IP
Address or Alias
Type
192.168.40.2
Enter the internal IP address of the server on which to map the ports. e.g.: 192.168.1.12 for IPv4
In case of IPv6 addresses, in must be from the same "scope",
i.e. it is not possible to redirect from link-local addresses scope (fe80:*) to local scope (::1)
Redirect target port
Other
Port
7777
Custom
Specify the port on the machine with the IP address entered above. In case of a port range, specify the beginning port of the range (the end port will be calculated automatically).
This is usually identical to the "From port" above.
Description
A description may be entered here for administrative reference (not parsed).
No XMLRPC Sync
Do not automatically sync to other CARP members
This prevents the rule on Master from automatically syncing to other CARP members. This does NOT prevent the rule from being overwritten on Slave.
NAT reflection
Use system default
Filter rule association

22 -port forward endereço 192.168.40.2





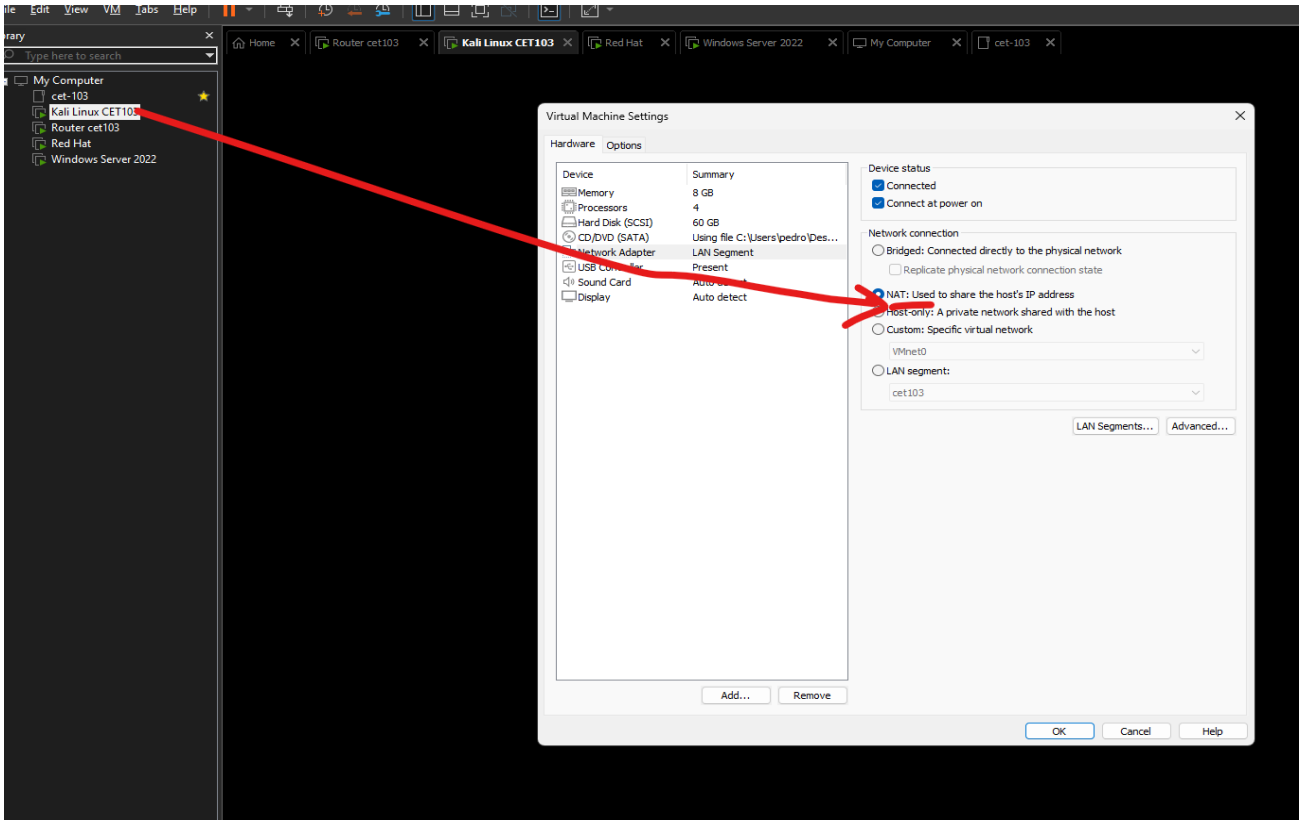




Resumo:

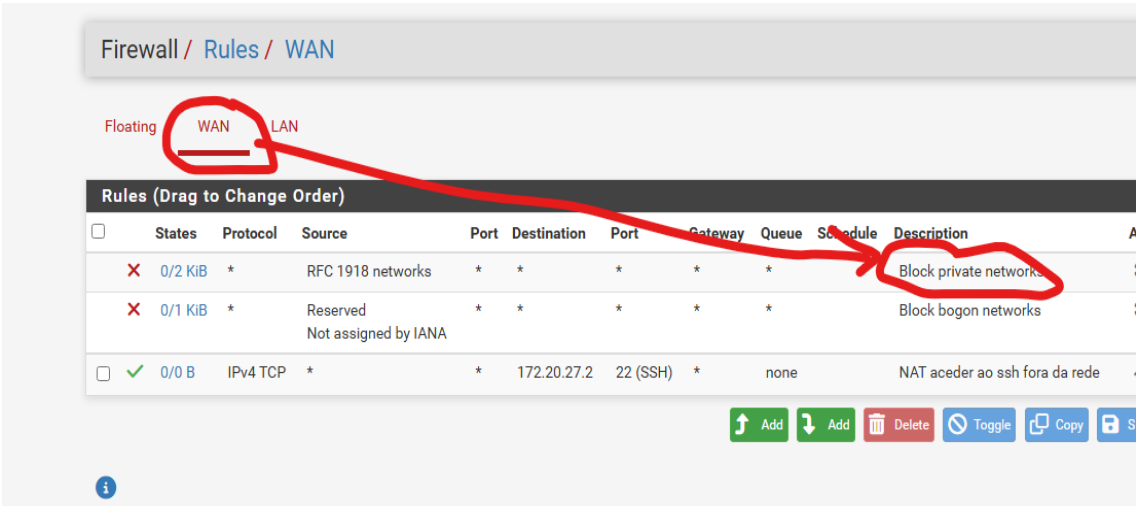
Qual o kali em,NAT-WAN ,requesita acesso á porta 22 ssh no router (pfsense) , este redireciona através da porta 7777 do mesmo para o redhat que se encontra em INtnet-LAN.O WinServer consegue igualmente aceder ao redhat pela porta 7777 e não mais pela porta 22

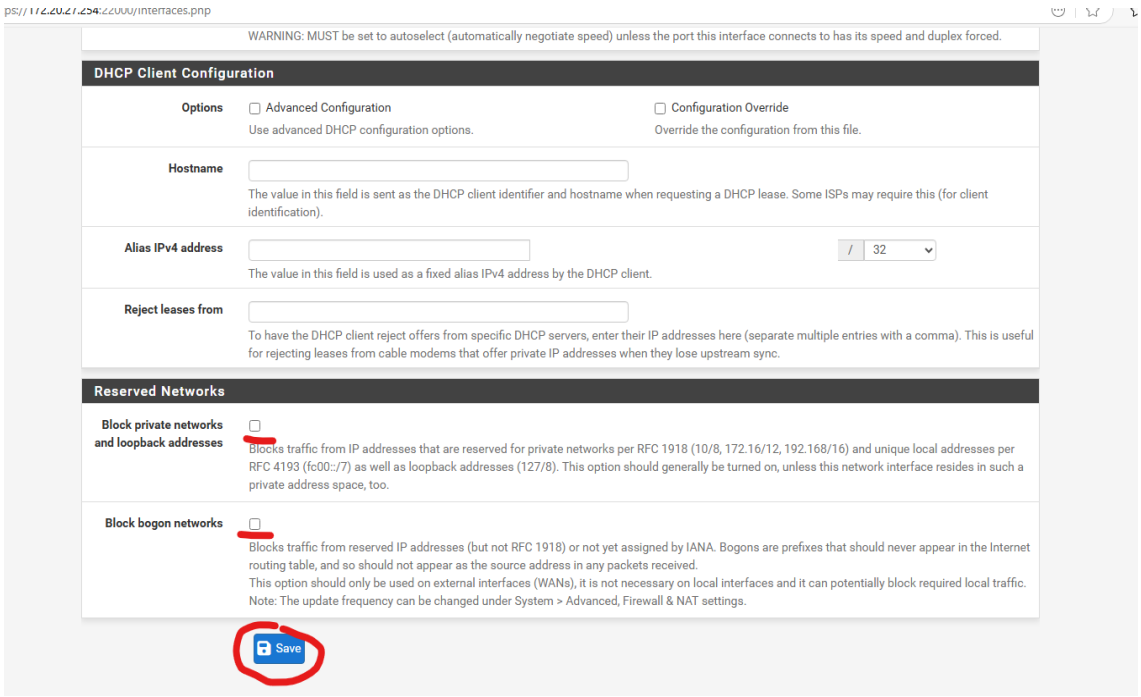
ar o kali em Nat

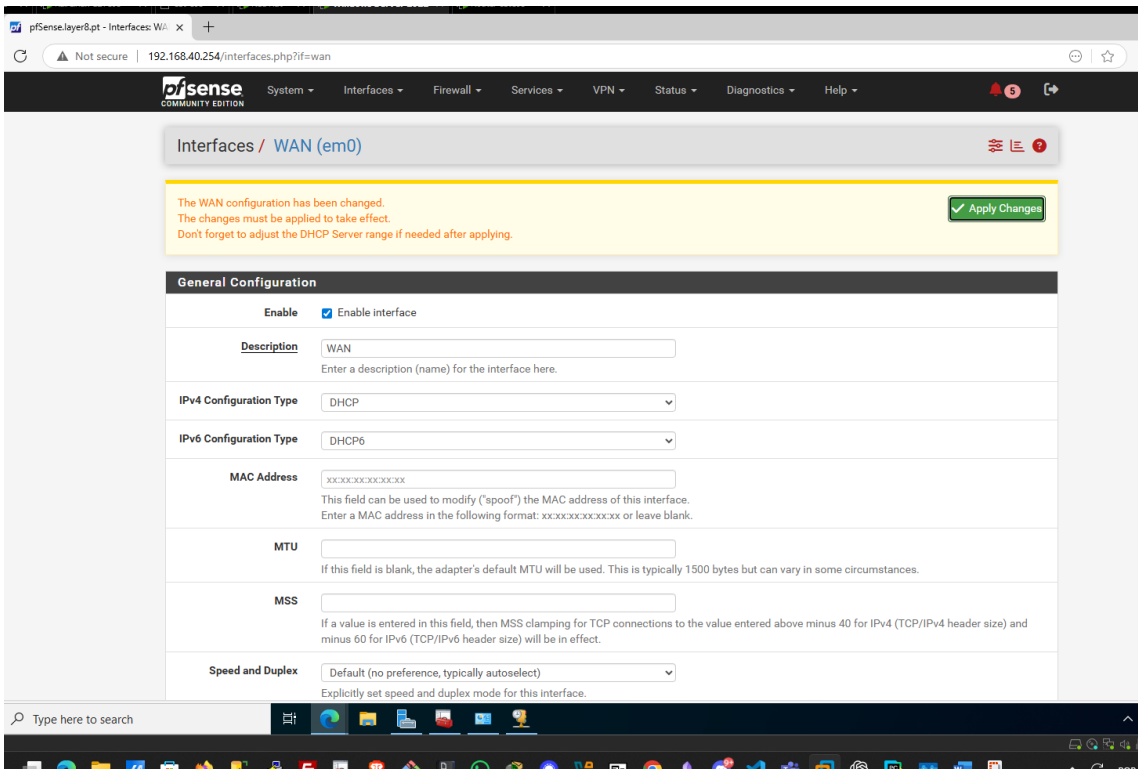


Clicar em –“Actions” .icon –“192roda dentada”-

Descelecionar os dois blocos indicados e gravar –“save “







Acesso por Wan no kali com o endereço da pfsense (WAN)

┌──(root㉿attacker)-[/home/pedro]

r Wan no kali com o endereço da pfsense (WAN)

┌──(root㉿attacker)-[/home/pedro]

└─# **ssh pedro@192.168.89.139**

Erro

nmap -p- -A -sV 192.168.89.139 -o router_layer8.pt.txt

Starting Nmap 7.95 (https://nmap.org) at 2025-07-01 19:44 WEST

Nmap scan report for 192.168.89.139

Host is up (0.00015s latency).

All 65535 scanned ports on 192.168.89.139 are in ignored states.

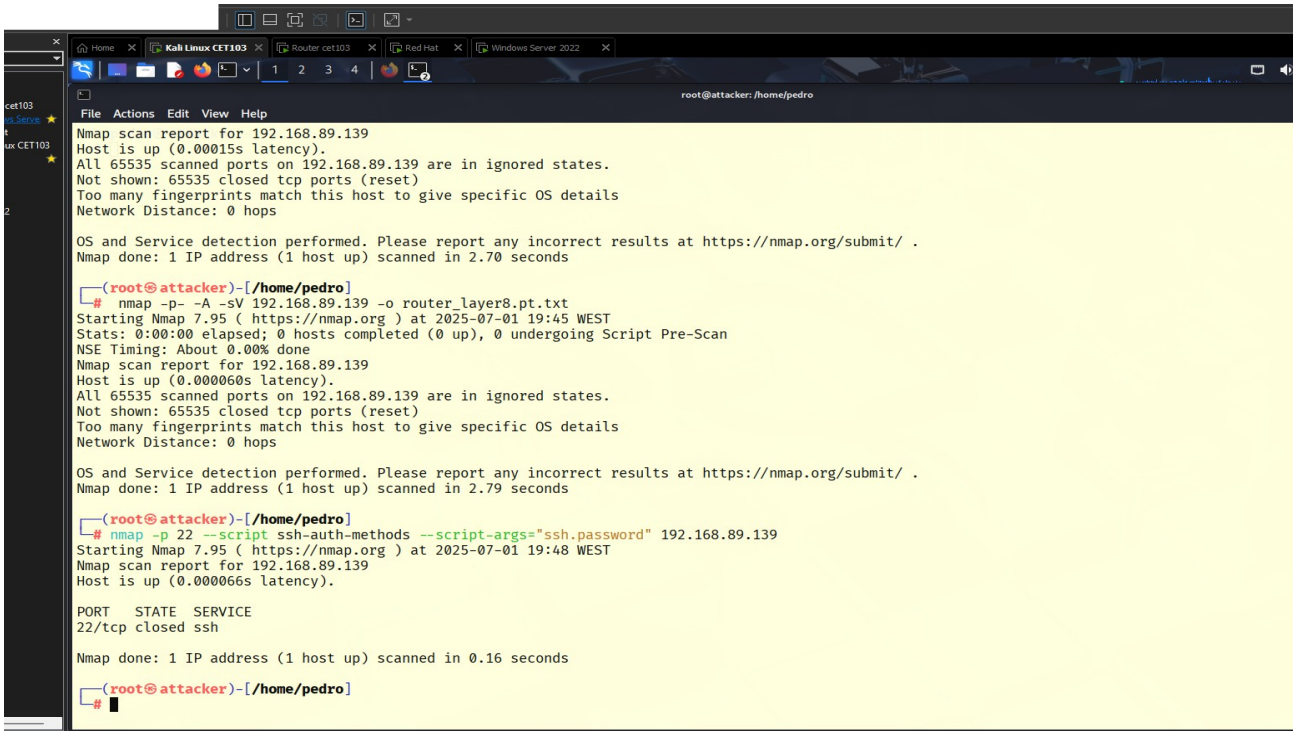
Not shown: 65535 closed tcp ports (reset)

Too many fingerprints match this host to give specific OS details

Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .

Nmap done: 1 IP address (1 host up) scanned in 2.70 seconds



Criação de ficheiros para Users

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# **touch username.lst**

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# **nano username.lst**

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# cat username.lst

Administrator

admin

root

pedro

Criação de ficheiros para Passwords

limpeza de ficheiro “password.lst”

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

password.lst

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# **nano password.lst**

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# **cat password.lst**

sporting

Sporting

root

pedro

sporting1906

---

script ssh-brute

- Tenta autenticar-se no serviço SSH usando todas as combinações possíveis entre os utilizadores em username.lst e as palavras-passe em password.lst

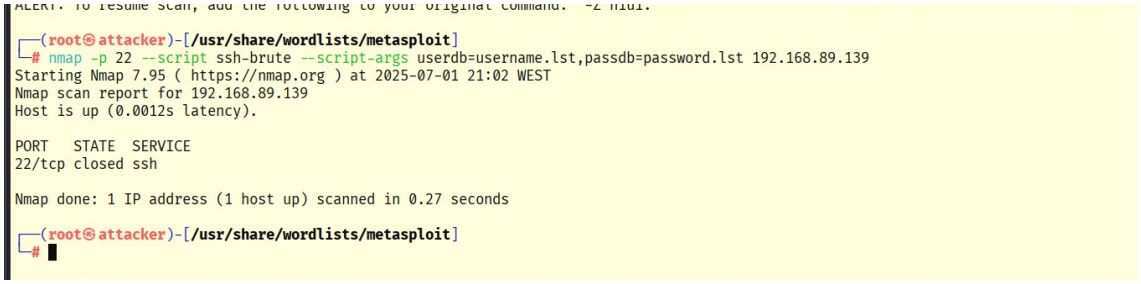
- **Parte do comando Significado**
  -

`nmap`	Ferramenta de varrimento de rede usada para descobrir hosts, serviços e vulnerabilidades.
`-p 22`	Especifica a **porta 22**, usada normalmente pelo **SSH**.
`--script ssh-brute`	Usa um **script do Nmap NSE** chamado `ssh-brute`, que realiza ataques de **força bruta SSH**.
`--script-args`	Define argumentos personalizados para o script.
`userdb=username.lst`	Indica o ficheiro com a lista de utilizadores a testar (`username.lst`).
`passdb=password.lst`	Indica o ficheiro com a **lista de palavras-passe** a testar (`password.lst`).
`192.168.89.130`	IP de destino — o servidor SSH que será alvo do teste de força bruta.

  -

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# **nmap -p 22 --script ssh-brute --script-args userdb=username.lst,passdb=password.lst 192.168.89.139**



Starting Nmap 7.95 (https://nmap.org) at 2025-06-24 18:08 WEST

Failed to resolve "-".

Bare '-': did you put a space between '--'?

Failed to resolve "passdb=password.lst".

NSE: [ssh-brute] Trying username/password pair: Administrator:administrator

NSE: [ssh-brute] Trying username/password pair: admin:admin

NSE: [ssh-brute] Trying username/password pair: root:root

NSE: [ssh-brute] Trying username/password pair: pedro:pedro

NSE: [ssh-brute] Trying username/password pair: Administrator:

NSE: [ssh-brute] Trying username/password pair: admin:

NSE: [ssh-brute] Trying username/password pair: root:

NSE: [ssh-brute] Trying username/password pair: pedro:

NSE: [ssh-brute] Trying username/password pair: Administrator:123456

NSE: [ssh-brute] Trying username/password pair: admin:123456

NSE: [ssh-brute] Trying username/password pair: root:123456

NSE: [ssh-brute] Trying username/password pair: pedro:123456

NSE: [ssh-brute] Trying username/password pair: Administrator:12345

NSE: [ssh-brute] Trying username/password pair: admin:12345

NSE: [ssh-brute] Trying username/password pair: root:12345

NSE: [ssh-brute] Trying username/password pair: pedro:12345

NSE: [ssh-brute] Trying username/password pair: Administrator:123456789

NSE: [ssh-brute] Trying username/password pair: admin:123456789

NSE: [ssh-brute] Trying username/password pair: root:123456789

NSE: [ssh-brute] Trying username/password pair: pedro:123456789

NSE: [ssh-brute] Trying username/password pair: Administrator:password

ying username/password pair: admin:password

Listagem de scripts do nmap

┌──(root㉿attacker)-[/usr/share/nmap/scripts]

└─# **ls**

acarsd-info.nse hostmap-crtsh.nse ip-geolocation-map-bing.nse rsa-vuln-roca.nse

address-info.nse hostmap-robtex.nse ip-geolocation-map-google.nse rsync-brute.nse

afp-brute.nse http-adobe-coldfusion-apsa1301.nse ip-geolocation-map-kml.nse rsync-list-modules.nse

afp-ls.nse http-affiliate-id.nse ip-geolocation-maxmind.nse rtsp-methods.nse

afp-path-vuln.nse http-apache-negotiation.nse ip-https-discover.nse rtsp-url-brute.nse

afp-serverinfo.nse http-apache-server-status.nse ipidseq.nse rusers.nse

afp-showmount.nse http-aspnet-debug.nse ipmi-brute.nse s7-info.nse

ajp-auth.nse http-auth-finder.nse ipmi-cipher-zero.nse samba-vuln-cve-2012-1182.nse

ajp-brute.nse http-auth.nse ipmi-version.nse script.db

ajp-headers.nse http-avaya-ipoffice-users.nse ipv6-multicast-mld-list.nse servicetags.nse

ajp-methods.nse http-awstatstotals-exec.nse ipv6-node-info.nse shodan-api.nse

ajp-request.nse http-axis2-dir-traversal.nse ipv6-ra-flood.nse sip-brute.nse

allseeingeye-info.nse http-backup-finder.nse irc-botnet-channels.nse sip-call-spoof.nse

amqp-info.nse http-barracuda-dir-traversal.nse irc-brute.nse sip-enum-users.nse

asn-query.nse http-bigip-cookie.nse irc-info.nse sip-methods.nse

etc...

as scripts NSE com nomes relacionados ao protocolo SSH, como

por exemplo:

┌──(root㉿attacker)-[/usr/share/nmap/scripts]

└─# **ls | grep "ssh"**

ssh2-enum-algos.nse

ssh-auth-methods.nse

ssh-brute.nse

ssh-hostkey.nse

ssh-publickey-acceptance.nse

ssh-run.nse

sshv1.nse

Redirecionar no ssh a “port per default -22 “ para “ port -777 “ - número aleatório”

root@redhatserver:/etc/ssh# **nano sshd_config**

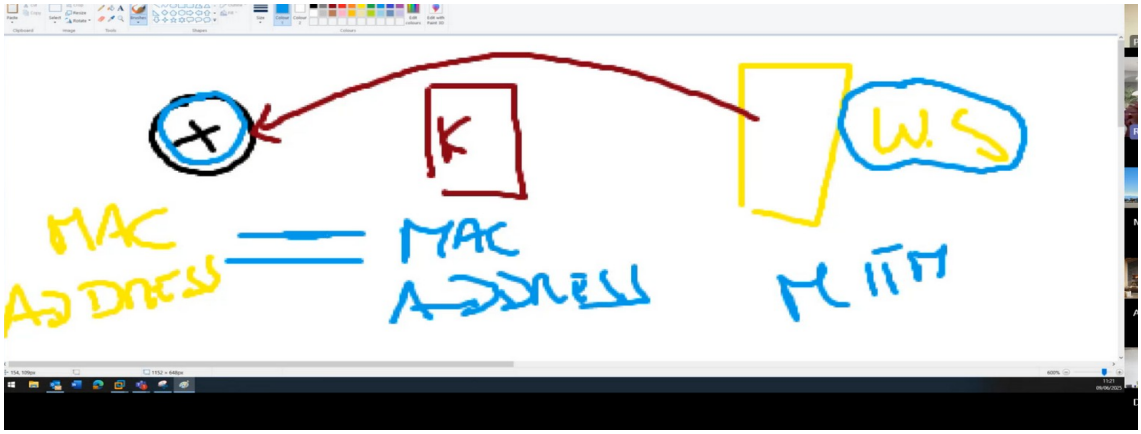
Kali Linux

Hostname: kali

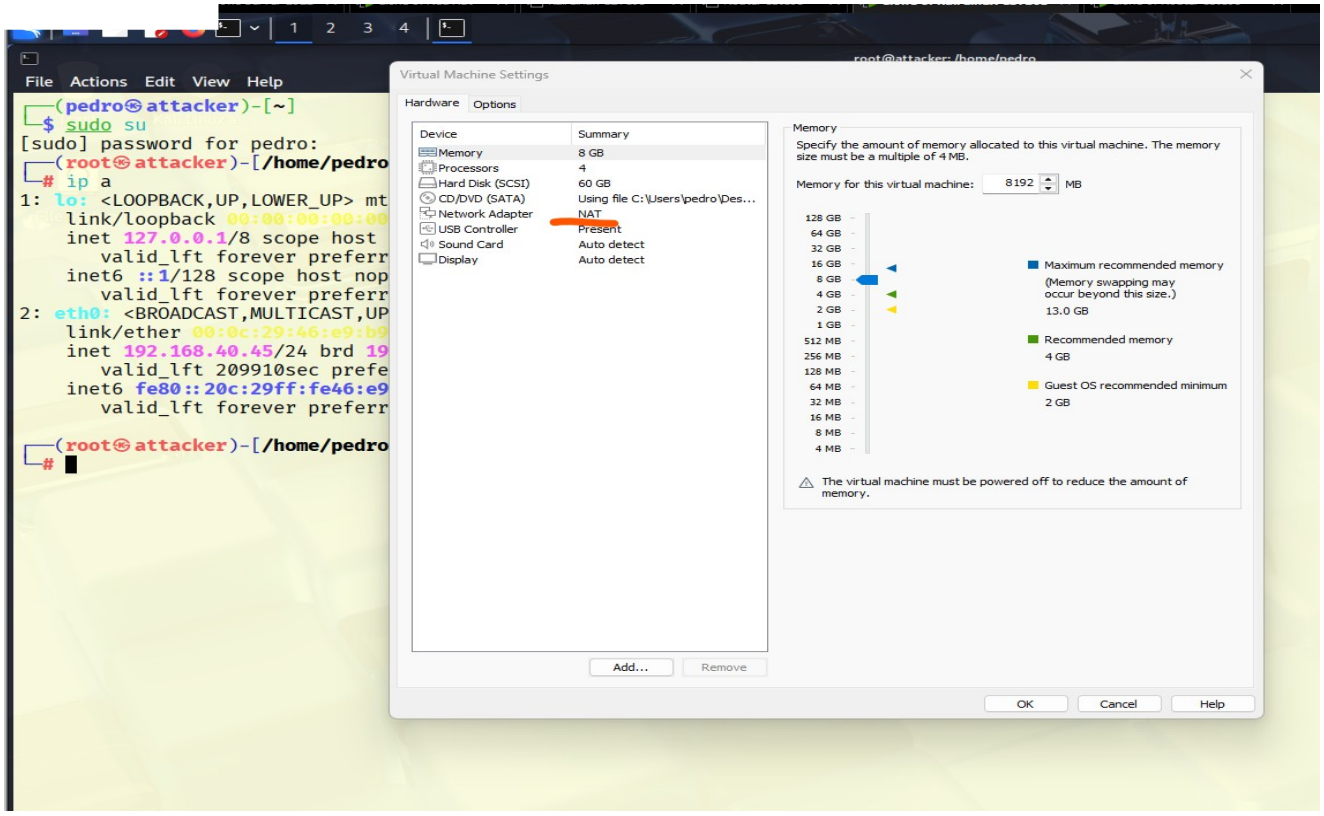
OS: Kali Linux

Interfaces IP: Ethernet0 (DHCP)

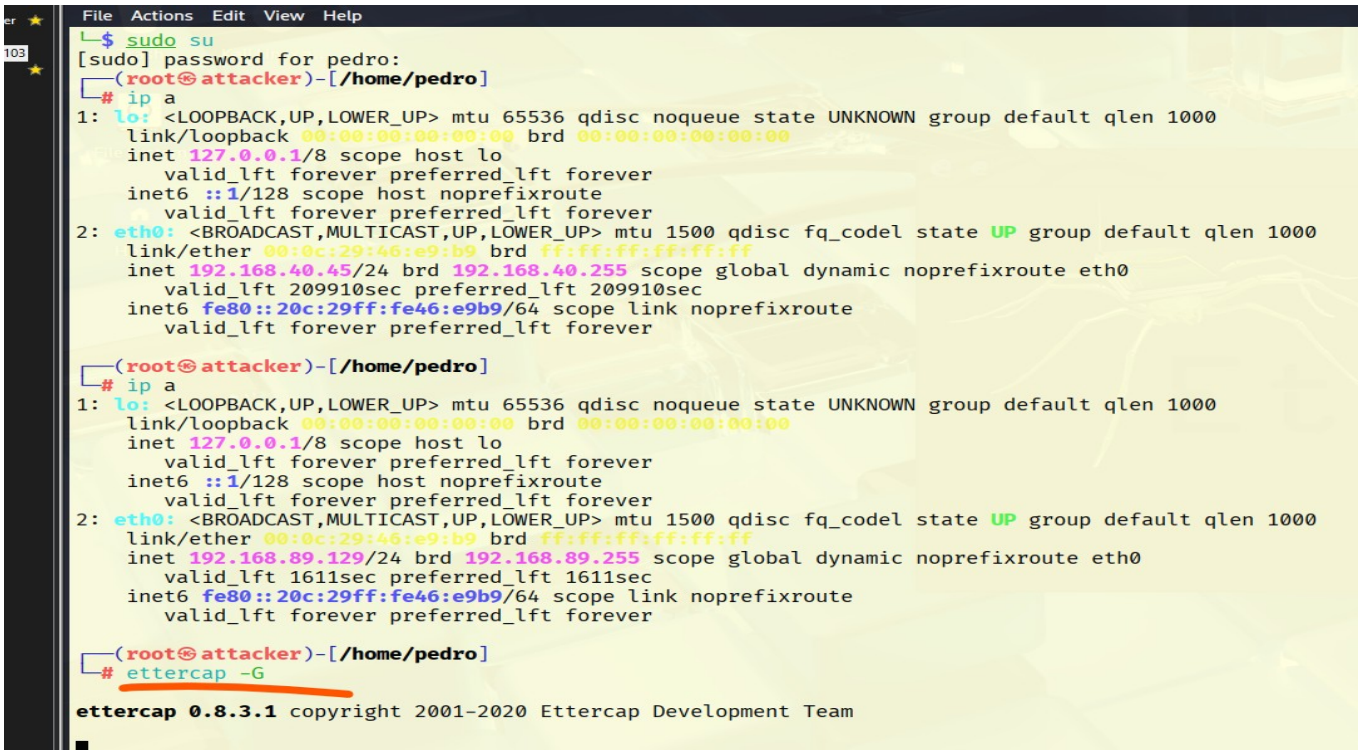
Descrição: Deve ser feito um ataque MITM capturando as credenciais do utilizador da pfsense.



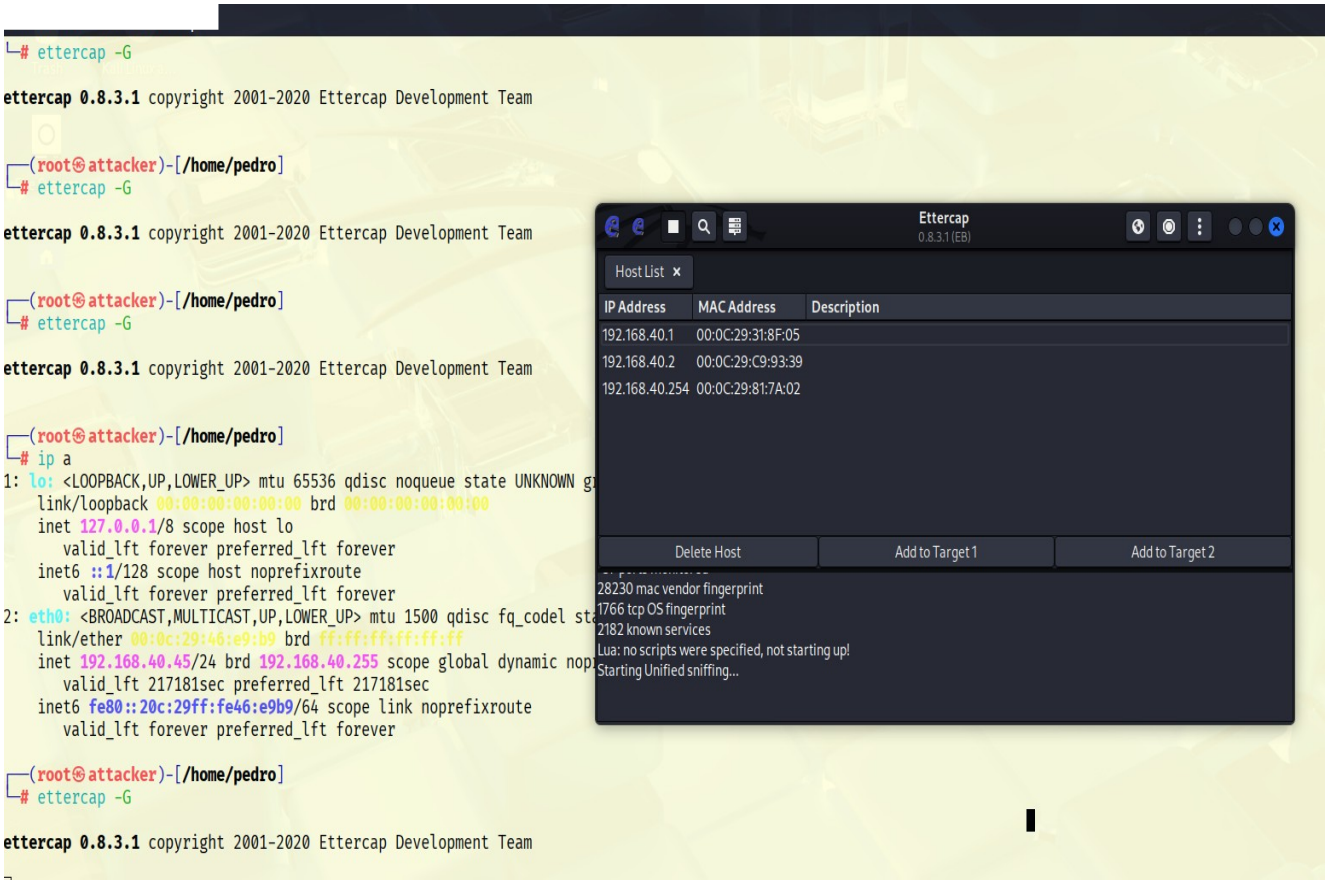
Colocar o Kali em Nat



ettercap -G

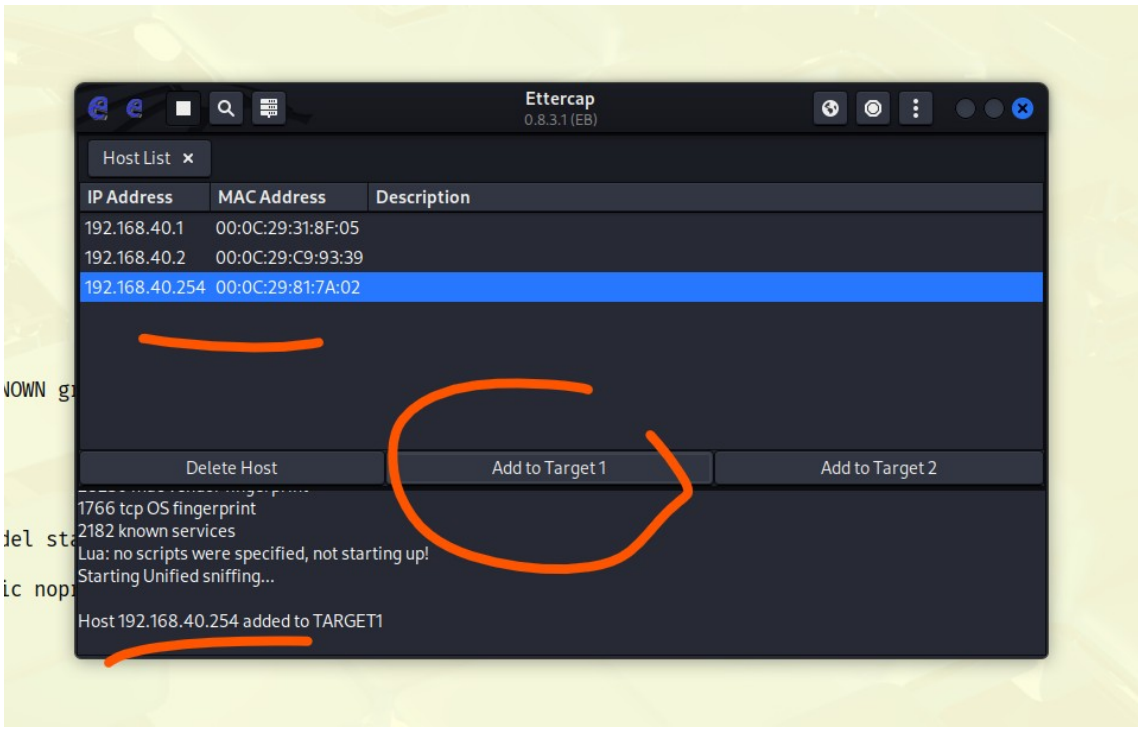


Host LIsts 192.168.40.1,192.168.40.2,192.168.40.254

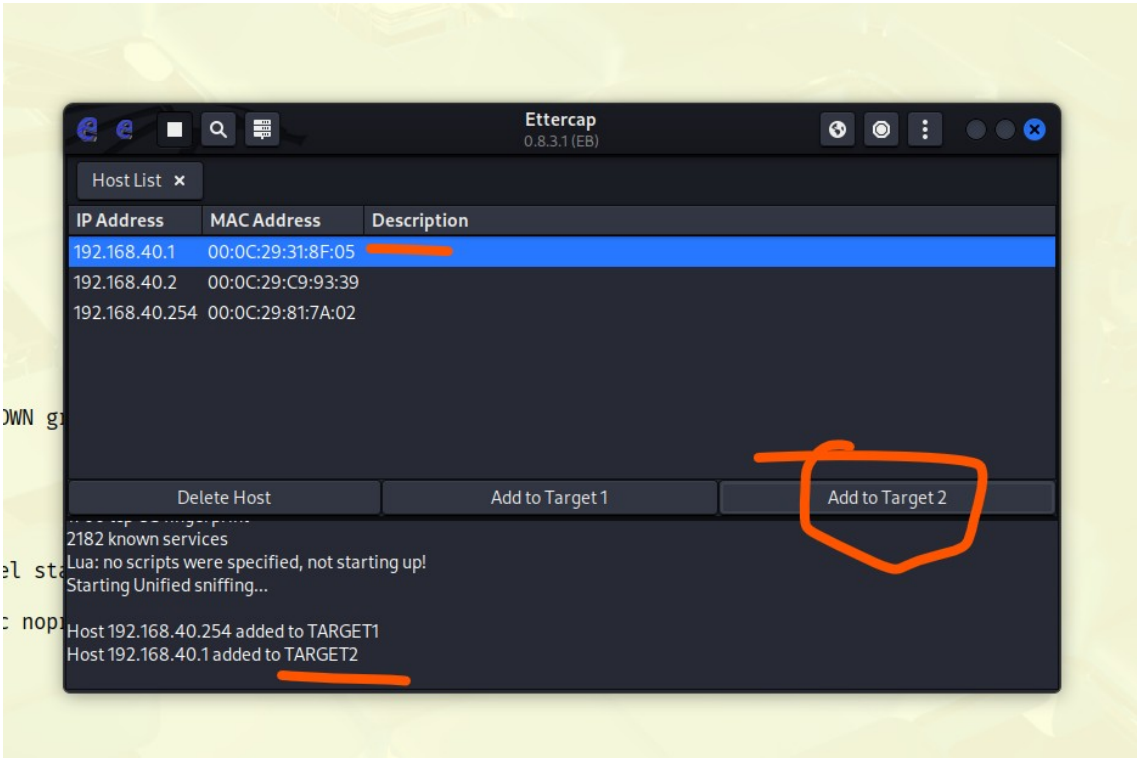


Ligar o router , pfsense -192.168.40.254

Adicionar o target 1 -pfsense no ethercqap



Definir Windows como target 2

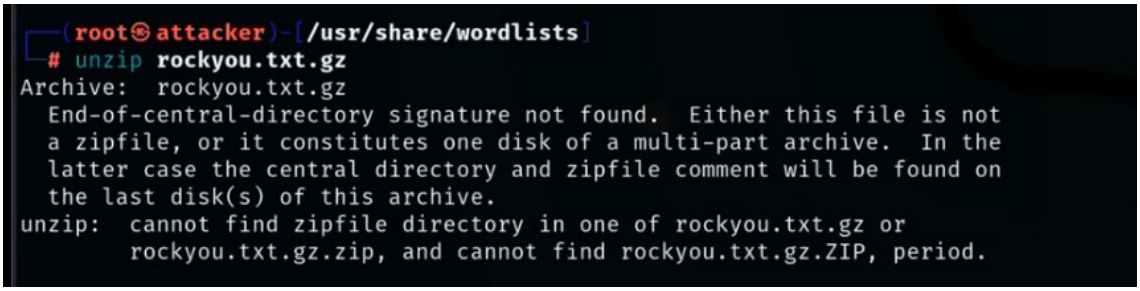


Ataque de Brute-force

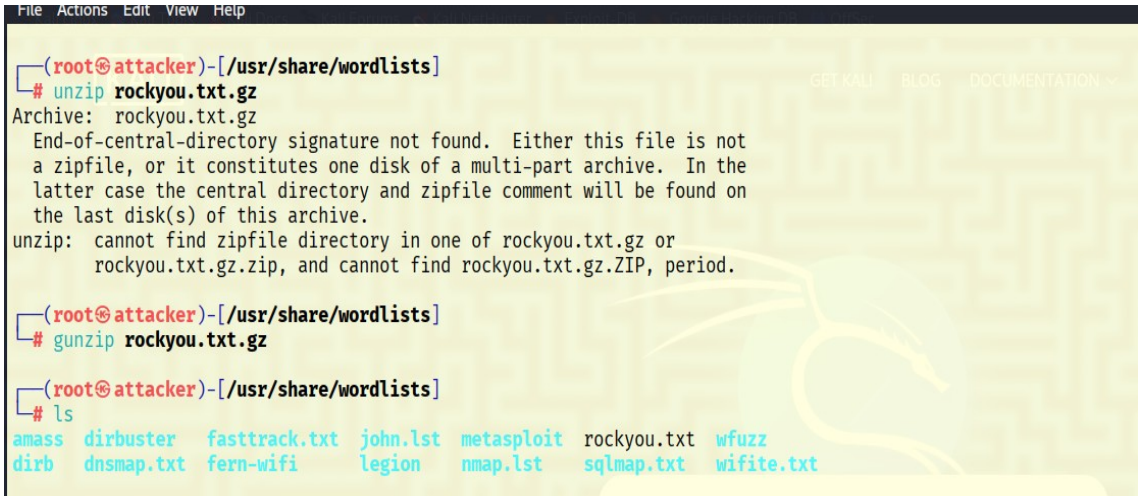
Deve ser feito um ataque de brute force ao serviço SSH estando o Kali fora da rede.



Fazer o Unzip do ficheiro “rockyou.txt.gz”



u.txt.gz



Aceder á directoria -metasploit :

┌──(root㉿attacker)-[/usr/share/wordlists]

└─# **cd metasploit**

(root㉿attacker)-[/usr/share/wordlists/**metasploit**]

└─# ls -la -→ **verificação de listagens de passwords**

total 5516

drwxr-xr-x 2 root root 16384 Jun 23 08:57 .

drwxr-xr-x 26 root root 4096 Jun 23 08:57 ..

-rw-r--r-- 1 root root 791 Jun 11 22:32 adobe_top100_pass.txt

-rw-r--r-- 1 root root 7706 Jun 11 22:32 av_hips_executables.txt

-rw-r--r-- 1 root root 754 Jun 11 22:32 av-update-urls.txt

-rw-r--r-- 1 root root 7418 Jun 11 22:32 burnett_top_1024.txt

-rw-r--r-- 1 root root 3585 Jun 11 22:32 burnett_top_500.txt

Etc.....

----Exemplo de ficheiro USERS

o – “http_default_users.txt “

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# cat http_default_users.txt

admin

manager

root

cisco

apc

pass

security

user

system

sys

wampp

newuser

xampp-dav-unsecure

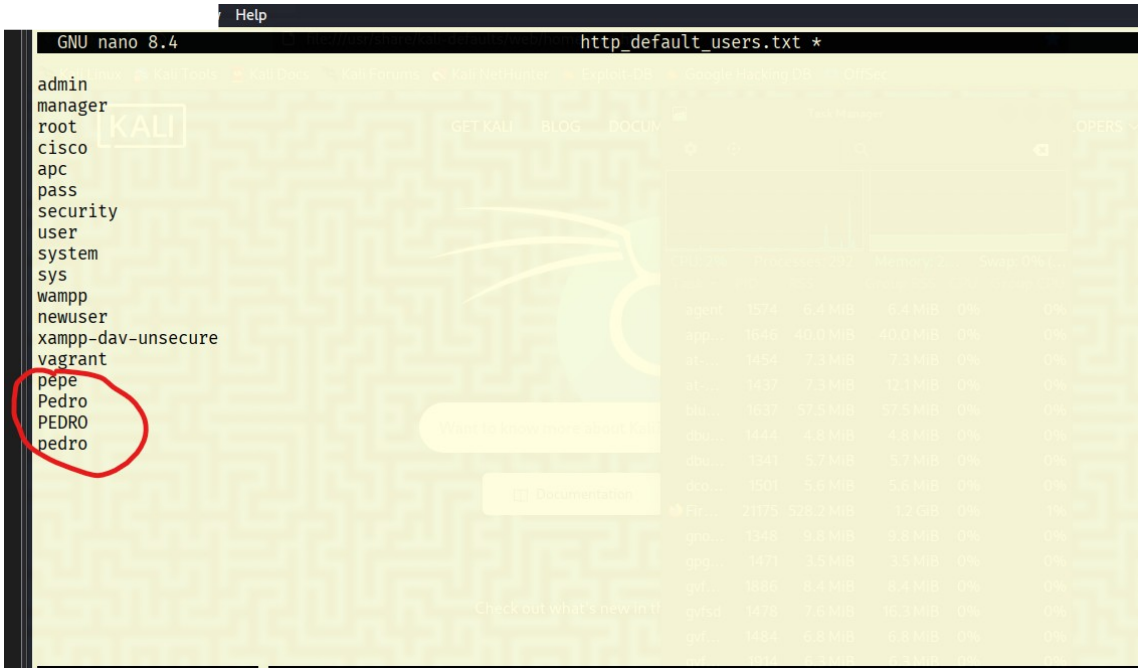
vagrant

Edição de ficheiro http_default_users.txt com comando nano

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# **nano http_default_users.txt**

Edição do ficheiro -linhas acrescentadas-a vermelho



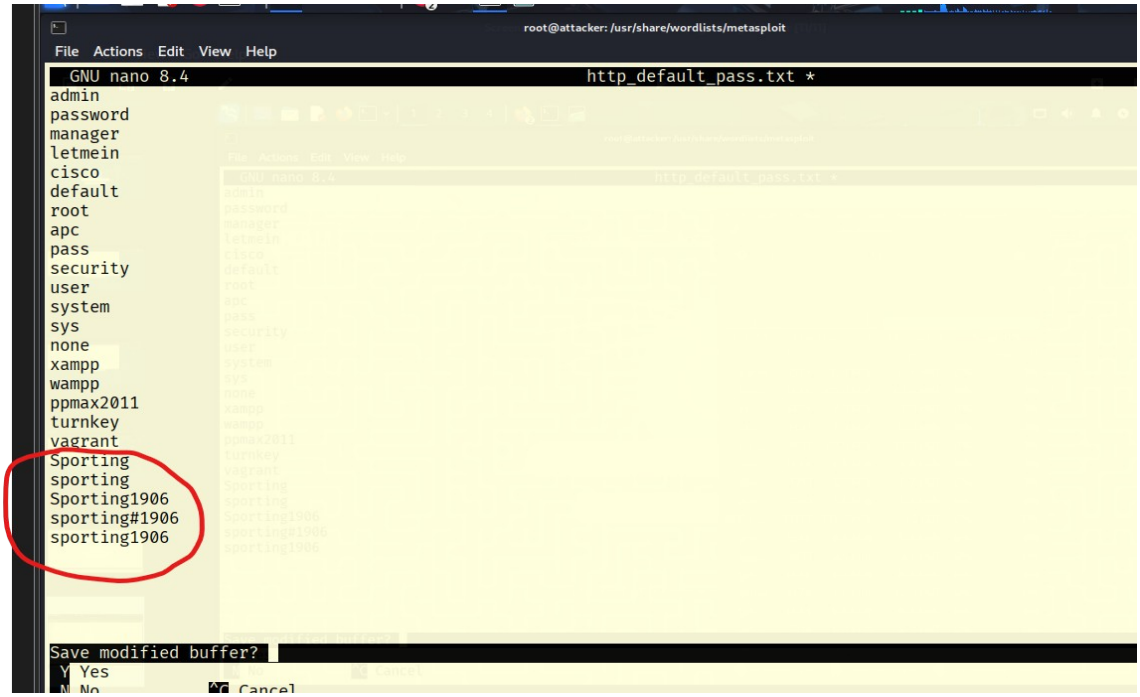
----Exemplo de ficheiro Passwords

Edição de ficheiro http_default_pass.txt com comando nano

┌──(root㉿attacker)-[/usr/share/wordlists/meta]

└─# **pepe**

Edição do ficheiro -linhas acrescentadas-a vermelho



aplicação Hydra para efectuar bruteforce

https://www.kali.org/tools/hydra/

-Edição de configurações ssh no RedHat

root@redhatserver:/home# cd /etc/ssh/

root@redhatserver:/etc/ssh# ls -la

-rw-r--r--. 1 root root 1916 Feb 18 00:00 ssh_config

root@redhatserver:/etc/ssh# nano sshd_config

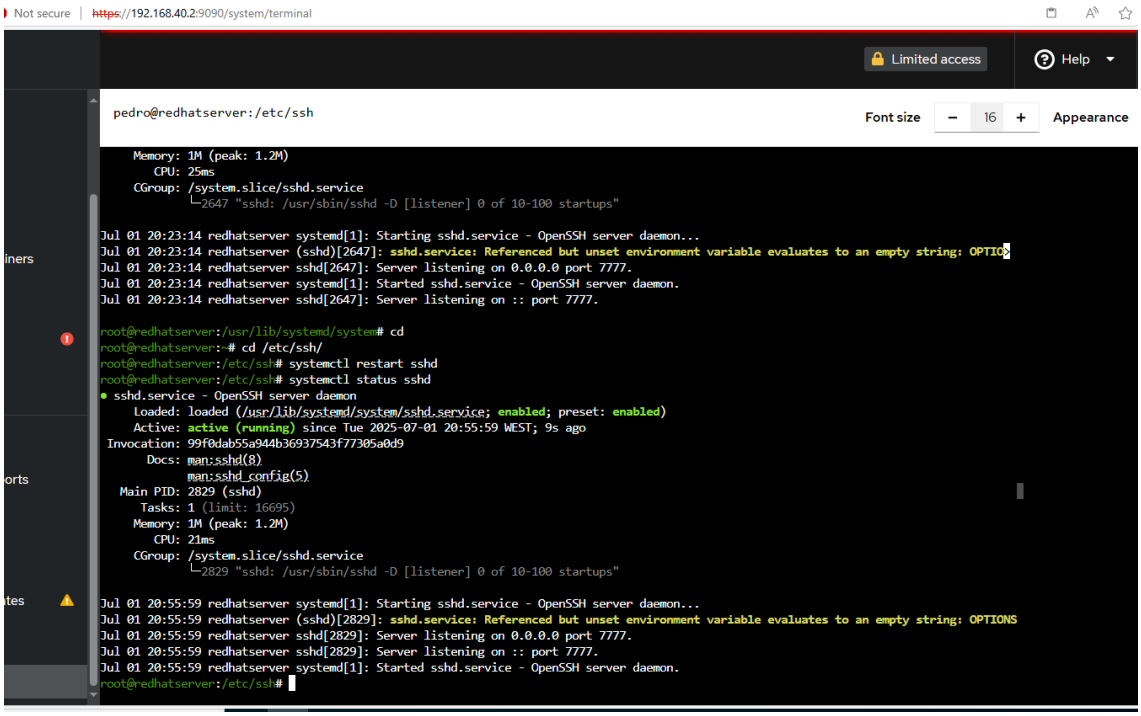
linha alterada

line **42 MaxAuthTries 10000** - → alteração de numerode tentativas para 2000

Restart do serviço ssh

root@redhatserver:/etc/ssh# systemctl restart sshd

root@redhatserver:/etc/ssh# systemctl status sshd



- OpenSSH server daemon

Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)

Active: active (running) since Mon 2025-06-23 20:30:55 WEST; 16s ago

-NO KALI

Teste com Hydra

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# hydra -l sporting -P http_default_pass.txt ssh://192.168.40.2 -t 4

Explicação detalhada:

Este comando é usado para realizar ataques de força bruta ao serviço SSH (porta 22) de uma máquina com o IP 172.20.27.2, tentando descobrir a palavra-passe do utilizador sporting, utilizando o ficheiro http_default_pass.txt como dicionário.

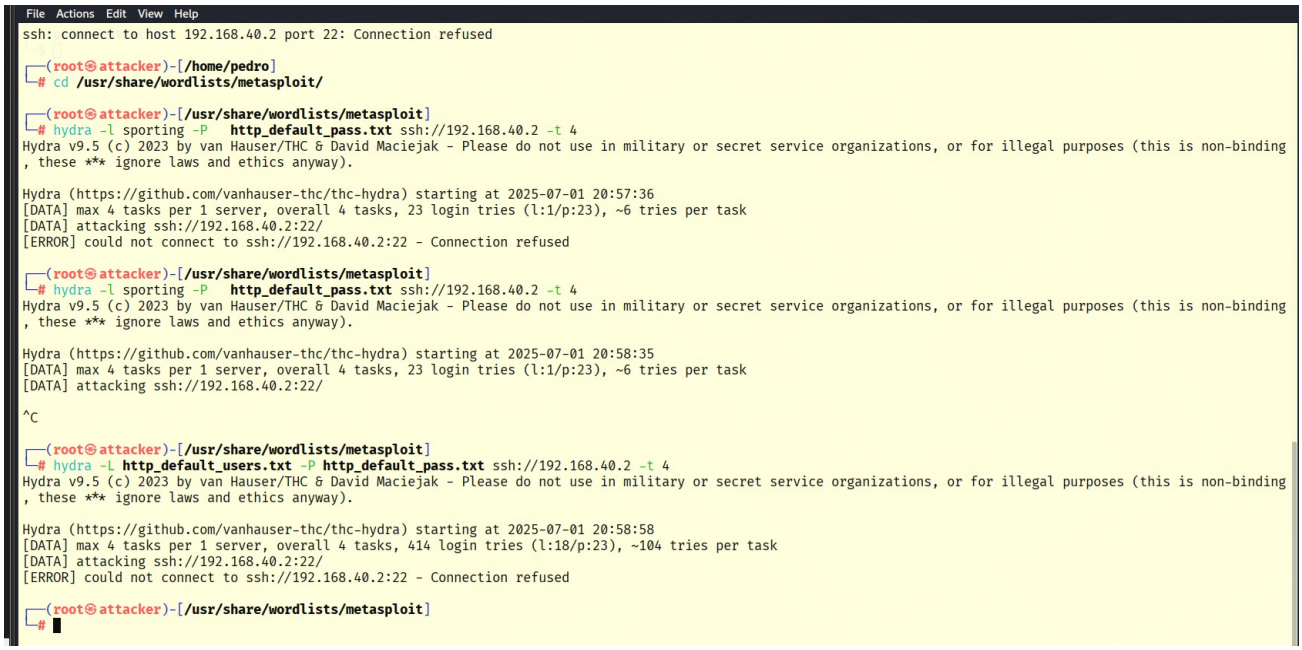
Explicação dos parâmetros

Parâmetro	**Significado**
hydra	Ferramenta usada para ataques de força bruta a serviços de login.
-l sporting	Login fixo a usar em todas as tentativas: sporting.
-P http_default_pass.txt	Ficheiro de texto com uma lista de palavras-passe (uma por linha).
ssh://172.20.27.2	Especifica o alvo, protocolo ssh, e IP de destino 172.20.27.2.
-t 4	Número de threads paralelas (nº de tentativas simultâneas): 4.

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

ttp_default_users.txt -P http_default_pass.txt ssh://192.168.40.2 -t 4

RESULTADO:



[INFO] Writing restore file because 2 server scans could not be completed

[ERROR] 1 target was disabled because of too many errors

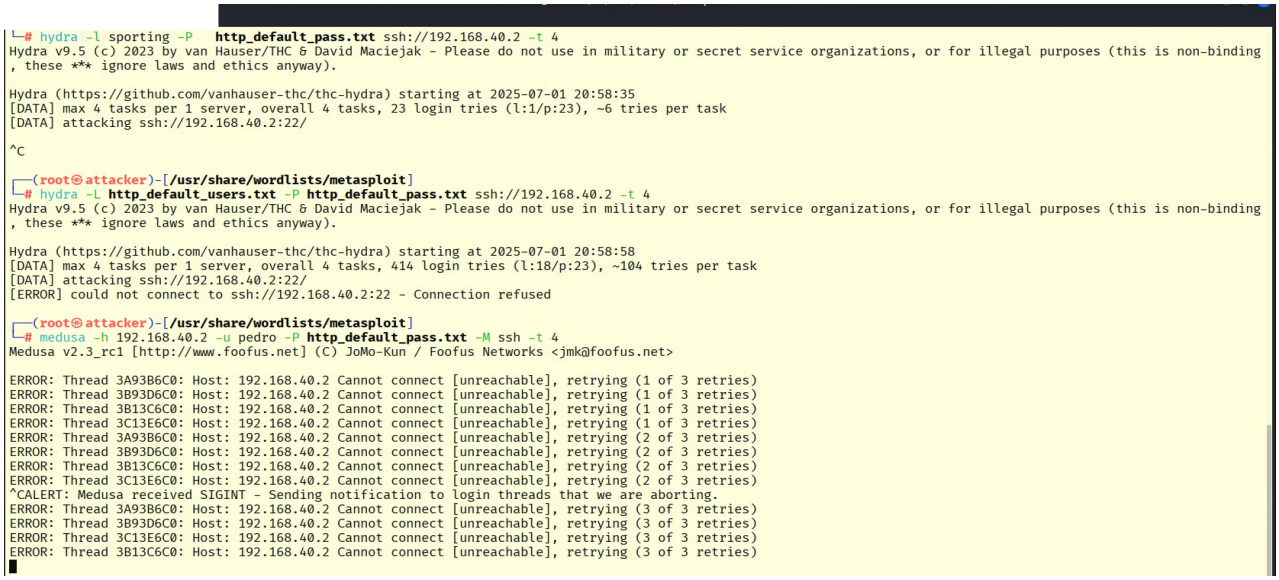
[ERROR] 1 targets did not complete

Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-06-23 20:14:47

Teste com Medusa:

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# medusa -h 192.168.40.2 -u pedro -P http_default_pass.txt -M ssh -t 4



Desbloquear o SELinux se estiver a bloquear os logins via SSH mesmo com a password correta.-OPCIONAL

root@redhatserver:/usr/lib/systemd/system# **nano /etc/selinux/config**

Agora que está em modo **permissive**, o hydra consegue autenticar corretamente, e os acessos passam — mas ainda são registados nos logs.

O que podes fazer a seguir:

✅ 1. Tornar o modo permissive permanente (opcional):

Edita o ficheiro de configuração do SELinux:

CopyEdit

nano /etc/selinux/config

Altera a linha:

SELINUX=enforcing

Para:

SELINUX=permissive

Depois guarda e fecha (Ctrl + O, Enter, Ctrl + X).

Criação de wordList

.org/tools/crunch/

exemplo :

┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# **crunch 1 3 abcdefghijlmnopqrstuvwxyz -O brute_force.txt**

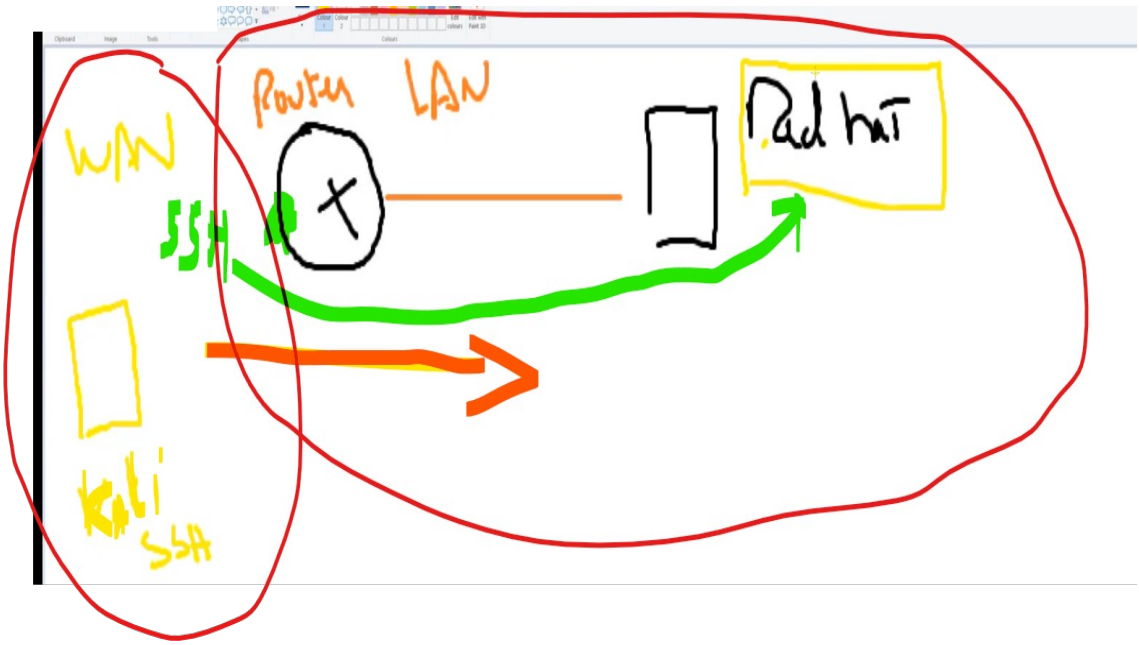
┌──(root㉿attacker)-[/usr/share/wordlists/metasploit]

└─# **crunch 10 10 0123456789 -t MyPass@@@@ -0 password_top.txt**

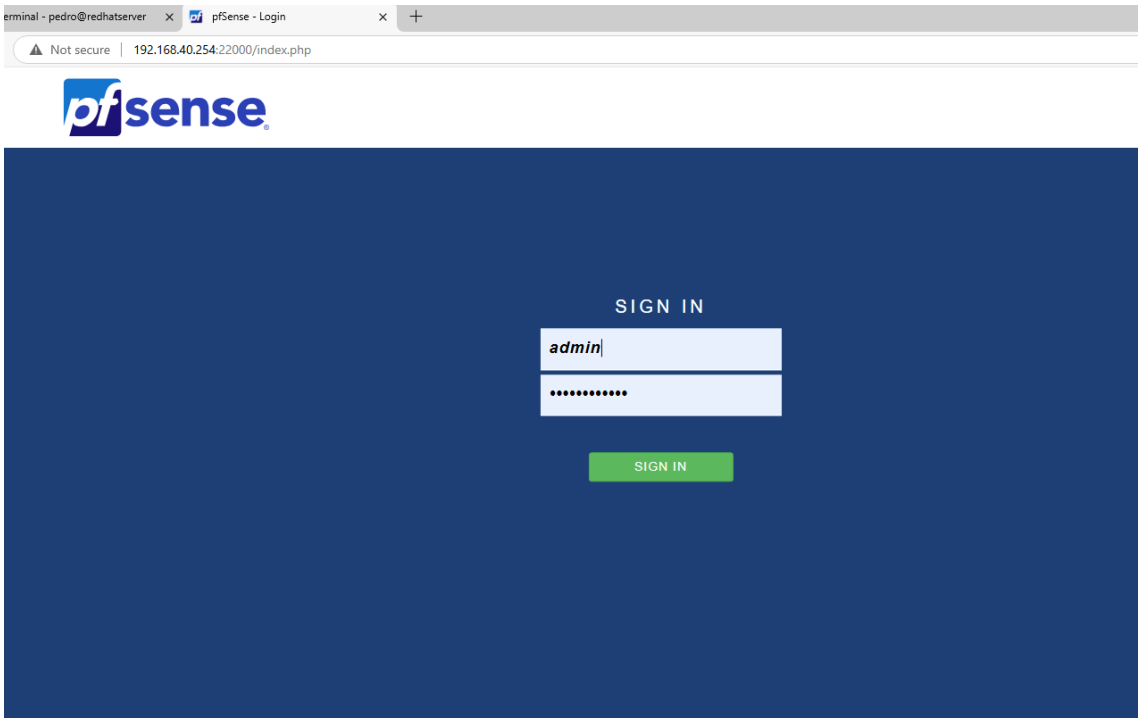
Comando **Explicação**

Explicação parte a parte:

Comando	Explicação
`crunch`	Ferramenta para gerar listas de palavras personalizadas
`10 10`	Define o **comprimento mínimo e máximo** das palavras gerar (neste caso, fixo: 10 caracteres).
`0123456789`	Define o **alfabeto a usar** na geração das partes variáve neste caso).
`-t MyPass@@@@`	Usa um **padrão fixo:** `"MyPass"` seguido de **4 caractere** substituídos por números (por causa do alfabeto).
`-o password_top.txt`	**Guarda** todas as palavras geradas no ficheiro `password`



Reinicializar o pfsense :



Resumo da tarefa (OSINT):



Esta é uma foto de um resort localizado numa ilha. Para execução desta tarefa é necessário que os formandos expliquem como chegaram às respostas através de printscreens.

a) Qual é o nome do resort?

b) Quais são as coordenadas da ilha?

c) Em que direção cardinal a câmara estava voltada quando a foto foi tirada?

Parte 3

efa (OSINT)

Este é um site propositado para recolha de informação e hacking.

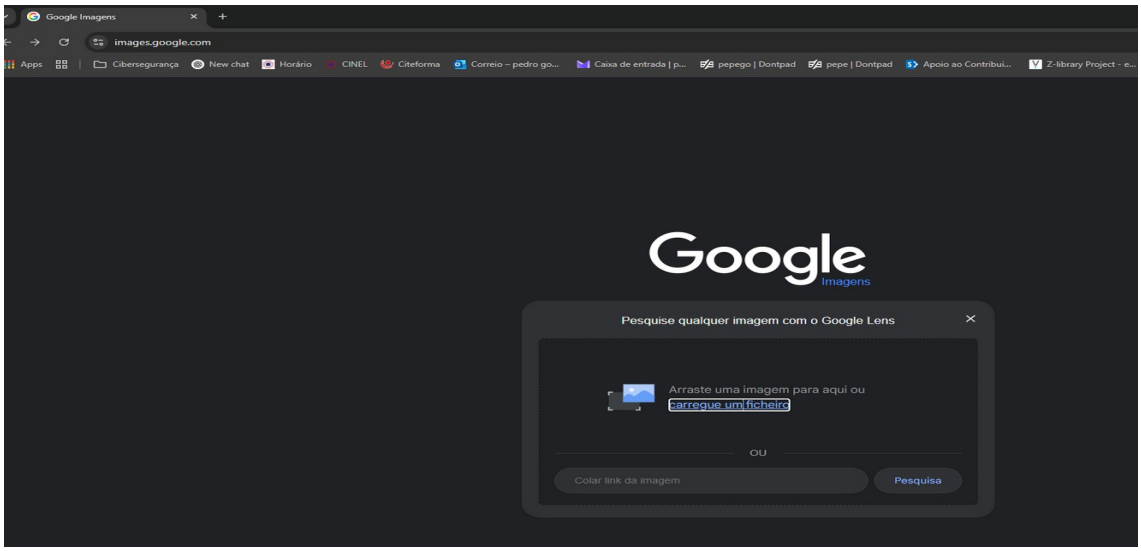
Hack This Site

De acordo com o feito em aula pede-se que sejam recolhidas o máximo de informações sobre o alvo que é o site acima.

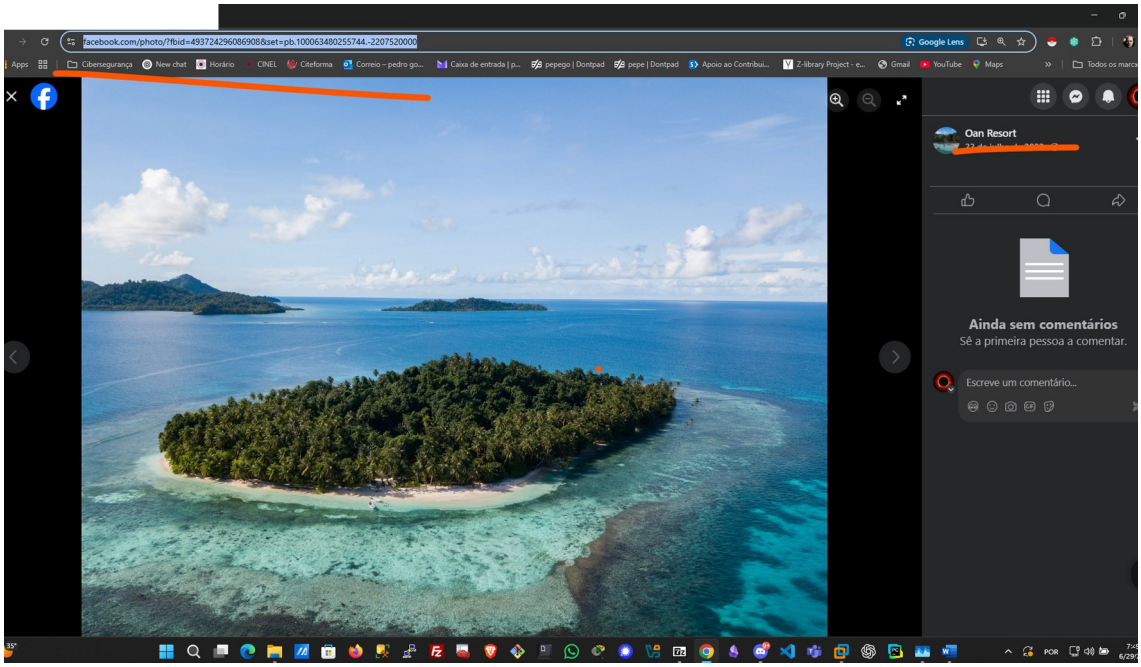
Tudo o que recolherem deve ser alvo de comprovativo pode ser mediante printscreens. (IP's,Domínios,Emails,Geolocalização, etc)

Pesquisa imagens em :

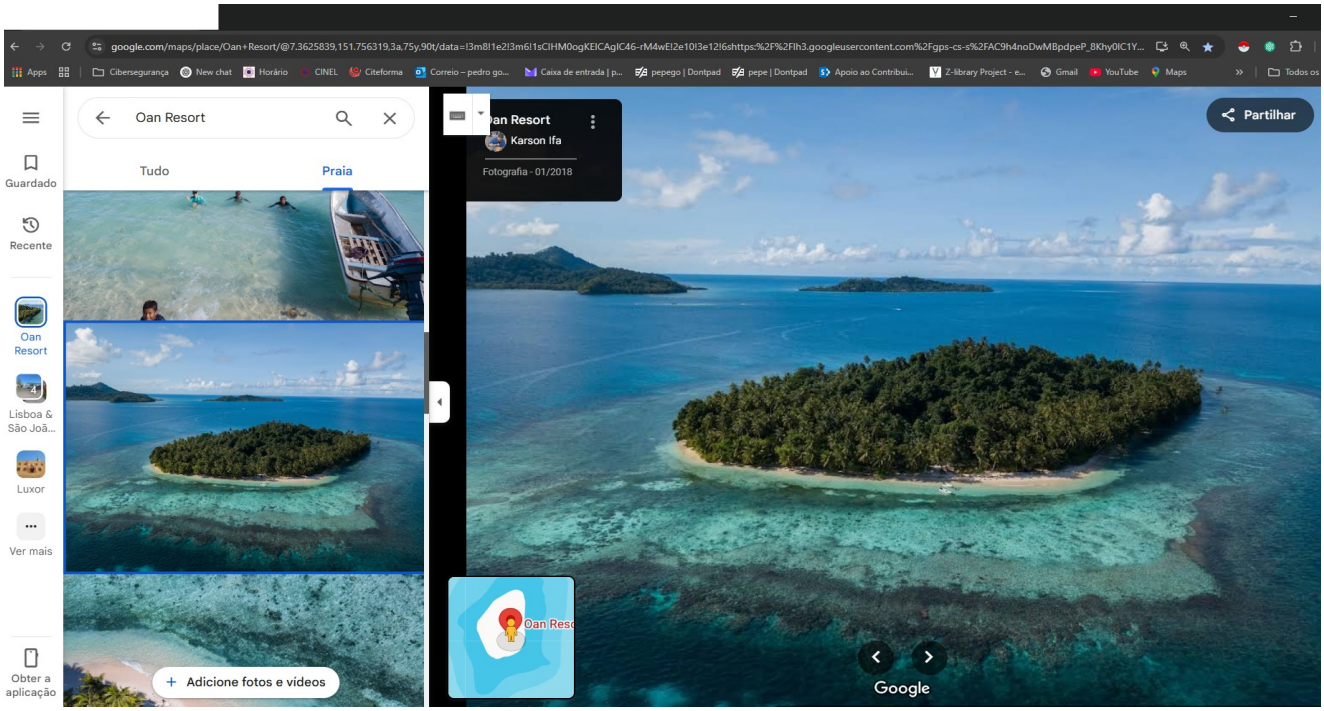
https://images.google.com/



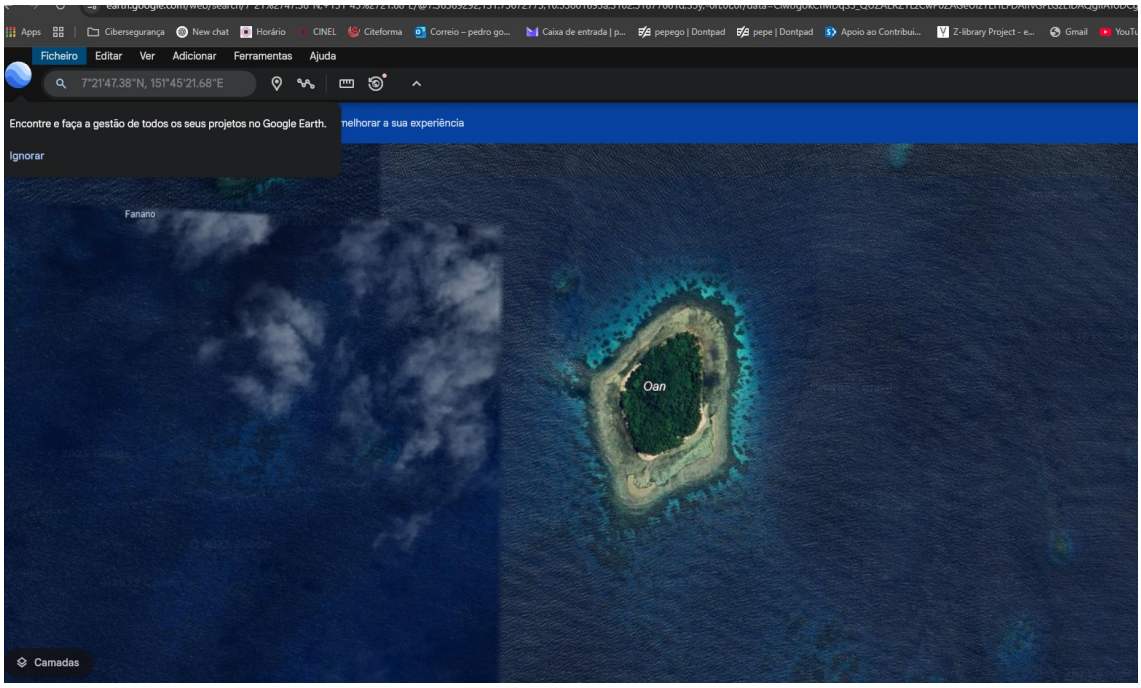
https://www.facebook.com/photo/?fbid=493724296086908&set=pb.100063480255744.-2207520000



https://www.google.com/maps/place/Oan+Resort/@7.3625839,151.756319,3a,75y,90t/data=!3m8!1e2!3m6!1sCIHM0ogKEICAgIC46-rM4wE!2e10!3e12!6shttps:%2F%2Flh3.googleusercontent.com%2Fgps-cs-s%2FAC9h4noDwMBpdpeP_8Khy0lC1YE8ub2lsBQq9XDtZlvY7yvbszoE8_r4wdQjx2TEk9WnUlq9hofYIiOC59CF0cF9LLr6NRl92lhzWmg0SNdrqe_E_ti0iEFAXgHng3zaVpwG8ogLMpHD%3Dw203-h152-k-no!7i3992!8i2992!4m9!3m8!1s0x6667a510cf78a285:0x675c6874bba0f42!8m2!3d7.3625845!4d151.7563188!10e5!14m1!1BCgwKCC9tLzBiM3lyMAE!16s%2Fg%2F11g10qwz9t?entry=ttu&g_ep=EgoyMDI1MDYyMy4yIKXMDSoASAFQAw%3D%3D



Island Coordinates **7°21'47.38"N, 151°45'21.68"E**



https://www.metadata2go.com/result#j=8e64dc49-0b80-4a1e-a323-49ec80bd42b3

METADATA 2 GO

The data shown is all the metadata we could automatically extract from your file. It may be neither complete nor adequate. Metadata could have been changed or deleted in the past. Please be aware that the metadata is provided without liability.

Metadata of Imagem1

Checksum	cc8d6672e472f5130fe0dd1e5f9fbe18
Filename	Imagem1.png
Filesize	351 kB
Filetype	PNG
Filetypeextension	png
Mimetype	image/png
Imagewidth	519
Imageheight	279
Bitdepth	8
Colortype	RGB with Alpha
Compression	Deflate/Inflate
Filter	Adaptive
Interlace	Noninterlaced
Srgbrendering	Perceptual
Gamma	2.2
Pixelsperunitx	3779
Pixelsperunity	3779
Pixelunits	meters
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Megapixels	0.145
Category	image